

# Observers Are All You Need

Bernhard Mueller<sup>1\*</sup> Alexander Osika<sup>2</sup> Mario Ponder Kai Xue Ben Cassie  
Peter Nguyen Maarten Antoine Visser Kale Arnav Anirudha David Matscheko  
Jonathan Hill William T. Glynn

August 6, 2026

**Paper release:** r2017 **Released:** August 6, 2026

**Author affiliations:** <sup>1</sup>Pragma Inc., Washington, USA; <sup>2</sup>SNRGY Inc., Gothenburg, Sweden.

## Abstract

Observer-Patch Holography (OPH) starts with finite observers that keep records, compare overlap readbacks, and repair disagreements. It treats public physical structure as the fixed point of those records and repairs. Finite repair gives normal forms and repeatable measurement records. On a declared signed-record branch, its normalized response metric has an exact rank-three quotient and completes to an abstract continuous Euclidean carrier. Its physical-space attachment is open. Modular screen geometry supplies Lorentz kinematics and a conditional Einstein branch. Complete reversible port response and endogenous observer transport force the local Standard Model gauge Lie algebra. Icosahedral incidence and target-blind port read-back derive a signed finite response. Under a conditional matrix current and fermionic matter contract, anomaly balance and tensor descent give exact charges, three colors, and the maximal faithful matter image  $(\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1))/\mathbb{Z}_6$  for the declared representation. Source selection of the matrix current, matter action, and physical  $\mathbb{Z}_6$  quotient enters as a premise. The local screen-cell closure has interval-certified roots on its declared maps. A bounded correctable-capacity completion class is non-identifying. The complete source antecedent and its physical verdict are unproved. A frozen primitive-port branch predicts three linked dispersion coefficients and one degree-six icosahedral angular shape; a qualifying mismatch rejects that branch. No physical sector, continuum, laboratory, or universe-level attachment is claimed.

**Keywords:** observer patch holography; quantum foundations; emergent spacetime; fixed-point consensus; Standard Model structure; quantum gravity

## What This Paper Contributes

This synthesis paper owns the observer interpretation and the two quantitative closures. The local equation

$$P = \varphi + \frac{\sqrt{\pi}}{A_T(P)}$$

is the stronger result: its fixed-point uniqueness schema and outward-rounded interval certificates give one root for each declared map on the full analytic domain. The direct global proposal

---

\*Corresponding authors: Bernhard Mueller, [bernhard@floatingpragma.ai](mailto:bernhard@floatingpragma.ai); Alexander Osika, [alex@snrgystudios.com](mailto:alex@snrgystudios.com).

$N = \log M_0(\mathfrak{U}_N)$  asks a trial universe to reproduce its logarithmic correctable-record capacity. The finite consensus and capacity algebra is established separately. A source-derived packet realizes the fixed-cutoff construction at  $D = 24$ . A bounded target-clean counterfamily has incompatible zero sets under shared base, positivity, and carrier controls. It does not establish universal all-rung membership in the complete A1–A3 capacity-source contract or an executable-to-Lean bridge. Direct  $N$  is not evaluable on that incomplete source antecedent, and the stronger source-class verdict is open. The source-derived physical hadronic transport for  $P$ , completion of the source antecedent and selection of one physical capacity for  $N$ , the universe-level carrier attachment, the horizon/record identification, and the common screen/electroweak load carrier are consumed as premises and are not constructed here. Numerical comparisons and falsifiers are stated in their owning sections.

OPH combines familiar mathematics under one operational constraint. Finite observers compare the records visible on overlaps, repair mismatches by declared recovery moves, and accept as public the normal form that survives those comparisons. The bounded observer patch is the primitive formal object. Its physical size and carrier realization are branch data. Spacetime, fields, and particles enter through additional physical readout maps. A machine-checked receipt makes all seven clauses theorem-level data at one fixed regulator. Its proper-overlap witness uses two operational observers whose local records differ before typed restriction, agree on a nonzero common section accessible to both, and fail after a controlled corner mutation. Cross-regulator naturality, source realization, and higher-overlap coherence are open. A companion common-origin theorem traces each visible event’s support region, public record observable, and Lorentz chart coordinate to one event packet (Section 2.2).

The declared twelve-port repair mean supports a conditional carrier construction. The normalized long-response kernel selects an intrinsic rank-three projector in port counting space. The limiting Gram form has a three-dimensional radical on the real six-control extension. The signed cumulative port-record/load module has no nonzero metric kernel. Its source readout is exactly onto  $\mathbb{Z}^6$ , and the conservative seam currents span the even-sum submodule  $D_6$ . Both modules embed densely in the quotient under the pullback response metric and complete to an abstract continuous local Euclidean translation carrier. The ordinary six-dimensional lattice metric on  $D_6$  is not used. The limit precedes the real quotient and integer-module completion; finite-step kernels retain full rank on the signed sector. The construction is canonical up to port-label-preserving isometry and chooses no Cartesian axes. Cumulative record addition acts simply transitively on the exact record carrier and by isometries on its completion. Under explicit A2-natural feasibility and objective premises plus an A3 unique-minimizer premise, the internal seam weights are forced to  $1/60$ . This gives an exact internal Markov average and the translation-invariant Dirichlet generator  $L = I - P$ . Plane waves diagonalize  $L$ , its carré-du-champ is a positive sum of squared seam differences, and response-metric normalization gives the complete thirty-edge cosine character. The source theorem fixes this internal record action under its named selection premises. It does not identify the action with a physical field, position, clock, or propagation law. A1–A3 do not by themselves provide a proved selection of the repair mean, the commutative signed-record module, or the Gram readback topology. Overlap/refinement gluing, physical scale, and the identification of record translations with physical space are open.

The consensus paper proves the same-source finite normal-form result. The presentation-invariant treatment in Ref. [1] separates hidden implementation choices from quotient-visible data. It does not identify carriers with different visible port incidence, topology, record processes, or repair laws. The spacetime and Einstein paper owns the controlled modular route to Lorentz and Einstein structure. The complete twelve-port response in A1 and the endogenous overlap holonomy in A2 force the abstract local compact Lie algebra  $\mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(3)$ . This result does not assume

an ambient continuous gauge group or a matrix current. The distinct transportable-sector/Tannaka route is conditional. Under the explicit inverse-port contract, incidence and target-blind inverse-port readback independently derive  $R = -J$ , without reconstructing the current generators or their bracket. Under the gauge paper’s conditional matrix-current and rank-15 matter contracts, anomaly balance and tensor descent give the Standard Model charge lattice, common  $\mathbb{Z}_6$  kernel, and maximal faithful matter image. Source selection of the matrix current, matter action, and physical global quotient, together with laboratory identification of the finite current and flux sectors, physical family attachment, and equality with the Tannaka current, enters the chain as a named premise set. The particle paper contains the downstream hierarchy, mass, and flavor calculations; the screen paper contains the finite carrier and record model. The common-domain Einstein tower and the physical closure packets required for  $P$  and  $N$  are consumed premises, not constructions of this paper.

The exact word algebra of the registered recurrence is  $\text{span}\{I, A, A^2, A^3\}$ . It is commutative and four-dimensional, so it does not supply the twelve generators, their bracket, or the nonidentity proper rechartings required by the physical current premise. This bounded result leaves order-sensitive port perturbations open. The target-free diagonal phase lift has first-order rank twelve and an abelian bracket. Adding the connected adjacency tangent generates  $\mathfrak{u}(12)$ , with derived rank 143. The required current therefore needs a non-diagonal source response with derived rank eleven. Other port actions are not covered by this control. For the canonical oriented carrier, the complete target-free  $A_5$ -equivariant alternating-bracket space over  $\mathbb{Q}$  has dimension fourteen and an exact rational basis. This is a search-space theorem. It does not select a Jacobi bracket, a compact current, source histories, or holonomy. The complete Jacobi condition has exact quadratic coefficient-row rank 38 and an  $11+27$  rowspace decomposition. The compact real locus of the resulting fourteen-parameter variety is exactly three families, two of them mirror  $\mathfrak{su}(3) \oplus \mathfrak{so}(3)$ -type cells, conditional on three named textbook lemmas (Section 3); the noncompact components away from the analyzed channel supports remain unclassified [12, 13].

This paper supplies the shared interpretation and closure language. Its claims are typed: finite theorem, controlled scaling result, conditional physical branch, numerical diagnostic, and interpretation are not interchangeable. A counterexample to a load-bearing theorem, a nonunique or target-leaking closure, failure of the physical readout premises, or a carrier-dependent change in the observable quotient falsifies the corresponding OPH claim. The axiom-forced local gauge Lie algebra, conditional matter image, controlled Lorentzian limit, and two dimensionless closure equations define the integrated claim. The typed premise sets above delimit the boundary between this theorem chain and full physical closure.

## One typed construction and its source gates

The finite carrier, accepted repair, support geometry, gravity, abstract compact current, and conditional matter belong to one typed construction. Accepted repair produces the quotient-visible normal form. A1’s complete twelve-port response and A2’s endogenous holonomy force the abstract local Standard Model gauge Lie algebra. A declared matrix current and matter packet provide conditional realizations, while the source producer derives only the inverse-port response. Each composition uses the objects and premises displayed here.

## Three meanings of screen

The word “screen” is used for three related objects that must not be identified without a receipt.

1. The *local carrier boundary* is the twelve-port oriented interface of one Echosahedral carrier on the declared branch. Its incidence has  $(V, E, F) = (12, 30, 20)$ .
2. The *federation screen* is the routed system of interfaces, records, repairs, and checkpoints of many carriers at finite cutoff.
3. The *support screen* is the observer-facing geometric chart. On the spherical branch it is the refined conformal  $S^2$  used for caps, collars, modular flow, and Lorentz reconstruction.

Local icosahedral incidence does not determine the topology of the federation nerve. A federation of identical local carriers can be routed as a path, a cycle, a higher-genus complex, or a spherical complex. The map from routed carriers to a support-visible spherical nerve is therefore a physical bridge, not a change of notation.

### Structure-sensitive, presentation-invariant physics

OPH is not neutral under arbitrary changes of substrate. It is invariant under changes of presentation that preserve the complete observer-visible carrier signature. On the Echosahedral branch that signature contains

$$\mathcal{C}_{i,r} = (\mathcal{A}_{i,r}, \rho_{i,r}, P_{i,r}, I_{i,r}^{\text{or}}, \mathcal{R}_{i,r}, \mathcal{U}_{i,r}, \text{Chk}_{i,r}, \text{Resp}_{i,r}, c_{sr}),$$

where  $P_{i,r}$  is the port set,  $I_{i,r}^{\text{or}}$  is oriented incidence,  $\mathcal{R}_{i,r}$  is the record algebra,  $\mathcal{U}_{i,r}$  is the repair or feedback interface,  $\text{Resp}_{i,r}$  is the visible response law, and  $c_{sr}$  is the refinement lineage. Hidden coordinates, port names, worker partitions, materials, and wiring presentations are silent when an isomorphism preserves this whole tuple and its error model. A change in port number, incidence, orientation, accessible algebra, response, repair law, clock, or refinement lineage need not be silent. A cube and an icosahedron are therefore different carrier contracts even when both are built from the same material.

A carrier body is not automatically an observer. It realizes an observer only when it supplies bounded access, self-readback, durable records, record-conditioned feedback, boundary prediction against controls, and checkpoint continuation. One carrier may pass that test. A connected sub-federation may pass it instead. No theorem fixes primitive observer size by counting carrier bodies.

### The common finite computation

At cutoff  $r$ , source-bound carrier data are routed into an observer-patch federation. Accepted repair then acts on the physical quotient:

$$\begin{aligned} \text{SourceCarrierTower}_r &\xrightarrow{\text{realize/route}} \text{ObserverFederation}_r \\ &\xrightarrow{\pi_r} \text{PhysicalQuotient}_r \xrightarrow{\text{Rep}_r} \text{PublicNormalForm}_r. \end{aligned}$$

The last arrow is the consensus result only under semantic-dependency-complete transactions, coherent union-collar payloads, repair completeness, local diamonds, protected records, and the stated endpoint conditions. A collection of oscillators with equal frequency does not supply those clauses.

Physical phase locking can instantiate one synchronization layer. For a routed edge  $e = ((i, a), (j, b))$ , a source-produced phase record may certify frequency entrainment and a stable relative phase,

$$\dot{\theta}_{i,a} - \dot{\theta}_{j,b} \longrightarrow 0, \quad d_{S^1}(\theta_{i,a} - \theta_{j,b}, \delta_e) \leq \varepsilon_e.$$

That certificate becomes a consensus parent only when the phase record fixes a commensurability map for the exposed packets and is tied to the accepted repair ledger, semantic records, an independently calibrated clock, and the confluence premises. Phase locking can synchronize an interface. It does not by itself make the interface an observer, settle semantic disagreement, or produce physical time.

## Two downstream projections of one source

The public normal form has two separately typed projections:

$$\begin{array}{c}
\text{PublicNormalForm}_r \xrightarrow{\text{carrier-to-support}} (\text{Support}_{S^2,r}, \text{FiniteCapBWCertificate}_r), \\
\left. \begin{array}{l} \text{FiniteCapBWCertificate}_r \\ \text{CompatibleModularStateTower}_r \end{array} \right\} \xrightarrow{\text{same tower}} \text{BW/KMS}_r \longrightarrow \text{Lorentz}/H^3_r \\
\longrightarrow \text{Events}_{3+1,r} \longrightarrow \text{Einstein}_r, \\
\left. \begin{array}{l} \text{CompleteResponse}_{12} \\ \text{EndogenousHolonomy}_{A_5} \end{array} \right\} \longrightarrow \mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(3) \\
\xrightarrow{\text{conditional matrix and matter packet}} \text{MaximalFaithfulMatterImage} \\
\dashrightarrow \text{PhysicalGauge/QFT}.
\end{array}$$

The dashed arrow is the open source and laboratory attachment.

The finite carrier-to-support leg is constructed from one oriented icosahedral incidence nerve: twelve carrier charts, thirty seam algebras, and twenty nonvacuous triple restrictions. The same bound artifact supplies confluent seam repairs, an operational observer receipt, and a refinement-natural oriented  $S^2$  support limit. The separate geometric  $2\pi$ -KMS comparison and  $\text{FiniteCapBWCertificate}$  are not outputs of that finite bridge. The state tower, common-comparison maps, compatible state/vector data, modular controls, and cofinal modulus form an independent compatible modular-state tower. The BW theorem consumes both inputs on the same refinement tower. Once the support leg produces a conformal  $S^2$ ,  $\text{Conf}^+(S^2) \cong \text{SO}^+(3,1)$  and  $H^3 = \text{SO}^+(3,1)/\text{SO}(3)$  is exactly three-dimensional.  $H^3$  is the observer-frame fiber. A 3+1-dimensional event manifold requires the population/realization, separation, rank-four affine-chart, overlap-cocycle, held-out quadratic-cone, and causal-reachability receipts (E1)–(E6), together with the MI/assembly premise. Operational-clock gluing separately requires observer-readable transitions, event correspondence, affine calibration, cycle identity, and normal-form invariance. The Einstein relation additionally requires the common-domain stress, entropy, vacuum, coupling, scale, and remainder packet. Hidden Cartesian coordinates of a finite carrier are ineligible as support-screen, event, or Lorentz data.

The second projection begins with an exact finite result on the certified Echosahedral lineage. The twelve-port module decomposes as

$$P_{12} \cong_{A_5} \mathbf{1} \oplus \mathbf{3} \oplus \mathbf{3}' \oplus \mathbf{5}.$$

The declared integer counting and normalized central-readback cost realization gives the twelve unit lines and exact gap. An append-only signed-event machine generates those integer loads. Conservative whole-unit seam repairs preserve total load and strictly decrease  $V(N) = \sum_i N_i^2$ ; minimum move count is natural under every carrier rotation and the declared refinements. Divisibility

of total load by twelve is necessary for consensus, while an explicit eighteen-move path settles the declared full-pile packet. A half-unit display rescales event values and the repair threshold together, so it is a units convention on the same move graph. Oriented incidence independently gives the antipodal pairing, proper  $A_5$  action, and rank-three Gram frame, and determines the antipode  $J$ . Complete reversible response and endogenous overlap transport force the abstract local Standard Model gauge Lie algebra. Under the explicit inverse-port contract, the signed central involutive responses are exactly  $\pm J$ , with common sign conventional. Conditional also on the matrix current and rank-15 matter contract with its unique charge-conjugate projector pair, anomaly and tensor-descent certificates give the hypercharge lattice, common  $\mathbb{Z}_6$  kernel, and maximal faithful matter image. The scalar scan fixes compatible charges and Yukawa channels, not scalar multiplicity. The target-blind producer derives the inverse-port response, without selecting the matrix current. Separate band premises select the rank-three response. Tensoring it with the declared generation table gives a conditional rank-45 candidate whose chirality and diagonal  $\mathbb{Z}_6$  action come from that table. A distinct local-domain receipt checks the declared tensor-identity operator and conditional gap inheritance without source-selecting the matter action or transporting the twelve-port Spin packet. Laboratory identification of the finite matter carriers, current, and line sectors, exclusion of extra light sectors, physical matter-pole and continuum family identification, scalar attachment and dynamics, and quantum-field construction require separate maps.

The compact sector-category/Tannaka route conditionally reconstructs a compact group by a logically independent route. Physical family identification and extra-sector exclusion require separate identifications; they do not enter the contract-conditional finite gauge implication. The physical family map remains an explicit premise on charged-lepton and quantitative selector/gap branches elsewhere. Physical unification requires a source-bound commuting square identifying its reconstructed compact group with the group acting through the Echosahedral current response:

$$\begin{array}{ccc} \text{A5PortResponse}_r & \longrightarrow & G_r^{\text{screen}} \\ \downarrow & & \downarrow \simeq \\ \text{TransportableSectorCategory}_r & \longrightarrow & G_r^{\text{DR}}. \end{array}$$

On those premises the abstract Lie-type agreement is exact. The physical vertical maps and the source identity of the two group actions are open. In the same way, the rank-three response band and declared generation table form a conditional complex rank-45 candidate, while three physical generations require matter-pole, continuum, seam-selection, persistence, and complement-complete refinement receipts. The rank-three result depends on the complete-band and cost-order premises; it is not a consequence of the icosahedral graph alone.

## Finite controls and claim boundaries

The finite  $A_5$  evaluator control has 60 reachable correctable public records on  $\mathcal{H}_k = \ell^2(A_5) \otimes \mathbb{C}^k$ :

$$M_0 = 60, \quad D_{\text{raw}} = 60k, \quad \Delta_{\text{raw}} = 60(k - 1).$$

Raw equality occurs only at  $k = 1$ . Publicly inert multiplicity makes  $D_{\text{raw}}$  implementation-dependent, so the result is an evaluator control rather than physical capacity closure.

The unified claim has a precise scope. Consensus and geometry/gravity are composable branches of one source-bound self-reading carrier tower. The finite gauge branch uses the same carrier architecture, and A1–A2 force its abstract local Standard Model gauge Lie algebra. The released matrix current and matter action are conditional realizations until a same-source bridge joins them to that tower. Local icosahedral incidence by itself constrains only the carrier module. The

physical maps that turn these constraints into one inhabited universe are named premises. Matching dimensions or symmetry labels does not supply them.

## Canonical Observer Language

An abstract observer patch is the algebraic object

$$\mathcal{O}_i = (\mathcal{A}_i, \rho_i, \mathcal{R}_i, \{(\mathcal{I}_e, \pi_{i,e})\}_{e \ni i}, \mathcal{U}_i, \text{Chk}_i).$$

Here  $\mathcal{A}_i$  is the finite or regulated accessible algebra,  $\rho_i$  the accessible state,  $\mathcal{R}_i$  the exact or approximate record algebra,  $\mathcal{I}_e$  the overlap-visible interface algebra,  $\pi_{i,e}$  the visible restriction,  $\mathcal{U}_i$  the allowed update and repair instruments, and  $\text{Chk}_i$  the checkpoint data used for continuation.

A support patch, such as a cap  $P_i \subset S^2$  or a causal diamond, is a geometric chart of the abstract patch on a geometric branch. A carrier patch is a physical or digital implementation that realizes the same visible interface and record statistics within a declared error. Echosahedral hardware is therefore a reference carrier architecture for a fixed-cutoff implementation surface. The physical claim lives at the quotient of visible records, interface statistics, repair maps, and checkpoint continuation.

The quotient removes presentation data such as coordinates, labels, worker layout, and hidden ancillas. Quotient-visible port number, oriented incidence, topology, and response maps are physical branch data when the observer can recover them. Implementation invariance therefore does not imply carrier neutrality.

The observer patch is therefore the formal access structure used by the proof. A human observer, a horizon cap, and a hardware module can realize or chart it, but none of those examples is the definition. Physical predictions attach to quotient data: visible interface statistics, record readout, repair maps, and checkpoint continuation. A carrier implementation is admissible only when those data are invariant within its declared error model.

**Definition 0.1** (Semantic observer history). *For an observer token  $O$ , the observer-readable history is a labeled causal poset*

$$\mathbf{H}_O = (E_O, \preceq_O, \ell_O).$$

*The elements of  $E_O$  are semantic record events,  $\preceq_O$  is physical or informational dependence, and  $\ell_O$  is the observer-visible event label. Event identity is derived from the canonical semantic payload, observer token, visible footprint, and semantic causal parents. It is not derived from worker ID, repair iteration, queue position, retry counter, timestamp, or packet latency metadata. Latency is execution gauge only when it does not alter semantic causality, delivered data, timeout-sensitive operations, or operational clock records.*

**Definition 0.2** (Observer registry groupoid and namespaces). *Observer identities live in a registry groupoid, not in a flat UUID table. Objects are observer tokens*

$$\text{RID}(O) = (\text{kind}(O), [\text{birth}(O)]_{\mathbf{H}}, \text{lineage}(O)),$$

*where  $\text{kind}(O)$  lies in the disjoint namespace*

$$\{\text{patch observer}, \text{cap observer}, \text{future observer}\}.$$

*The birth event is a semantic event class in  $\mathbf{H}$ , and lineage is carried by explicit continuation, split, and merge arrows. Local registries on overlapping presentations must preserve observer kind, birth*

event, lineage arrows, visible observer structure, and checkpoint cuts. The global registry is the descent groupoid or colimit of those local registries when the cocycle and trivial-monodromy checks pass. A split creates child tokens with parent links; it does not silently assign one identity to multiple descendants.

**Definition 0.3** (Operational observer clock). *The modular parameter of an observer-facing cap pair is not, by itself, an arbitrary operational clock reading. A clock claim supplies an observer-accessible instrument*

$$\Theta_O : \text{Chk}_O(D) \rightarrow \text{Prob}(\mathbb{R})$$

and a declared affine calibration law. For a corresponding observer  $O'$ , the required finite or exact certificate has the form

$$\Theta_{O'}(\Phi_{O*}\rho) \approx (A_{a,b})_*\Theta_O(\rho), \quad A_{a,b}(\tau) = a\tau + b,$$

with an explicit residual bound from state mismatch, calibration error, and record-readout error. Execution counters are provenance; observer record order, modular parameter, and clock uncertainty are separate semantic or operational fields.

**Remark 0.4** (Observer-clock naturality boundary). *This paper proves observer-first quotient and readback claims only where the cited theorem surface supplies the required data. Scheduler-independent observer history and clock naturality require a history-augmented quotient relation with history-coherent diamonds, registry descent, state-preserving observer-algebra extraction, support-cap chart naturality, and the operational clock instrument above. A public evidence bundle can falsify the claim by showing counter-dependent semantic event keys, duplicate or namespace-colliding registry identities, broken registry cocycles, non-isomorphic semantic history DAGs across executors, or clock residuals exceeding the declared bound. Implementation field names alone are not the proof.*

The formal time-and-order ledger enforces this boundary at the type level. Universe closure, repair execution order, observer record order, modular parameter, worldline realization, clock readout, proper time, and an optional global time function are eight separate structures. In the committed source environment, the complete matrix of 56 ordered transitive coercions between distinct ledger layers fails to elaborate. Interpretations pass through an explicitly named realization map without a function coercion. More strongly, composition with any strictly increasing real function preserves the bare clock interface. A three-record control sends readings (0,1,2) to (0,1,8) and proves that the resulting clock is not related to the original by any positive affine map. Record order therefore supplies order data and an order-compatible scalar readout, while fixing neither affine structure, origin, nor physical unit.

The bounded observer-time continuation keeps the missing receipt visible. If a supplied event atlas and record map satisfy an affine coordinate law along a supplied future-unit timelike direction, every precedence has future-timelike displacement in every overlapping chart. On this calibrated branch the positive clock increment along that same supplied affine history is additive, and its square equals the chart-invariant Lorentz quadratic interval. One shared public event cannot determine affine rate and origin. Two ordered shared events determine the unique positive-affine interpolation of their four readings. At a third shared event, affine consistency is equivalent to an exact cross-multiplication equation. Distinction of the event and both readings from the two anchors makes this a nondegenerate check with no new fit parameter. Calling it held out requires a separate predesignation and custody protocol. The affine and cubic controls synthesize separate atlases from their clocks; they pass and fail this algebraic check without constructing two physical clocks



on one calibrated geometry. A separate literal three-record, two-chart example proves the geometric interface is inhabited. These are bounded conditional algebraic theorems with finite controls: the source atlas, affine unit-speed law, shared-event correspondences, refinement transport, physical instrument, common physical clock geometry, predesignation protocol, three-clock network, and SI unit are not constructed, and no global time function or modular-to-clock identity follows.

The companion formal tower interface packages the next structural layer in one object: finite observer and record fibres, observer-indexed record orders, private matrix algebras, commutative public subalgebras, selected states and generators, and functorial refinement maps with explicit compatibility laws. An exact constant adaptor packages one existing projective partition and certified state using a discrete order and zero generator. This is an inhabited packaging witness, not a nonconstant source-derived tower, repair endpoint, causal net, physical evolution, geometry, clock, or continuum limit.

This definition does not fix the primitive observer size. The axioms require finite or regulated access at fixed cutoff and permit connected support regions with different finite or regulated algebras. A primitive observer may therefore be a variable finite federation of screen cells. The stronger claim that elementary carriers are isomorphic fixed-capacity cells would require a separate branch: homogeneous UV cellulation, isomorphic local cell algebras, equivariant overlap maps, refinement preservation of the cell type, and a proof identifying one cell or a fixed block of cells with the primitive carrier. One Echosaedron is therefore a candidate primitive carrier on the homogeneous branch. It is not the definition of an observer patch.

## Failure Tests

OPH is wrong as stated if a load-bearing theorem admits a counterexample under its declared assumptions, if the controlled modular/Einstein branch cannot be instantiated under physically realizable conditions, if the exact rank-three screen-band theorem fails under its named premises, if a claimed physical family or sector attachment is contradicted inside its stated class, if a quantitative closure is nonunique or target-leaking, or if a physical implementation shows branch-essential dependence on hidden carrier presentation after visible quotient data are held fixed. Concrete falsifiers include failure of a displayed Maxwell, pure-Yang–Mills, or pure-Einstein quadratic kernel to have its stated transverse or transverse-traceless (TT) classical mode under all of that theorem’s action/background/phase hypotheses; failure of a claimed quantum pole after its full quantization and spectral receipt is supplied; an observed low-energy  $X/Y$ -type  $(3, 2, \pm 5/6)$  gauge generator on the realized product branch; a fourth light matter generation, which would falsify a claimed three-family attachment if that attachment were supplied, while leaving the finite rank-three band theorem and the three-to-five window as separate statements; nonunique  $P_\star$  or  $N_{\text{CRC}}$  closure; a failed fixed-cap entropy-to-Einstein tensor upgrade; or carrier-dependent observables after the visible interface is held fixed. A failed continuation does not erase the recovered core unless that continuation is one of the stated parents of the core claim.

## Contents

|          |                                                                 |           |
|----------|-----------------------------------------------------------------|-----------|
| <b>1</b> | <b>Introduction</b>                                             | <b>11</b> |
| <b>2</b> | <b>Main Results</b>                                             | <b>24</b> |
| 2.1      | Theorem and source-object map . . . . .                         | 24        |
| 2.2      | The observerhood receipt and the common-origin bridge . . . . . | 35        |

|          |                                                                                   |           |
|----------|-----------------------------------------------------------------------------------|-----------|
| 2.3      | Conditional history weights and the variational bridge boundary . . . . .         | 37        |
| 2.4      | Thermodynamics from conditional repair in an observer modular parameter . . . . . | 40        |
| <b>3</b> | <b>Physical Implementation and <math>A_5/E_8</math> Significance</b>              | <b>43</b> |
| <b>4</b> | <b>Consensus, Defects, and Implementation Hiding</b>                              | <b>54</b> |
| <b>5</b> | <b>Particle-Spectrum Branch</b>                                                   | <b>55</b> |
| <b>6</b> | <b>Screen Microphysics, Records, and Observer Continuation</b>                    | <b>72</b> |
| <b>7</b> | <b>Worksheet/String Branch</b>                                                    | <b>74</b> |
| <b>8</b> | <b>Theorem Boundaries and Conditional Continuations</b>                           | <b>75</b> |
| <b>9</b> | <b>Conclusion</b>                                                                 | <b>78</b> |
| <b>A</b> | <b>Candidate Microphysics Finite Reference Checks</b>                             | <b>79</b> |
| <b>B</b> | <b>Interpretive Epilogue: State-and-Law Habitat</b>                               | <b>80</b> |
| B.1      | Additional Conditional Results . . . . .                                          | 82        |
| <b>C</b> | <b>Observer Continuation and Backup</b>                                           | <b>85</b> |
| C.1      | Observer as Algebraic Pattern . . . . .                                           | 85        |
| C.2      | Markov Collar Factorization . . . . .                                             | 85        |
| C.3      | Checkpoint and Restoration Map . . . . .                                          | 85        |
| C.4      | Physical Meaning . . . . .                                                        | 86        |
| <b>D</b> | <b>Cosmology, Horizons, and Modular-Anomaly Continuations</b>                     | <b>86</b> |
| D.1      | Observer Registry and Clock-Naturality Certificate Contract . . . . .             | 86        |
| D.2      | Cosmological Principle . . . . .                                                  | 87        |
| D.3      | Horizon-Problem Control . . . . .                                                 | 88        |
| D.4      | Conditional Screen-Spectrum Continuation . . . . .                                | 89        |
| D.5      | Cosmological-Constant / Screen-Capacity Closure . . . . .                         | 90        |
| D.6      | Black-Hole Structural Package and Continuation Boundary . . . . .                 | 92        |
| D.7      | Modular-Anomaly Continuation . . . . .                                            | 93        |

# 1 Introduction

Observer-Patch Holography asks how much of physics is forced once the world is described from the inside. No observer has access to a view from nowhere. In the idealized description there is a shared screen net, often charted by  $S^2$ , but each finite observer has access only to a local patch  $P_O$  of that net. The observer carries records inside that patch and can compare only the observables shared with neighboring patches. OPH takes that comparison rule literally: where descriptions overlap, they must agree. Thus an "observer screen" is an operational access cut, not a private physical sphere assigned to the observer.

The formalism used here is deliberately quantum-algebraic. Patches carry algebras and states, record events use the trace/Born rule on their declared operator surface, and the gravity branch uses generalized entropy. OPH's theory-of-everything claim is that the observed effective universe can be recovered from this axiom set and its stated branch conditions if the framework is correct. The reconstruction may use declared standard mathematics and branch hypotheses without treating those ingredients as hidden empirical constants.

The computational interpretation has a sharp boundary. A conventional simulation evolves surrogate universe states,

$$U_t \longrightarrow U_{t+1} \longrightarrow U_{t+2} \longrightarrow \cdots .$$

OPH uses a fixed-point computation, but the equation alone is not the definition of simulation. In the fundamental description there is no global timeline on which the contents of spacetime are updated. The readback-and-repair operator acts on observer-readable world candidates and selects a stable solution

$$\mathcal{T}(W) = W.$$

The companion consensus paper [4] defines an OPH simulation by five independent clauses: recovery-derived endogenous update, nontrivial quotient-readable records, strict overlap-repair descent with a schedule-independent normal form, elimination of a proper candidate basin in the selected boundary/sector fiber, and implementation/clock closure. Its simulation theorem proves that the selected finite OPH packet meets those clauses under the stated boundary and record hypotheses and then derives the fixed-point equation. The same theorem gives an identity-map/constant-functional counterexample, so an ordinary equilibrium is insufficient as a simulation.

The local readout of the selected normal form supplies an internal record history, not by itself a physical clock. Operational time requires an observer-readable transition process, event correspondence, and affine calibration; it is not the external update parameter. In OPH, "simulation" means this certified observer-facing self-consistency structure, not a bare fixed point and not a frame renderer. The executable OPH-FPE simulator and its finite-run receipt bundles are a separate public evidence surface for finite OPH experiments [12]; they do not replace the theorem-level definition above.

The effective bulk is reconstructed from compatible screen data. Locality comes from the way collars control recovery across patch boundaries. Gauge freedom is the freedom to change the local description without changing any shared observable. Records are the stable parts of a patch that can be checked by other patches. Inside the declared tensor category, matter is the charge and excitation content that survives the certified transport and fusion tests. Completeness of the physical light sector is a separate source-grammar and laboratory attachment premise.

The first major result is spacetime kinematics. Modular flow on screen caps becomes geometric, and the compact cap-normal theorem identifies a sky direction with  $q(\Omega) = (1, \Omega)$ , represents an oriented round cap by  $n_C = (\cot \alpha, \csc \alpha \mathbf{c})$ , proves  $\eta(n_C, q(\Omega)) = 0$  exactly on the cap boundary

with the interior/exterior sign rule, and gives  $n_{gC} = \Lambda_g n_C$ . Future unit observer frames then form  $H^3 \simeq \text{SO}^+(3,1)/\text{SO}(3)$ , exactly three-dimensional. A cap determines an  $H^3$  plane/half-space, not an event position or populated bulk. On the conditional frame-response branch, quotient-visible record tokens whose calibrated modular cap responses factor through frame-local  $H^3$  data admit exact frame identification and stable finite-noise frame estimates. The finite frame radius is  $R_H[(L/\alpha)\varepsilon + (2/\alpha)\sigma]$  for exact net minimization, with a positive residual gap required before a unique finite frame value is reported. Event location requires the separate affine translation and ancestry receipts of the event-manifold branch. Generalized entropy at fixed cap then gives a tensor first-variation relation in the large-scale regime. The absolute Einstein equation additionally uses one source-derived common-domain tower with uniform asymptotics, universal coupling, a vacuum reference, and independent scale readouts; that tower enters as a premise and is not constructed here. The proposed cosmological capacity relation is  $\Lambda_{\text{CRC}} = 3\pi/(GN_{\text{CRC}})$ . The Gibbons–Hawking entropy and the Planck-2018 late-time de Sitter benchmark give a concrete normalization [32, 34]: with  $R_{\text{dS}} \simeq 1.66 \times 10^{26}$  m and  $\ell_P \simeq 1.616 \times 10^{-35}$  m, the bare horizon ratio is  $N_{\text{patch}} = (R_{\text{dS}}/\ell_P)^2 \simeq 1.05 \times 10^{122}$ , the entropy capacity is  $N_{\text{scr}} = S_{\text{dS}} = A_{\text{dS}}/(4\ell_P^2) = \pi N_{\text{patch}} \simeq 3.31 \times 10^{122}$ , and  $\Lambda \ell_P^2 \simeq 2.85 \times 10^{-122}$ . OPH uses  $N_{\text{scr}}$  for the Gibbons–Hawking entropy capacity and  $N_{\text{patch}}$  for the bare horizon area ratio. The proposed input-free closure begins at finite cutoff with a frozen carrier dimension  $D$ , the unclosed terminal quotient fiber, local and interface record atoms, compatible public global sections, endogenous reachability, a frozen publicness policy, and the globally coupled semantic checkpoint kernels. Their compound confusability graph  $G_q$  gives the exact finite readback

$$M_0(q) = \alpha(G_q).$$

This correctable-code capacity includes joint checkpoint compatibility and confusability. A cyclic checkpoint can preserve every label while fixing only constant functions, and equal local marginals can hide different joint capacities. The first producer is the set

$$\mathfrak{F}_{r,\varepsilon}(D) = \{M_\varepsilon(q) : q \in \tilde{\Omega}_{r,D}\}.$$

A scalar exists only when the whole nonempty terminal fiber agrees. A faithful capacity-carrier representation gives  $M_\varepsilon \leq D$ ; exact equality means a complete rank-one public record basis. If the scalar exact map is total, monotone, and deflationary on a declared finite chain, top-down iteration reaches its greatest fixed point. The universe-level cosmic record-closure equation and its stable finite closure condition are

$$N = \log M_0(\mathfrak{U}_N), \quad \mathfrak{F}_{r,0}(D_\star) = \{D_\star\}, \quad N_{\text{CRC}} = \log D_\star.$$

The finite public-section, correctable-code, boundary, scalarization, order, extension, refinement, and sewing implications can be proved from the stated data. A source-derived packet realizes the fixed-cutoff construction at  $D = 24$  inside its declared finite source category. An executable all-rung counterfamily shares base agreement, positivity, the carrier bound, and finite incidence, action, projection, composition, extension, and sewing controls. Its reversible, copy-collapse, two-class, and hidden-spectator completions have different exact slack-zero sets. Lean proves the corresponding all-rung arithmetic result. This establishes nonidentifiability for the bounded completion class. The complete A1–A3 source contract has not been shown to contain those completions at every rung, and no executable-to-Lean bridge establishes that membership. Direct  $N$  is not evaluable on the incomplete capacity source antecedent; the stronger source-class verdict is open. A positive universe-level result requires a complete source antecedent, one physical zero, and proof that both sides of the strange loop read the same invariant quantity. Its slack would be

$$s(D) = \log D - \log M_0(D)$$

with one physical zero.

The source-only closure audit covers the declared seventeen-row menu for the executed capacity families, two declared  $P$  maps, one direct fixed-cutoff control, the common-load baseline, reserve maps, RC-LOAD, the archived construction aggregate, and five hierarchy packets. Fifteen of these thirty entries carry exact fixed-point progress. None supplies target-independent selection, a constructed same-quantity identification, and a complete source return map together. The input catalogue, nuisance quotient, coordinate and scheme catalogue, and candidate grammar are not frozen, so no entry qualifies as a source-only candidate in this registered inventory. Both declared  $P$  maps have certified unique stable roots while their selection, same-quantity construction, and complete return maps remain unproved. The four multiplicative CAP-P maps fix whichever positive normalization seed is supplied and therefore do not select that seed; the two additive rows have no positive fixed point. The conditional RC-LOAD equation has positive discriminant, maps its declared macroscopic domain into itself, is contractive there, has one macroscopic fixed point, and excludes its microscopic root from that domain. Its pricing law, same-quantity identity, and target-independent selection remain premises. The audit excludes laboratory, cosmological, and absolute-scale attachment from its qualification rule. It neither exhausts other source laws nor opens an observational comparison.

Two further physical identifications would be required afterward: the correctable-record carrier must be identified with the de Sitter horizon record for the de Sitter area law, and the screen load must be identified with the electroweak load by a positive, unital, refinement-natural map for the electroweak bridge. An independently produced operational scale is a commuting-square test of the code capacity, not its definition. The electroweak bridge supplies a mathematical candidate of  $N_0 = 3.5321315434 \dots \times 10^{122}$  on the source-forward branch only after the common-load receipt. The weighted Planck base- $\Lambda$ CDM read-off is  $3.3129270981 \dots \times 10^{122}$ ; the uncorrected bridge candidate is about 6.6 percent above it. The separately premised finite-presence and Poisson carriers give  $3.2920978773 \dots \times 10^{122}$  and  $3.3000722254 \dots \times 10^{122}$ . Exact positive composition counter-models select neither action from the finite source. Their comparisons are retrospective. The joint Planck posterior, propagated through the  $\Lambda$ -to- $N$  map across three degeneracy routes, places the gap at a propagated, retrospective 2.4 to 2.5 one-dimensional standard deviations (the public posterior predates this comparison) under the consumed likelihood combination and 3.8 to 3.9 under Planck+BAO, so the combination freeze is part of the declared comparison rule. The physical statement requires the source-derived record carrier and common-load premises. Thus neither the physical fixed-point identity nor the cosmological display  $\Lambda_{\text{CRC}} = 3\pi/(GN_{\text{CRC}})$  is established by the bridge calculation.

The second result is a finite Standard Model recognition package. Complete reversible response and endogenous overlap transport force the local Standard Model gauge Lie algebra. Incidence expresses  $J$  as a polynomial in adjacency. A target-blind impulse and port readback derive  $R = -J$ , whose relative sector pattern on  $\mathbf{1} \oplus \mathbf{3} \oplus \mathbf{3}' \oplus \mathbf{5}$  is exact. Under the conditional matrix current and declared fermionic Spin category, the anomaly and tensor-descent calculations give the maximal faithful matter image

$$\text{SU}(3) \times \text{SU}(2) \times \text{U}(1)/\mathbb{Z}_6.$$

This conditional matter conclusion uses the stated premises alone. The matter theorem fixes exact hypercharge and a three-color carrier. The scalar scan fixes compatible charges and Yukawa channels, not scalar multiplicity. The physical matrix current, matter action, and global quotient need source receipts. No laboratory interpretation of them is claimed. Separately, the CKM and weak-sector conditions give  $3 \leq N_g \leq 5$ . The target-free matter reduct does not fix the count. The packet also gives integer charge for color singlets and the absence of the simple-GUT  $X/Y$

gauge channel. Inside the screen an exact selection is certified: among single complete faithful in-window multiplicity objects of the coefficient space, the operational seam-cost order  $5 - \sqrt{5} < 6 < 5 + \sqrt{5}$  has the rank-three band as unique strict minimizer under two named premises: realization of one complete multiplicity object inside the screen, and comparison by the operational cost order. Tensoring this response band with the declared fifteen-state table gives a conditional rank-45 candidate. The declared finite simulator realizes the band comparison. Its per-band adjacency channels have the frame triplet as strict minimizer, and the response resolvent of the declared Laplacian generator has four pole clusters at the band costs. The rank-three frame residue sits at the lowest positive generator frequency. The channel is unitary, so every mode preserves its norm and this statement supplies no relaxation rate. The nondegenerate chirality grading and diagonal  $\mathbb{Z}_6$  action are properties of the generation table. A separate 8,662-node local-domain receipt checks the declared operator  $D_\sigma \otimes I_{45}$  and conditional gap inheritance. The source does not select this matter action, and no certified source, domain, or transport bridge joins the twelve-port Spin packet to the local operator domain. Matter-pole identification, the continuum Spin/locality limit, physical seam selection, refinement persistence, and laboratory attachment are separate premises; external copy completions stay grammar-indistinguishable, and the canonical rank-three screen band is a family-fiber candidate conditional on those receipts. On a declared trace-balanced block carrier  $V = C \oplus W$ , with dimensions  $3 + 2$  and hypercharges  $(-1/3, 1/2)$ , the exact exterior package  $\Lambda^2 V \oplus \Lambda^4 V$  branches as one fifteen-state Standard Model generation, supports the three one-Higgs invariant lines, cancels all five listed perturbative gauge and mixed anomalies, and has weak-doublet multiplicity  $3 + 1 = 4$  with even Witten parity. This is an axiom-forced Lie-type theorem and a conditional finite representation theorem. Laboratory identification of its current and finite flux sectors, physical line-spectrum attachment, equality with the independently reconstructed Tannaka current, extra-light-sector exclusion, physical interpretation of the conditional rank-45 candidate, selection of the local matter action, cross-domain Spin transport, scalar dynamics, genuine 1PI realization, and continuum QFT are separate premises. The subsequent QFT landing is a typed graph: exact finite quantization and formal perturbative quantization are parallel descendants of the finite local action, while the nonperturbative continuum requires its own observable tower; the conditional implications are explicit, and the OPH-native producers enter as premises. The selected exterior package is not the full even Clifford module, because  $\Lambda^0 V$  is omitted. On separately stated Maxwell, perturbative/deconfined Yang–Mills, and pure-Einstein action/phase branches, the quadratic reductions have the corresponding classical null carrier modes. Photon, gluon, or graviton particle claims require the additional physical-Hilbert-space and spectral-pole receipts.

The finite exterior statement has a direct algebraic realization. The basis of  $\Lambda(\mathbb{C}^3 \oplus \mathbb{C}^2)$  consists of the 32 subsets of five coordinate modes. Removing the vacuum and top line leaves the ten color/weak bidegrees in the component table, with exact charge, parity, conjugation, square-zero, and anticommutation identities. A supplied map from those rows into nonzero sectors of a projective record partition, together with supplied central-weight labels, preserves the diagonal  $\mathbb{Z}_6$  weight-kernel arithmetic. A separate supplied selection mask is forced to one of the two parity rows. No group action on the partition projectors or physical attachment of that mask follows. Source selection of the maps, physical matter dynamics, continuum spin–statistics, and laboratory charge remain separate.

The third result is quantitative and sits outside the recovered core. Its primary equation defines a fixed point between a declared outer detuning and a declared electromagnetic endpoint map:

$$P = \varphi + \frac{\sqrt{\pi}}{A_T(P)}, \quad A_T(P) = \alpha_{\text{em}}^{-1}(0; P).$$

The outer coordinate is the normalized detuning  $(P - \varphi)/\sqrt{\pi}$ . The inner coordinate is the Thomson-

limit coupling returned at the same trial input. The contraction-based theorem gives existence and uniqueness on a closed interval. Outward-rounded interval certificates verify the concrete hypotheses, enclose one root for each declared map, and exclude a second root across the full analytic domain. No measured value is inserted into either solve. Physical identification requires a source-derived same-scheme hadronic transport, target-independent selection of the applicable map, and a typed bridge proving that its two coordinates read the same physical cell.

The secondary global extension uses the exact finite correctable-record capacity  $M_0(q) = \alpha(G_q)$  and the universe-level equation  $N = \log M_0(\mathfrak{U}_N)$ . Its finite public-section, carrier-bound, scalarization, order, and refinement implications are conditional theorems. The fixed- $D = 24$  check-point packet and whole-fiber scalar readback are exact inside their declared finite source category. A bounded all-rung counterfamily has incompatible zero sets without carrying the full observer packet across capacity. Universal all-rung membership in the complete A1–A3 source contract and its executable-to-Lean bridge are unproved. Direct  $N$  is not evaluable on that incomplete antecedent. A complete source antecedent with one physical finite-size slack zero, the universe-level carrier attachment, the horizon identification, and the common screen/electroweak carrier are premises this construction consumes and does not supply. The weighted Planck base- $\Lambda$ CDM capacity  $3.3129270981 \dots \times 10^{122}$  and the conditional source-forward electroweak value  $3.5321315434 \dots \times 10^{122}$  are comparison coordinates. Their 6.6-percent difference does not enter the construction of the direct capacity map. The finite-presence and Poisson values require separate carriers, and their comparisons are retrospective.

The local pixel coordinate and independent scale certificate are

$$P := \frac{a_{\text{cell}}}{\ell_\star^2}, \quad \gamma_\star = \frac{\ell_\star \nu_{\text{Cs}}}{c}, \quad B_\star = \frac{3\pi}{\ell_\star^2}.$$

If the two declared-map outputs acquire their missing physical identifications,  $P_\star$  and  $N_{\text{CRC}}$  fix  $\Lambda_\star \ell_\star^2 = 3\pi/N_{\text{CRC}}$  and  $\Lambda_\star a_{\text{cell}} = 3\pi P_\star/N_{\text{CRC}}$ . They fix the dimensionless geometry. The selected scale certificate supplies the SI scale product  $B_\star = \Lambda_\star N_{\text{CRC}} = 3\pi/\ell_\star^2$ . The selected scale certificate does not construct the cosmic readback map or remove the uncorrected 6.6-percent central-value mismatch. Under those same identifications, the pixel  $P_\star$  together with the  $\Lambda$ -located working capacity  $N_\Lambda = 3.31 \times 10^{122}$  fixes a canonical geometric pixel area and a total geometric cell count on an equal-area screen chart,

$$K_{\text{cell}} = \frac{A_{\text{screen}}}{a_{\text{cell}}} = \frac{4N_\Lambda}{P_\star} \simeq 8.12 \times 10^{122}.$$

This count is a measured-side comparison display and does not enter either closure. At the conditional bridge capacity the same chart count reads  $8.66 \times 10^{122}$ . The result supplies a geometric chart count. It does not choose a primitive observer capacity. The quantity  $P_\star/4 \simeq 0.408$  nats, or about 0.588 bits, should not be read as  $\log \dim \mathcal{H}_{\text{cell}}$  for an autonomous tensor factor. OPH capacity lives on shared cuts, edge centers, and constrained overlap data; adjacent geometric pixels are not assumed to factor into independent Hilbert components. No fixed primitive observer algebra follows from  $P_\star$  and  $N_{\text{CRC}}$  alone. The no- $G$  burden of the scale proof record is the clock hierarchy. The first load-bearing object must be a strict source-root proof for  $\alpha_U(P_\star)$ ; the source-check branch is only a witness. After that, the cesium gap  $\varepsilon_{\text{Cs}} \sim 10^{-33}$  must be emitted by source-side electroweak transmutation, QCD/hadronic data, flavor and nuclear data, and a cesium hyperfine spectral readout, with no dependency path from measured gravity or any calibrated electroweak or gravity scale. The public  $\alpha_U$  record is a one-dimensional comparison-branch fixed-point certificate,

evaluated at the CODATA-located comparison pixel  $P_C \simeq 1.6309682094$ :

$$\alpha_U(P_C) = 0.041124336195630495,$$

$$\frac{v}{E_\star} = 2.0199803239725553 \times 10^{-17},$$

and the Krawczyk image of  $[0.041123336195630494, 0.041125336195630496]$  lies strictly inside that interval. The source-side diagnostic branch excludes the public Thomson endpoint upstream. Its certified forward pixel is  $P_{\text{fwd}} = 1.630972095858897 \dots$ . That source branch is a witness, not a full endpoint proof; it does not supply the strict interval root for the Ward-projected self-map. The dimensionless formula is the strongest clean hierarchy statement. It does not supply an independently physical  $E_\star$ , a mass in GeV, or a complex propagator pole. The full SI gravity row requires the full set of cesium-clock source records. The QCD and hadronic weakpoint is narrow. The low-energy Thomson endpoint and the cesium-clock display require confined-quark and nuclear source payloads that the first-principles trunk in this paper does not emit. A source-only Thomson endpoint requires a target-independent choice of the physical map, a typed same-quantity bridge, and a source-derived hadronic spectral backend: source QCD parameters, a quotient ensemble, finite Euclidean vacuum transfer, Ward-normalized electromagnetic currents, the two-current spectral marginal, same-scheme remainder, systematics, and no-target-leak receipts.

Four fine-structure coordinates must be kept separate. The source/root analysis gives the undressed declared-map witness  $\alpha_{\text{root}}^{-1} = 136.994835177413 \dots$  (interval-certified unique fixed point of that numerical map, with enclosure width  $7.2 \times 10^{-24}$ ). The declared map is incomplete: no derivation identifies this root with a physical fine-structure endpoint. Adding the finite-screen unified gauge-width contribution at the CODATA-derived comparison pixel gives the mixed-provenance, no-hadron diagnostic

$$A_{\alpha_U}^{\text{fp}} = 137.0359595136 \dots$$

The certified self-consistent gauge-width fixed point is  $\alpha^{-1} = 137.035660136946577 \dots$ . The mixed diagnostic  $137.0359595136 \dots$  combines the inner value at  $P_{\text{fwd}}$  with  $\alpha_U$  at the CODATA-derived comparison pixel, is not a fixed point of any single declared map, and is excluded from the physical output set. An executed empirical closure, using external  $e^+e^- \rightarrow \text{hadrons}$  spectral data, records

$$\Delta\alpha_{\text{had}}^{(5)}(M_Z) = 0.027609 \pm 0.000112, \quad \alpha_{\text{emp}}^{-1}(0) = 136.3827548175$$

with interval  $[136.3670480603, 136.3984651934]$ . The measured endpoint value  $137.035999177(21)$  for  $\alpha^{-1}(0)$  lies outside that interval. The resulting same-scheme anchor correction is  $[0.6198609041, 0.6505569679]$  inverse-alpha units; the standard on-shell reference deficit 0.631 and the exact closure value 0.6379 both lie inside it, and their difference is the scheme term carried by the anchor-bridge premise. This empirical payload tests the endpoint map and exposes the missing contribution; it does not supply a source-only OPH hadron law. The two-current spectral measure covers the running- $\alpha$ /HVP marginal. HLbL and rare-decay rows require higher-point and transition spectral sectors from the same source law. For that reason the compact quantitative stress test is the hierarchy row above: it avoids the public Thomson endpoint,  $G$ ,  $\Lambda$ ,  $W/Z$ , Higgs mass, hadronic payloads, and the cesium clock.

The same hierarchy exponent has a local/global capacity form:

$$\frac{v}{E_{\text{cell}}} = \left( \frac{N_{\text{CRC}}^{\text{EW}}}{\pi} \right)^{-P_\star/12}, \quad N_{\text{CRC}}^{\text{EW}} = \pi \exp \left[ \frac{6\pi}{P_\star \alpha_U(P_\star)} \right].$$

This is the electroweak bridge capacity, about  $3.53235 \times 10^{122}$  on the declared comparison branch at  $P_C$ ; the interval-certified value at the forward pixel  $P_{\text{fwd}}$  is  $N_0 = 3.5321315434 \times 10^{122}$ . Identifying



it with the cosmic capacity requires the source-derived correctable-public-record producer and the common screen/electroweak load-carrier identification. The weighted Planck base- $\Lambda$ CDM capacity  $3.3129270981 \dots \times 10^{122}$  is the comparison coordinate; the source-forward bridge value is about 6.6 percent above it. The conditional finite-presence and Poisson candidates are  $3.2920978773 \dots \times 10^{122}$  and  $3.3000722254 \dots \times 10^{122}$ , respectively. Their physical attachments are distinct, and exact countermodels select neither one from the finite source. Their comparisons are retrospective. The factor 12 has a branch-specific geometric reading. On the declared echosahedral carrier lineage, twelve equal-trace primitive central port atoms are part of the finite architecture. The named counting realization additionally declares the integer defect fiber  $\sum_p q_p = 12$  and the normalized central-readback Hilbert–Schmidt cost  $H(q) = \sum_p q_p^2$ . The identity  $H(q) = 12 + \sum_p (q_p - 1)^2$  uniquely selects one unit per port with exact gap two. An append-only signed-event machine realizes the integer load: writes contribute +1, retractions contribute -1, and each port readback sums its atomic events. A conservative repair transfers one whole unit across a seam with mismatch  $d \geq 2$ . It preserves total load and changes  $V(q) = \sum_p q_p^2$  by  $-2(d - 1)$ , so every repair path terminates. Divisibility of the total load by twelve is necessary for consensus and is not sufficient on its own; the declared full-pile packet has an explicit eighteen-move settling witness. Minimum move count is natural under all sixty carrier rotations and the declared refinements. The half-unit display rescales the event values and repair threshold together, leaving the event graph and move-count cost unchanged. It is a units convention rather than a competing half-event mechanism. The positive readback scale is a units convention. For the separately declared uniform twelve-port reference, the local Kullback–Leibler Taylor coefficient is  $6I$  and its Hessian, equivalently the Fisher matrix, is  $12I$ . This infinitesimal curvature does not determine the exact discrete cost. Oriented  $(12, 30, 20)$  incidence then derives the unique graph-distance three antipode, six axes, proper automorphism group  $A_5$ , and the canonical rank-three icosahedral Gram frame. Edge-center collars expose the defects as central ports. Reversible write/check orientation gives an oriented 24-slot screen register. Independently, the product adjoint has count  $m_{\text{rep}} = 2(8 + 3 + 1) = 24$ . Their equality is bookkeeping and supplies no load or clock. The exterior matter package gives the exact four-copy weak multiplicity; attaching that multiplicity to the physical screen load is an explicit hypothesis.

The oriented coefficient module is

$$P_{12} \cong_{A_5} \mathbf{1} \oplus \mathbf{3} \oplus \mathbf{3}' \oplus \mathbf{5}.$$

Let  $\mathfrak{g} = D(P_{12}) \subseteq \mathfrak{u}(H)$  be the complete faithful unitary response supplied by A1. Then  $\dim \mathfrak{g} = 12$ , and equivariance gives  $\dim \mathfrak{g}^{A_5} = 1$ . A2 makes the proper recharting action internal to this same response, so its centre is fixed pointwise and has dimension at most one. If the centre vanished, compact classification would leave only  $\mathfrak{su}(2)^4$ . Each simple factor has  $A_5$ -fixed dimension zero or three under an internal action, contradicting the one-dimensional fixed space. The centre therefore has dimension one, while the semisimple part has dimension eleven. The unique compact simple-dimension split is  $11 = 3 + 8$ , which forces

$$\mathfrak{g} \cong \mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(3).$$

No ambient continuous gauge group, factor dimensions, or matrix current enter this argument.

A conditional matrix witness uses the six antipodal axes. Its exact frame maps split the even and odd modes, while the outward faces orient the complementary triplet. Pulling the block commutator back through

$$\Theta(b + d_G + d_W) = (\kappa_E(\tfrac{1}{2}Ud_G) + i\Phi(b), \kappa_W(d_W))$$

constructs

$$(P_{12}, [\ , \ ]_{\Theta}) \cong \mathfrak{u}(1) \oplus \mathfrak{su}(3) \oplus \mathfrak{su}(2)$$

with center **1**, derived dimension eleven, and a noncentral **5**. This bracket acts on coefficient fluctuations; the central record projectors commute. Its four nonzero equivariant coefficients give an injective realization with a positive Hilbert–Schmidt pullback,  $A_5$  covariance, inner action, and algebraic refinement naturality.

Separately, incidence expresses the antipode  $J$  as a polynomial in adjacency. Under the explicit inverse-port contract, target-blind port readback derives  $R = -J$ . The simulator producer executes the maximal-distance impulse/readback protocol, and the port-current certificate recomputes the incidence polynomial and sector signs independently. It does not reconstruct  $D$ , the matrix generators, their bracket, or the same-current holonomy. Source realization of the abstract response, physical carrier attachment, laboratory identification with measured Standard Model currents, and the intertwiner to the independently reconstructed Tannaka current are separate premises.

Anomaly freedom of the declared fifteen-state module forces the determinant balance, and primitive integrality fixes the charge pair  $\pm(-1/3, +1/2)$ ; the common sign is charge conjugation. This completes the conditional map to  $\mathfrak{s}(\mathfrak{u}(3) \oplus \mathfrak{u}(2))$ , whose connected group is

$$S(U(3) \times U(2)) \cong \frac{SU(3) \times SU(2) \times U(1)}{\mathbb{Z}_6}.$$

The exhaustive tensor-kernel computation and Smith invariants independently fix the same  $\mathbb{Z}_6$  kernel on those tensors and compute its conditional character/cocharacter arithmetic. The determinant balance, the measured non-split Spin lift, the central embedding, and the refinement descent are receipt-checked; the rank-15 matter object is selected by the exhaustive 1024-subset anomaly scan with the fermionic-parity grading as an output, and the declared deck and six-axis relations reproduce the  $\mathbb{Z}_6$  descent. They do not source-select the complete character lattice or physical quotient. Hypercharge, chiral content, and this tensor quotient come from the stated premises within the chain. Scalar existence and multiplicity, four-dimensional instanton sectors and theta periodicity, laboratory current identification, physical matter-pole attachment of the conditional rank-45 candidate, selection of the local matter action, cross-domain Spin transport, and continuum QFT are separate premises.

The positive twelve-term sum-to-twelve theorem establishes the all-one split inside its declared integer counting fiber. The finite algebra separately establishes the abstract matrix commutator, dimension  $9 + 3$ , a matrix noncentrality witness, the six-axis lattice quotient, and the sixfold central-axis arithmetic. On the declared echosahedral lineage, the normalized central-readback Hilbert–Schmidt cost establishes the strict gap, while oriented incidence establishes inverse pairing, proper  $A_5$  action, the rank-three port frame, and refinement/relabeling naturality. A derivation of the integer fiber and physical discrete cost from the complete three-axiom schema is not claimed. This result also does not derive the echosahedral carrier from arbitrary OPH data. The noncentral current lift, trace-balanced group, Spin/deck descent, and one-generation matter calculation are exact conditional on the response and matter contracts. Physical source binding, laboratory current identification, equality with the Tannaka current, scalar multiplicity, physical three-family attachment, and continuum QFT are separate premises. The face- $C_3$  representation theorem gives a three-dimensional minimal extension; physical family attachment is a separate premise. See Refs. [2, 3].

The Standard Model gauge paper packages this quantitative surface as a dependency-audited compression test. A row counts in the accidental-hit estimate only when its declared source map has no dependency path from the measured target or a calibrated proxy. If  $p_i$  bounds the conditional

accidental-hit probability for row  $i$  after the preceding accepted rows, then

$$P_{\text{acc}} \leq \prod_i p_i.$$

Only rows whose source maps have no dependency path from a measured target or calibrated proxy enter the accidental-hit bound. A compression statement must include the declared settings and basins. The proof load is the fixed-point maps, uniqueness certificates, source records, and falsifiers.

The declared map for  $P$  places an outer detuning coordinate and an inner electromagnetic coordinate on one trial domain. The outer coordinate sits above the exact self-similar entropy balance point  $\varphi = (1 + \sqrt{5})/2$ . The inner coordinate is returned by the electromagnetic map. Each certified map has one mathematical root on its declared interval. The measured fine-structure constant tests the resulting endpoint and does not enter the solve. Interpreting a root as the local physical closure coordinate requires the unconstructed same-scheme hadronic transport, target-independent map selection, and the same-quantity bridge.

The named  $P$  branches organize the electroweak carrier chart, the low-energy electromagnetic endpoint, the conditional Higgs/top surface, the rejected reciprocal-ray and register-Clebsch quark candidates and the six-coordinate physical quark interface, a target-informed weighted-cycle neutrino comparison candidate, and the local cell/edge consistency used by the gravity readout. The hierarchy theorem fixes  $v/E_*$ . An independent physical  $E_*$  and the separate pole receipt govern any mass in GeV. The gravity normalization itself comes from  $\ell_*^2$ , and  $P$  cancels in the Newton area-law readout. The quantitative branches distinguish the electroweak chart, conditional law, and inverse target adapter; the target-informed charged-lepton candidate; the NuFIT 6.1 rejection of the neutrino candidate; its tautological declared-basis transport and missing physical charged basis; and the hadron construction [19].

The screen-microphysics construction makes patches, overlaps, edge observables, records, repair maps, and observer checkpoints explicit at finite cutoff through a federated patch-carrier architecture. This regulated observer-and-record surface carries the basic consistency machinery directly.

### The three axioms.

**Axiom 1 (oriented twelve-port observer screen).** There exists an observer patch net on an oriented spherical screen: at every finite resolution each local carrier has twelve primitive boundary ports forming the vertices of an oriented triangular boundary with 30 edges and 20 faces, combinatorially the boundary of an icosahedron, and carriers join through typed seams and coherent triple overlaps, refine to an oriented spherical support, and expose local state, readback, records, repair moves, and checkpoints. Their complete infinitesimal port response is represented faithfully on a finite-dimensional unitary response space and is closed under the commutators generated by ordered response composition. Formally, for every regulator  $r$  in a directed system there is a typed object  $\mathfrak{N}_r = (\mathcal{P}_r, \mathcal{A}_r, \mathcal{R}_r, \mathcal{I}_r, \mathcal{U}_r, \mathcal{C}_r, N_r, S_r, b_r)$ : a finite patch/overlap category with an isotone local algebra net and central record algebras; a federation of finite carriers, each with twelve primitive pairwise-orthogonal central port projections summing to one and a boundary packet  $K = (P, E, F, o)$  with  $|P| = 12$ ,  $|E| = 30$ ,  $|F| = 20$ , degree-five ports, five-cycle links, and coherently oriented two-face edges, isomorphic to the icosahedral boundary complex with no preferred labels; seam algebras with unital restrictions and coherent triple-overlap cocycles forming the nerve  $N_r$ ; a finite edge-midpoint refinement support  $S_r$  realized orientation-preservingly in  $S^2$  with mesh tending to zero; and a source-bound degree-one bridge  $b_r : N_r \rightarrow S_r$ , all commuting with refinement. Each carrier also has  $V_{r,i} = \mathbb{R}[P_{r,i}]$ , a finite-dimensional complex Hilbert space  $H_{r,i}$ , and an injective real-linear map

$$D_{r,i} : V_{r,i} \longrightarrow \mathfrak{u}(H_{r,i}).$$

The image  $\mathfrak{g}_{r,i} = D_{r,i}(V_{r,i})$  is closed under commutators and complete for the declared quotient-visible infinitesimal port response. The primitive port probes span  $V_{r,i}$ , no additional public response direction is omitted, and  $-\text{Tr}(D_{r,i}(v)D_{r,i}(w))$  is positive definite. The response construction is natural under admissible presentation equivalence and refinement. The axiom does not select a particular response table, inverse-port law, Lie type, global group, particle interpretation, or metric content.

**Axiom 2 (observer agreement).** Observers operating on the screen agree on the meaning of the data they jointly interpret. Formally, the interpretation map  $\mathcal{J}_r : \text{Data}_r \rightarrow \text{Meaning}_r$  from observer-accessible data to operational meanings is natural with respect to every visible overlap restriction, recharting, seam translation, higher-overlap map, federation map, and refinement map: every declared data-access diagram for accepted public data commutes after interpretation, and  $\mathcal{J}_r \circ c_{s \rightarrow r} = C_{s \rightarrow r} \circ \mathcal{J}_s$  across resolutions. The axiom constrains accepted shared data only, and the domain it quantifies over is supplied by the Axiom 1 interfaces rather than by the axiom itself.

Accepted reversible overlap transports form a groupoid  $\mathcal{O}_r$ . For a complete carrier chart  $o$ , its closed paths  $\text{Hol}_r(o)$  map onto the orientation-preserving incidence automorphisms  $\text{Aut}^+(K_{r,i})$ . For every proper carrier automorphism  $a$ , Axiom 2 supplies a closed path  $\gamma_a$  and one projective implementer  $[U_a] \in \text{PU}(H_{r,i})$  satisfying

$$\text{Ad}_{[U_a]} D_{r,i}(v) = D_{r,i}(a \cdot v).$$

Writing

$$G_{D,r,i}^0 = \langle \exp(tD_{r,i}(v)) : t \in \mathbb{R}, v \in V_{r,i} \rangle^0,$$

the implementer is endogenous:

$$[U_a] = [g_a c_a], \quad g_a \in G_{D,r,i}^0, \quad c_a \in C_{U(H_{r,i})}(\mathfrak{g}_{r,i}).$$

Closed paths compose projectively and the construction commutes with refinement. A unitary lift need not be unique. Scalar phases and centralizer actions are permitted because they act trivially on the response algebra. An independent spectator carrying a matching projective action cannot replace the response-generated factor or contribute a public response direction. The closed path, port action, and implementer must arise from the same accepted source history in an executable realization.

The two response clauses support an abstract target-free compact Lie-type theorem. They do not certify that an executable producer emitted a particular current fixture. Such a certification must reconstruct the port generators, commutator, and projective implementers from ordered source histories without a named current model upstream. Axiom 2 does not imply global state extension, repair termination, confluence, unique normal forms, record durability, Byzantine safety, dissemination, or commutation of state optimization with refinement; raw mismatch and repair precede acceptance.

**Axiom 3 (conditional maximum randomness).** Everything that observer agreement leaves unconstrained is maximally random. Formally, at finite regulator  $r$  a state is a compatible family of local normalized states on the accessible algebra net;  $\mathcal{K}_r$  is the nonempty convex set of such families satisfying the finite observer-visible constraints supplied by Axioms 1 and 2 through an A1-generated constraint grammar with a factorization theorem; and, relative to an exact reference family  $\tau_r$ , an A1-generated observer cover  $\mathcal{G}_r$ , and strictly positive exact weights  $w_{r,p}$  constructed

from quotient-visible A1 data by a rule natural under admissible presentation equivalence, the realized state is the information projection. The cover restriction map is injective on  $\mathcal{K}_r$ ; equivalently, at smooth points no nonzero feasible tangent is invisible to the whole cover. The realized state is

$$\rho_r = \arg \min_{\rho \in \mathcal{K}_r} \sum_{P \in \mathcal{G}_r} w_{r,P} D(\rho_{r,P} \| \tau_{r,P}).$$

The same projection rule applies separately to each declared optimizer object type (ontic state, inference state, or transition distribution), and statements about one type do not transfer to another. This is equivalent to weighted local entropy maximization when every local reference density is identity-proportional in its declared trace. "Random" means least informative relative to the declared reference and the complete agreement constraint set. The axiom selects one state inside one fixed feasible space. It does not compare unrelated state spaces or field contents, does not imply optimizer compatibility across refinement, and does not imply recovery, alignment, or mixing; each such statement carries its own theorem, interface, or countermodel.

The complete formal basis, including its constraint map and non-implications, is given in the [canonical three-axiom reference](#).

**The closure principle.** The axioms describe the observer screen. The global closure principle states that the simulating and the simulated description are one system. Every quantity that has both a construction-side reading and a readback-side reading must take the same value once a typed bridge proves that the two readings denote one invariant. The equality is forced by self-identity. Constructing that bridge and the return map remains part of the physics. Existence, uniqueness, and stability are separate determinacy tests on the resulting equation. The name given to such an equation carries no mathematical weight: after the typed identification, unequal readings would describe two systems rather than the single self-referential universe.

The local screen-grain proposal

$$P = \varphi + \sqrt{\pi}/A_T(P),$$

with  $A_T(P)$  the inverse coupling returned by the declared source map, has one interval-certified fixed point on its declared analytic domain. Interpreting that root as the laboratory coupling additionally requires a target-independent choice of the physical map, a typed bridge identifying the two sides as readings of one quantity, and same-scheme hadronic spectral transport to the Thomson endpoint. The displayed map's certified fixed point sits at  $P = 1.6309720959$  and returns  $A_T = 136.9948352$ ; the certified gauge-width variant's fixed point sits at  $P = 1.6309682414$  and returns  $137.0356601$ . Through the same outer equation the measured 2022 CODATA coupling  $\alpha^{-1} = 137.035999177(21)$  reads back the grain value  $P = 1.6309682094$ . The comparisons stand at  $3.0 \times 10^{-4}$  relative for the displayed map and at  $2.5 \times 10^{-6}$  for the gauge-width variant, as diagnostics of the declared source maps.

The direct global proposal is

$$N = \log M_0(\mathfrak{U}_N).$$

The bounded completion class defined by base agreement, positivity, and the carrier bound does not select a unique cosmic value. Universal all-rung membership in a complete A1–A3 capacity-source contract and the corresponding executable-to-Lean bridge are absent. Direct  $N$  is not evaluable on that incomplete antecedent, and the stronger source-class verdict is open. A positive result must complete the source antecedent and prove one physical zero, either within the three-axiom source

contract or through a separately named stronger source law. A separate common-load branch starts from

$$N_0 = \pi \exp\left(\frac{6\pi}{P \alpha_U(P)}\right).$$

On the finite collar branch, the declared total reserve expectation  $P/4$  and six-class equidistribution give presence probability  $P/24$  for each declared class. If one class is physically selected as the blocked event, its scalar-weighted presence receipt is discharged, and that normalized collar-survival factor is proved to act on the global capacity, the corresponding candidate is

$$N_{\text{pres}} = N_0 \left(1 - \frac{P}{24}\right), \quad \ln \frac{N_{\text{pres}}}{\pi} = \frac{6\pi}{P \alpha_U(P)} + \ln \left(1 - \frac{P}{24}\right).$$

The exponential alternative  $N_{\text{Pois}} = N_0 e^{-P/24}$  requires a separate mean-count or continuum carrier. Exact neutral, one-class, and six-class-total actions share the declared local datum, obey positive composition and finite cut-count regrouping, and disagree globally. The finite source therefore selects no action or blocked-event semantics. The named-law branch is not evaluable on this source class, and its horizon branch has no capacity object. A stronger source-derived action would require the screen/electroweak bridge, physical common-load bridge, physical  $\mathbb{Z}_6$  seam action, scalar-weighted receipt, and horizon-record identification. On the source-forward numerical branch,

$$N_0 = 3.5321 \times 10^{122},$$

so the finite-presence candidate is  $3.2921 \times 10^{122}$  and the exponential candidate is  $3.3001 \times 10^{122}$ . The full weighted Planck base- $\Lambda$ CDM chain gives the comparison coordinate  $3.3129 \times 10^{122}$ . The respective residuals are  $-0.63$  percent and  $-0.39$  percent. Both comparisons were exposed before the branch choice and carry no predictive weight.

The gauge consequence shows how the axioms cooperate. Carrier incidence gives the twelve-dimensional  $A_5$ -module and its one fixed line. A1 supplies a faithful compact response that is complete under reversible composition. A2 makes the proper recharting action internal to that same response. Geometry alone leaves many equivariant laws, and the response conditions without the internal action leave a centreless compact alternative. Together they exclude  $\mathfrak{su}(2)^4$ , whose fixed dimension under such an action is divisible by three. The centre must therefore be one-dimensional, leaving the unique semisimple split  $11 = 3 + 8$ . The resulting local algebra is  $\mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(3)$ .

The descent repair law serves a different purpose. Given termination and the local diamond, it turns compatible local corrections into a schedule-independent public record. Simulator runs can test candidate repair laws and their relation to the carrier response. The Lie-type theorem itself uses the complete response and internal transport clauses. Source tomography, its matrix realization, matter representation, physical global quotient, and laboratory attachment are separate.

For the declared mean  $T = I - (5I - A)/60$ , let  $P_3, P_5, P'_3$  denote the three nonconstant spectral projectors, of ranks  $(3, 5, 3)$ , ordered from the slow to the fast response. The centered complete port-probe kernel is

$$C_n = QT^{2n}Q = r_{\text{slow}}^{2n} P_3 + (9/10)^{2n} P_5 + r_{\text{fast}}^{2n} P'_3, \quad r_{\text{slow}} = \frac{55 + \sqrt{5}}{60}, \quad r_{\text{fast}} = \frac{55 - \sqrt{5}}{60}.$$

The strict order  $0 < r_{\text{fast}} < 9/10 < r_{\text{slow}} < 1$  makes the normalized kernel converge to  $P_3$ . Unit-diagonal normalization gives the exact twelve-port Gram form  $G = 4P_3$ . Intrinsically,  $H = \text{ran } P_3$  and  $v_p = 2P_3 e_p$  satisfy  $\langle v_p, v_q \rangle = G_{pq}$ . Equal port weights fix these angles; strictly positive unequal weights preserve the leading rank-three range and generally change its Gram form. The conditional signed cumulative port-record/load module injects densely and completes to  $H$  only after the

normalized limit has produced the degenerate form and its real radical has been quotiented. The exact source readout is onto  $\mathbb{Z}^6$ , while the thirty conservative seam boundaries have image  $D_6$  with Smith invariants  $(1, 1, 1, 1, 1, 2)$ . Their pullback response metric has the same completion. Record addition supplies homogeneous internal translations. Ordered histories, repair costs, and exact record distance remain separate data, and the finite state-dependent repair kernel does not descend to position alone.

Finite measures follow the same pattern. Counting and trace give exact measures on the carrier algebra. Agreement and repair decide whether those measures survive on the public quotient without schedule dependence. A physical readout must preserve the resulting measure before it can be identified with a laboratory or spacetime quantity.

Collar recovery, generalized-entropy structure, and sector completions are named premises at the results that consume them. Quantum mechanics and quantum field theory are effective descriptions carried by the observer-patch architecture.

**The quantitative selection chain.** Self-reading, overlap consensus, modular geometry, transportable charge, entropic gravity, and stable records define the structural branch. They do not prove that the branch is inhabited, force the icosahedral screen-to-Standard-Model attachment, or construct the global capacity packet. The two closure equations test independently produced local and global readback maps. If both equations hold with the declared selection rules, the branch admits at most one coordinate pair  $(P, N)$ . This statement does not prove that exactly one physical universe exists. The local uniqueness certificate gives existence and uniqueness on its certified interval, and the domain-global certificate bounds  $\sup |g'| < 1$  on every piece of a 256-piece subdivision of the declared numerical domain ( $\alpha^{-1} \in [100, 200]$ , both readout maps, empty exceptional set), so each declared readout map has exactly one fixed point on that domain. This certificate does not derive a physical endpoint relation for the incomplete declared map. Public global sections, zero-error graph capacity, approximate stability, the carrier bound, whole-fiber scalarization, greatest-fixed-point order theory, and exact refinement stabilization are finite implication theorems. A source-derived fixed-cutoff simulator packet at  $D = 24$  verifies the finite evaluator contract inside its declared source category. A bounded capacity-indexed counterfamily shares base agreement, positivity, the carrier bound, and the executable finite controls while giving different exact zero sets. This proves nonidentifiability for the bounded completion class. The complete A1–A3 source contract has not been shown to contain those completions at every rung, and no executable-to-Lean bridge establishes that membership. Direct  $N$  is not evaluable on its incomplete source antecedent, and the stronger source-class verdict is open. A physical universe-level map requires a complete antecedent and one physical zero, together with the universe-level carrier attachment. The de Sitter and electroweak consequences then require the horizon–record identification and the common screen/electroweak load-carrier identification, respectively. The operational resolution experiment is an independent test, not the producer. The certified value  $3.5321315434 \times 10^{122}$ , evaluated at the forward pixel  $P_{\text{fwd}}$ , is a solution of the declared bridge equation, not of the unconstructed universe-level public-record producer; it is about 6.6 percent above the weighted Planck base- $\Lambda$ CDM coordinate  $3.3129270981 \dots \times 10^{122}$ . The separately premised finite-presence and Poisson expressions give retrospective residuals of  $-0.63$  and  $-0.39$  percent. Exact neutral and multiplicative global-capacity completions share the same local survival datum and satisfy positive disconnected-cut composition and finite cut-count regrouping, but they disagree after one cut. The finite source therefore selects neither expression as a global action, and its horizon branch has no capacity object to attach. The two electromagnetic endpoint residuals have no valid target-blind score: the exposed target conflates total and residual coordinates, and no active decision threshold exists. A

valid test requires a complete method fixed before genuinely withheld data are examined, or an independently verified clean-room producer.

The synthesis surface gives the operational picture first, then places the technical proof machinery after the reader has the construction in hand.

## 2 Main Results

The OPH papers support the following unified picture.

### 2.1 Theorem and source-object map

The proof architecture separates three distinct jobs. A paper theorem proves that stated finite or analytic premises imply a conclusion. A simulator or source construction produces the premises on a declared refinement tower. A receipt records enough public data for an independent verifier to check that the produced object satisfies them. A successful finite run does not replace an implication proof, and an implication proof does not construct its physical source object.

The receipt discipline has an executed bounded source run whose gate contract was committed before execution. Its extracted counts and class labels are mirrored as kernel-decided Lean literals, with each collapse of the truncated data stated as a theorem instead of repaired by synthetic data. The paired state and transition artifacts do not supply one common reference: the exact spectral and empirical-denominator obstructions in Section 2.4 rule out that binding for this source packet. The run is therefore reproducible negative source evidence rather than an end-to-end common-object realization, and its integers carry no physical claim [13].



| <b>Mathematical domain</b> | <b>Result</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Required source object and boundary</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Consensus protocol         | Semantic-dependency-complete transactions with one coherent canonical union-collar aggregate per conflict component satisfy the local diamond. Descent, this diamond, and repair completeness give a schedule-independent quotient normal form. Coherent recovery squares and pentagons give parenthesization-independent collar gluing. At one finite regulator, equality of complete public readbacks defines a literal kernel quotient equivalent to the range of realized signatures. Finite termination, terminal completion, confluence, fixed-point completeness, and repair-output and enabledness congruence imply one public endpoint for every completed schedule from every representative of one public class. | The engine supplies functional supports, read/write closures, aggregate payloads, protected and history data, repair completeness, and concrete finite peaks. Atomicity without semantic closure fails on a two-bit protected-observable countermodel; pairwise collar compatibility without coherent union data fails on parity and GHZ countermodels. The endpoint theorem assumes its confluence, completion, and quotient-congruence packet. No exhibited source repair family supplies the whole packet. Terminal completion is a finite postcondition rather than scheduler fairness, and the result supplies no cross-regulator endpoint or physical world. |

| Mathematical domain            | Result                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Required source object and boundary                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Compact relativity and gravity | <p>Real Pauli coordinates on Hermitian <math>2 \times 2</math> matrices give the quadratic form <math>t^2 - x^2 - y^2 - z^2</math>. Its positive future-null rays modulo positive scale are set-equivalent to <math>S^2</math>; its future unit-timelike hyperboloid has three-dimensional positive-definite rest spaces. An explicit real-linear chart into the Einstein tensor coordinates sends this form to minus the repository's <math>(-+++)</math> form and preserves the null cone exactly. Given an actual coincidence setoid, a coincidence-invariant raw readback descends uniquely. Separately, one supplied time-oriented affine Lorentz overlap cocycle and a chart-coordinate family satisfying its overlap law induce covariant displacements, invariant intervals, celestial directions, observer frames, and metric-preserving local rest-space readbacks. The formal stack does not identify the quotient-descended readback with that supplied coordinate family. The rank-three source Gram quotient is exactly isometric to the standard internal rest fiber as a candidate readout, not as absolute space. Event germs form the base while <math>H^3</math> is the local frame fiber. Finite oriented four-ball degree and two-sided metric control imply open four-dimensional charts. A convergent symmetric form with a spectral margin and two-sided causal classification has Lorentzian inertia. <math>C^{1,1}</math> tetrads give the metric regularity used by the curvature branch. Proper time is chart invariant, while clock comparison between worldlines requires an explicit event correspondence and calibration cocycle.</p> | <p>The Pauli and soldering results are internal algebraic geometry and a bounded conditional contract. The latter does not construct its coincidence setoid, chart-coordinate family or overlap law, source atlas, population, certified separation, open charts, physical cone attachment, refinement, causal reachability, frame selection, clock, spacetime, or dynamics. A reflexive symmetric pairwise-overlap relation is explicitly insufficient because it need not be transitive. The source tower supplies event equivalence and pseudometric data, finite chart maps and degree, quadratic-form samples, tetrads and inverses, causal tests, clock correspondences, calibration cycles, and their refinement moduli. The mixed Gelfand–Naimark–Segal common-comparison packet is an independent premise on the same tower; neither local maximum entropy nor refinement supplies it.</p> |

| Mathematical domain        | Result                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Required source object and boundary                                                                                                                                                                                                                                                                                            |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Finite local action domain | <p>One target-clean source capture gives an exact finite causal complex. Its declared chart contains one ancestry-depth coordinate and three spectral embedding columns, all nondegenerate on the sample, and its global held-out form has inertia <math>(1, 3)</math>. The capture also gives six observer neighborhoods with induced affine wiring transitions, an exact seam-sign layer, and typed scalar, chiral, and gauge sections. The declared unit-counting realization has a sign-twisted local derivative. Exact deterministic-section evaluations test its adjoint, kinetic, covariance, gluing, refinement, and boundary relations, while the signed-graph theorem fixes its kernel. Separate band premises select the rank-three response, and its tensor product with the declared fifteen-state table is a conditional rank-45 candidate. The table carries the chirality and diagonal <math>\mathbb{Z}_6</math> action. This local receipt checks the separately declared operator <math>D_\sigma \otimes I_{45}</math> and conditional inheritance of the signed seam operator's positive gap. It does not source-select the matter action or bridge the twelve-port Spin packet to this domain.</p> | <p>Five local quadratic fits have inertia <math>(1, 3)</math> and one has <math>(0, 4)</math>; the cone margins are negative. Continuum chart topology, Stiefel–Whitney identification, matter-pole identification, physical seam selection, laboratory attachment, and continuum operator limits require separate inputs.</p> |

| Mathematical domain | Result                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Required source object and boundary                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Screen microphysics | <p>Exact finite coarse graining sends modular potentials through log-sum-exp. A faithful equilibrium state leaves the reversible generator underdetermined. Rate affinities, the weighted graph Hodge split, entropy production, and a convention-fixed finite KMS jump theorem give the equilibrium operator layer. The <math>A_5 \times \mathbb{Z}_2</math> register has an eight-parameter self-adjoint commutant, its twelve-port commutant is generated by icosahedral adjacency, and it admits no equivariant directed 24-cycle. A source-derived 24-channel realization carries the edge modular grading exactly when its channel-closure residual vanishes; with <math>A_5</math> and reversal covariance, the compressed spectrum has sign-paired 1, 3, 3, 5 degeneracies, while the four gap values are source-rate data. On the certified echosahedral lineage, the declared integer counting and normalized readback-cost realization supplies the twelve unit lines. Local port incidence independently derives inverse pairing, proper <math>A_5</math> action, and the rank-three frame.</p> | <p>The source supplies coarse states or channels, conductances or rates, KMS frequency operators, positive rate matrices, the oriented channel map and its provenance, quotient lumpability, cycle-affinity, channel-closure, commutant, and refinement receipts. Register symmetry classifies finite operators only. It supplies neither a clock nor a physical gauge-current lift. Local icosahedral incidence does not produce the global federation nerve or its <math>S^2</math> support receipts. Physical phase locking is a candidate implementation bridge to overlap repair; no theorem identifies it with transactional confluence, modular flow, or an observer clock.</p> |

| Mathematical domain  | Result                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Required source object and boundary                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Finite event algebra | <p>The span of a projective record partition is a commutative star subalgebra exactly star-isomorphic to complex functions on its nonzero projector labels. A common linear isometry can copy two distinct sharp states from one normalized blank only when they are orthogonal. Partition pinching and partition averaging have explicit normalized Kraus families and preserve trace. The two-scale public relaxation and Poissonized repair families obey exact semigroup laws, and on bounded endomorphisms of a real Banach space the latter is exactly <math>\exp[t\gamma(E - I)] = E + e^{-\gamma t}(I - E)</math>. Positive unital maps of the active-record function algebra are exactly real row-stochastic kernels. Every star automorphism of the public function algebra is uniquely pullback by a label permutation, so every pointwise-continuous real-parameter group of arbitrary public star automorphisms is trivial. On one finite full private matrix block, every star automorphism is unitarily inner, while a fixed self-adjoint Hamiltonian gives a unitary real-parameter flow obeying the von Neumann equation. For nonselective projective pinching, <math>S(\mathcal{E}\rho) - S(\rho) = D(\rho\ \mathcal{E}\rho) \geq 0</math>; this identity is separate from a normalized selective Lüders branch.</p> | <p>The projective partition and input state are finite source data. The full mixed-state no-broadcasting implication enters through an explicit adapter. The finite complete-positivity predicate is defined in a separate module that proves the averaging, pinching, and nonnegative-parameter relaxation channels completely positive and trace preserving; the packet's own formal claims are Kraus normalization, Kraus form, and trace preservation. The operator-exponential theorem supplies no source rate or physical clock. The public automorphism classification is complete only for the exact finite active-record function algebra and reaches the matrix public algebra through the B1 equivalence. The private theorem covers one simple full block; the converse derivation of a coherent Hamiltonian from a pointwise-continuous automorphism group on that block is Theorem 2.2. The entropy identity is an analytic matrix proof outside the Lean export. No physical rate or clock is attached.</p> |

| Mathematical domain                                                | Result                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Required source object and boundary                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|--------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Finite Born-frame boundary and the effect-valuation representation | <p>The twelve declared central port atoms form one classical context with an eleven-dimensional weight simplex. A distinct qubit spinor adapter has six disjoint antipodal binary contexts whose additive weights have affine dimension six; the trace-one Hermitian/Born slice is three-dimensional and obeys three exact golden-ratio relations. Representation there is unique when it exists, but exact weights show that neither existence nor positivity follows. On the full C1 celestial sphere, the continuous weight <math>F(n) = (1 + n_z^3)/2</math> is normalized on antipodes yet non-affine. The positive bridge is a theorem: every valuation on the effects of a finite matrix algebra that is nonnegative, normalized, and additive on coexisting effects equals <math>E \mapsto \text{Tr}(\rho E)</math> for a unique density operator <math>\rho</math>, in every finite dimension including two, with no continuity axiom. Finite receipts locate the missing hypothesis rather than deriving it: sharp binary webs admit the cube valuation as a non-Born extension, and one unsharp three-effect context excludes that particular extension. The same finite battery nevertheless admits the planar cubic response <math>F_y(n) = (1 + n_y^3)/2</math>, which agrees with the maximally mixed Born weight on every displayed axis while remaining non-affine away from them. The realized real <math>S_3</math> source web is also blind to the Pauli-<math>Y</math> direction: two distinct complex states give equal weights on every source projector.</p> | <p>The spinor projector family is a declared mathematical adapter, not a source-produced public instrument. The celestial countermodel has a sharp role: it proves that projector webs alone underdetermine the weight, and the planar control proves that one finite unsharp battery does not force a global valuation. Positive closure requires a source-realized complex-tomographically-complete effect/instrument web, an operational coarse-graining or equivalence theorem deriving noncontextual additivity, and only then preregistered frequencies as validation. Frequencies alone do not prove the universal valuation law.</p> |

| Mathematical domain          | Result                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Required source object and boundary                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Finite regional observer net | A proof-carrying enrichment of one finite consensus tower packages a finite region order, isotony, exact commutation on declared-disjoint regions, covariant refinement, compatible regional expectations, and idempotent local repair. A declared repair that fixes remote observables gives exact Heisenberg nondisturbance for every matrix state. Compatible local sections have a unique glue only on nonempty finite subregion families carrying the explicit restriction and uniqueness receipts. The API name <b>FiniteCover</b> does not include a joint-coverage law. A separately declared basis split and Kraus completeness give a generic reindexed marginal identity; no theorem identifies the regional algebras with its matrix factors. | Contravariant restriction maps and their retraction of isotony are strong extra premises, not consequences of an ordinary isotonic net. The consistency construction is parameterized by a supplied partition and state, assigns the same commutative public-record algebra to every region, and uses identity restrictions. It checks consistency of the interface but supplies no closed concrete witness, genuine coverage theorem, regional factorization, or noncommutative source-attached model. Positive quantum channels, adaptive-repair scheduler locality, spacetime causality, a time-slice property, continuum field theory, sectors, and laboratory attachment are open. |
| Boltzmann transport          | Convergence of the discrete generators to the null Hamiltonian vector field, vanishing carré du champ, tightness, and uniqueness of the limiting ODE imply deterministic Liouville transport. A limiting collision generator gives Boltzmann transport; nonzero limiting quadratic variation gives a diffusive limit.                                                                                                                                                                                                                                                                                                                                                                                                                                     | The simulator supplies quotient-lumpable transition kernels, physical time steps, first and second conditional moments, mass-shell and boundary defects, and schedule/refinement-natural convergence. Path confluence alone supplies none of these transport limits.                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Particle and gauge sectors   | Complete reversible response and endogenous overlap transport force the abstract local Standard Model gauge Lie algebra from the twelve-port module. The released charged-double-triplet matrices give an exact full-rank conditional witness. The simulator producer derives the inverse-port response constraints, not the matrix generators or their bracket.                                                                                                                                                                                                                                                                                                                                                                                          | The source supplies no ordered current tomography, same-current holonomy, laboratory identification with Standard Model currents, or intertwiner to the independently reconstructed Tannaka current; the identification consumes these as premises. A register adjacency or modular generator cannot be relabeled as this current.                                                                                                                                                                                                                                                                                                                                                      |

| Mathematical domain | Result                                                                                                                                                                                                                                                                                                                                            | Required source object and boundary                                                                                                                                                                                                                                                                                                   |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Yang–Mills gap      | For the source-defined finite collar family, the finite-type Dobrushin/minorization theorem gives one positive projection/transfer-gap modulus uniform in location, boundary condition, system size, and the declared cofinal refinement tower. Removing locality or the stated mixing control admits finite countermodels with a collapsing gap. | The physical compact-gauge branch must produce the admissible collar types, transition matrices, uniform constants, refinement maps, reflection-positive vacuum, and continuum regularity packet. The finite theorem closes the repair-gap implication. The Clay-facing continuum construction is conditional on that source receipt. |

This map fixes the mathematical type of each result. The cited specialist papers give the detailed definitions and proofs. A finite operator, receipt, or simulation trace retains its type when used in another argument.

1. **Observers and records become physical objects.** Finite patches, shared overlaps, record algebras, repair maps, and observer checkpoints are represented on the screen as part of the construction.
2. **Spacetime is reconstructed.** On the controlled Einstein-branch geometry readout, the finite cap-normal Bisognano–Wichmann (BW) certificate (support-order faithfulness, BW framing, held-out oriented cross-ratio rigidity, and independently normalized geometric  $2\pi$ -KMS (Kubo–Martin–Schwinger) convergence), together with an independent mixed Gelfand–Naimark–Segal common-comparison premise on the same tower, makes the support-visible cap modular automorphism geometric and supplies Lorentz kinematics. Recoverable generalized entropy together with the null modular bridge, bounded-interval transport, fixed-volume small-ball area variation, admissible fixed-cap stationarity, and timelike tensor upgrade then supplies the Einstein relation on one source-derived common-domain tower with certified tails and independent physical identifications. The finite artifacts test algebra, evaluator logic, manifests, deletion rules, and synthetic controls. They do not certify such a tower; the tower enters the Einstein relation as a premise. Bare finite consensus supplies only the upstream quotient-normal-form input. The screen-capacity branch proposes the dimensionless cosmological-constant relation below. The identification of a cosmic record-closure value with the electroweak bridge requires the common screen/electroweak load-carrier identification. A fixed-cutoff source-derived simulator packet exists at  $D = 24$ . A bounded capacity-indexed counterfamily has different exact fixed sets under shared base, positivity, carrier, and executable finite controls. The exact verdict applies to this bounded completion class. Universal all-rung membership in a complete A1–A3 capacity-source contract and the executable-to-Lean bridge are open. Direct  $N$  is not evaluable on the incomplete source antecedent:

$$\mathfrak{F}_{r,0}(D_\star) = \{D_\star\}, \quad N_{\text{CRC}} = \log D_\star, \quad \Lambda_{\text{CRC}} \ell_\star^2 = \frac{3\pi}{N_{\text{CRC}}}, \quad \Lambda_{\text{CRC}} = \frac{3\pi}{GN_{\text{CRC}}}.$$

The last display uses the selected scale certificate  $G_{\text{geom}} = \ell_\star^2$ . The Einstein-branch antecedent is instrumented end to end: a typed common-domain source tower with hash-bound provenance and cross-source splice rejection, and fail-closed instruments with adversarial negative



controls for geometric modular normalization, GNS cyclicity and modular intersections, the ancestry-derived event cone, and same-source stress/coupling, together with semantic counter-models proving each receipt family load-bearing [12]. Two clauses are theorem-grade: coupling universality holds with zero spread for every icosahedrally equivariant source law, and generator positivity holds by construction on the declared law family. The Einstein-cone convergence ladder, measured at 16,384, 65,536, and 262,144 carriers with constant cross-observer coupling density, carries Lorentzian inertia  $(1, 3)$  at every rung with the cone margin halving per rung and the coupling spread decreasing beside it; a density-control run degrades the signature on cue, isolating coupling density as the mechanism of cone merging. Primary data, per-rung configurations, and provenance commits accompany the numerical evidence. The cap-state modular temperature and the projected margin zero crossing have fixed acceptance rules and lack qualifying measurements. A separate target-clean capture inhabits the finite local action domain: 2,304 causal events, six observer neighborhoods with exact induced affine wiring transitions, and an observer-visible seam complex with 8,662 carriers, 11,816 seams, and 38 frustrated triangles. Two coordinate routes through the same  $\mathbb{F}_2$  rank implementation agree with the exact rank identity and give lift-ambiguity rank 3,117. The section, operator, boundary, and replay certificates are exact [12]. One local chart has inertia  $(0, 4)$ , and the measured cone margins are negative. The finite object therefore inhabits the typed domain without establishing the continuum event-manifold or Einstein antecedent. The source-forward bridge formula has the certified mathematical value  $3.5321315434 \dots \times 10^{122}$ ; the weighted Planck base- $\Lambda$ CDM comparison coordinate is  $3.3129270981 \dots \times 10^{122}$ . The bridge value is about 6.6 percent above it. On the declared one-class branch, physical class selection and a scalar-weighted presence-survival attachment would give  $3.2920978773 \dots \times 10^{122}$ ; the separate exponential mean-count branch would give  $3.3000722254 \dots \times 10^{122}$ . Exact positive composition countermodels select neither action on global capacity. The named-law branch is not evaluable on this source class, and both comparisons are retrospective. A uniqueness certificate for the bridge formula does not establish the unique zero of the public-record slack, select one physical capacity, or construct its universe-level carrier.

3. **The finite gauge chain forces the abstract local Standard Model gauge Lie algebra.** Complete reversible response and endogenous overlap transport supply the load-bearing premises. Under the explicit inverse-port contract, incidence and target-blind port readback derive the signed response. Under the conditional matrix current and matter representation, anomaly-forced determinant balance and exhaustive tensor descent fix exact hypercharge, a three-color carrier, a common  $\mathbb{Z}_6$  kernel, and the maximal faithful matter image  $(\text{SU}(3) \times \text{SU}(2) \times \text{U}(1))/\mathbb{Z}_6$ . The cover and its three nontrivial central quotients carry the same local tensors. The six-axis menu matches the  $\mathbb{Z}_6$  quotient after its relations are declared, so the physical global form is not source-selected. The scalar scan fixes compatible charges and Yukawa channels, not scalar multiplicity. Within this gauge chain, physical family interpretation and extra-sector exclusion require separate identifications; charged-lepton and quantitative branches retain their separately stated declared premises. The same product structure gives the structural absence of the simple-GUT  $X/Y$  gauge channel. General proton stability is not implied. Conditional on the declared one-generation matter table and baryon and lepton labels, an exact dimension-six census admits  $QQQL$ ,  $QQu_R e_R$ ,  $QLu_R d_R$ , and  $u_R u_R d_R e_R$ . Their Wilson coefficients, physical emission, decay amplitudes, and lifetimes are not supplied. On the ordinary electromagnetic branch, the unbroken  $\text{U}(1)_Q$  factor reduces the compact-gauge curvature to  $F_Q = dA_Q$ ; after separately assuming the low-energy Maxwell action and its nonzero kinetic term, variation gives  $dF_Q = 0$  and  $d \star F_Q = g_Q^2 \star J_Q$ , i.e. Maxwell's

equations after canonical electromagnetic normalization. The rank-three screen band and declared fifteen-state generation give a conditional rank-45 tensor candidate. The table carries its chirality and diagonal  $\mathbb{Z}_6$  action. On a separate finite local domain, a receipt checks the declared operator  $D_\sigma \otimes I_{45}$  and conditional inheritance of the signed seam operator's positive gap. The source does not select this action, and the local domain is not certified as the domain of the twelve-port Spin packet. Interpretation as three physical families requires matter-pole identification, a continuum Spin/locality limit, physical seam selection, refinement persistence, and laboratory-current receipts. A target-clean finite local-domain capture separately supplies scalar, chiral, and gauge section spaces and a sign-twisted kinetic Hamiltonian. The exact kinetic identity makes the Hamiltonian positive semidefinite, and the frustrated seam layer makes its kernel trivial. Its dimensionless gap is therefore strictly positive. A sparse numerical refinement gives  $0.1175367\dots$  with a relative residual below  $1.5 \times 10^{-14}$ . This number is not a certified gap bound, a selected physical transition, or an SI energy. Physical fiber selection and clock attachment are separate [12]. The transportable-sector/Tannaka route is a separate conditional compact-group classification. The target-blind producer derives  $R = -J$ , without deriving the matrix current. Physical matter typing, identification of the finite current with the Tannaka current, and laboratory-current attachment are separate premises.

4. **The measured endpoint tests the local closure.** The comparison pixel  $P_C \simeq 1.6309682094$  is inferred from the measured fine-structure constant solely for comparison:  $\alpha^{-1}(0) = 137.035999177(21)$ ,  $\alpha(0) \simeq 0.00729735256433$ . The forward model with the source anchor and Ward-projected electromagnetic transport measures the loop-closure residual:  $3.0 \times 10^{-4}$  relative on the source chain and  $2.5 \times 10^{-6}$  relative for the certified numerical gauge-width fixed point  $\alpha^{-1} = 137.035660136946577\dots$ , a certified fixed point of the declared map. Physical promotion additionally needs source-derived Ward-projected transport, target-independent map selection, and the typed same-quantity bridge. The mixed-provenance diagnostic  $137.0359595136\dots$  is a fixed point of no single declared map and is excluded from the physical output set. The separate root  $136.994835177413\dots$  is an interval-certified output of an incomplete declared map; no derivation relates it to the physical fine-structure endpoint. The comparison distinguishes the source-side diagnostic from the endpoint residual.
5. **Edge sectors also admit two-dimensional Yang–Mills and worldsheet-style effective descriptions.** The heat-kernel edge partition function reorganizes the same boundary data into the two-dimensional Yang–Mills form, and on the declared large-edge branch this yields a controlled worldsheet effective description. This 2D bridge sits beside the Standard Model gauge paper's controlled compact-gauge repair-gap theorem. The string-vacuum selector uses this bridge only as the effective edge-language input; critical worldsheet closure, Bouchard–Donagi cohomology, operator-safety realization, threshold matching, and moduli locking are separate premises. The rank certificate gives proxy rank zero on the specified incomplete slice and supplies no physical slice or Jacobian. The Bouchard–Donagi construction is therefore a structural benchmark rather than a selected string witness.
6. **The compact-gauge branch carries the 4D Euclidean Yang–Mills form and isolates the gap theorem.** On the declared controlled compact-gauge branch, with the four-dimensional scaling chart, reflection-positive ordinary vacuum, no additional gauge-invariant relevant dimension-four pure-gauge operator on that branch besides curvature squared, repair completeness, and the Standard Model gauge paper's explicit finite cylinder system in force, OPH proves projective weak-\* / GNS extraction of a support-visible cylinder cluster family. Identification of that family with the Euclidean Yang–Mills action additionally

uses the renormalized four-dimensional regularity/universality receipt. Weighted observation-fiber resampling is the finite conditional-expectation projector, with the noncircular matrix-recognition criterion of Ref. [1]. A concrete active repair law is identified with that projector only after its independently extracted transition matrix passes the support, equal-fiber-row, and weighted detailed-balance receipt. Identification of Euclidean transfer with the repair generator, vacuum persistence, and passage of a uniform finite-stage gap use the separate finite ground-state-transform, transfer/vacuum, OS-regularity/noncollapse, and uniform-gap receipts. The checked 244-type Ising collar table is an exact calibration of that finite receipt format, not evidence for the physical compact-gauge source family. Only on that certified branch is  $\Delta_{\text{YM}} = \Delta_{\text{rep}}$ . Weak-\* extraction alone is not the Clay-facing axiomatic construction.

## 2.2 The observerhood receipt and the common-origin bridge

The observer named in the title has a theorem-level bounded receipt. Over any instance of the companion formal tower interface, its base structure packages clauses 1 through 6 and the own-observer half of clause 7 as typed data: a bounded accessible star subalgebra of the private algebra containing the observer's own public record algebra; a self-readback map sending accessible elements into that public algebra and fixing each of its elements; durability, with every record element annihilated by the observer's declared generator; record-conditioned control maps preserving the accessible interface; a prediction map whose output never strictly precedes its input in the observer's own record order; label and prediction families commuting with the declared refinement maps; and a declared readout equal to the trace pairing of each record element against the observer's own state, with every record element fixed by every declared control. Readback idempotence and refinement stability of the readout follow. An explicit witness inhabits the receipt away from the constant adaptor: private algebra  $M_2(\mathbb{C})$ , a nonzero generator given by the scaled commutator with  $\text{diag}(1, 2)$ , which annihilates every diagonal record element while moving the raising matrix unit, control by conjugation with the involution  $\text{diag}(1, -1)$ , three chain-ordered records, and a nonconstant non-identity prediction map. Four negative controls (an erasing readback, an identity generator, a resetting prediction, and a constant misreading) each violate one attained field on the same data. The base structure is operational and finite: it selects no unique observer over a given tower, attaches no physical instrument, makes no consciousness claim, and supplies no source realization of the tower data. A separate fixed-regulator mixin closes the cross-observer half on an access cut. It requires two distinct labels, exact agreement of both accessible interfaces with the cut, equality of every committed record after the two typed restrictions to the owner-region meet, equality of own readouts, and a nonzero common restriction. The common section is thereby accessible to both observers. Its proper-meet witness uses two distinct diagonal owner algebras, a scalar bottom algebra, and corner-character restrictions: observer-indexed records differ before restriction and agree nontrivially only after restriction. A one-corner mutation makes the overlap receipt false. The E6 event consumer takes two operational observers owning two charts in which one event is visible and proves equality of the two owner-region restrictions of its packet record together with two-sided self-readback. Thus all seven operational clauses are inhabited at one finite cut. Cross-regulator naturality, triple-overlap coherence, and source realization remain E2 obligations [13].

An access cut on the finite regional observer net above assigns each observer one declared support region and one accessible star subalgebra squeezed between the observer's public record algebra and the regional algebra of that region, stable under the declared restrictions below it; the observer-local algebra at a region is the meet of the regional algebra with the accessible algebra, and the net receipts transport to this meet. A finite event world over the supplied Lorentz soldering

binds one packet  $\mathbf{p}_e$ , with packet carrier fixed independently of the quotient state, to each event  $e$ , with declared chart ownership. The common-origin theorem states that for every active visible chart/event pair  $(i, e)$ ,

$$\text{supp}(\mathbf{p}_e) \preceq R_{O(i)}, \quad \text{rec}(\mathbf{p}_e) \in \mathcal{L}_{O(i)}(\text{supp}(\mathbf{p}_e)) \cap \mathcal{P}_{O(i)}, \quad x_i(e) = \text{geo}_i(\mathbf{p}_e),$$

where  $O(i)$  is the declared owner of chart  $i$ ,  $R_{O(i)}$  its observer region,  $\mathcal{L}_{O(i)}$  its observer-local algebra, and  $\mathcal{P}_{O(i)}$  its public record algebra: one event packet determines the event support region below the owner's region, the public record observable in the meet of the observer-local algebra at that support with the owner's public algebra, and the Lorentz chart coordinate as a readout of the same packet. A selected coarse-to-fine refinement transports all three conjuncts. The witness chain is strict, with the public algebra strictly inside the accessible algebra strictly inside the private algebra, and the witness records separate packets. Charts and observers are distinct types, ownership is declared data, and no physical region, coverage, instrument, spacetime, causal cone, event actualization, or continuum object is claimed [13].

Coexistence of regional operator algebras on this net interface has an exact boundary.

**Theorem 2.1** (Scalar-character obstruction and character-block coexistence). *For every  $n \geq 2$  there is no unital complex-algebra homomorphism from the full matrix algebra  $M_n(\mathbb{C})$  into the scalars, and no unital complex-algebra homomorphism into the scalars accepts an anticommuting pair of units. On this net interface, at any regulator of nonzero dimension, a region whose local algebra consists of scalars admits above it no region whose local algebra contains a unital copy of  $M_n(\mathbb{C})$  with  $n \geq 2$ : the declared star-homomorphic restriction retraction would compose with the copy to a scalar-valued unital homomorphism on a full matrix algebra. At any such regulator, a region lying above a region with commutative local algebra carries no anticommuting pair of units: the declared restriction would send the pair to commuting units whose product equals its own negation. No full matrix factor of dimension at least two sits above a scalar overlap region, and the standard tensor-factor local-algebra picture with trivial overlap is impossible on this interface. The boundary is inhabited: over a dimension-four private algebra, the block algebras of matrices  $\text{diag}(a, a, A)$  and  $\text{diag}(B, b, b)$  with arbitrary two-by-two blocks  $A$  and  $B$  are each noncommutative, commute elementwise, neither contains the other, and each retracts onto the scalar overlap algebra through the character evaluating its one-dimensional corner block.*

*Proof.* The matrix ring  $M_n(\mathbb{C})$  is simple, so the kernel of a unital homomorphism into the scalars is trivial and the homomorphism is injective, while the off-diagonal matrix unit is a nonzero element whose square is zero and whose scalar image would be a nonzero complex number whose square is zero. For the anticommuting form, a unital homomorphism into the scalars sends units to nonzero scalars, and applying it to the anticommutation relation forces the product of the two image scalars to vanish. The interface statements compose these arguments with the declared restriction into the scalar or commutative region. The inhabitant is finite linear algebra over one declared regulator with identity repair maps; the general matrix obstruction, the interface compositions, the interface laws, both noncommutativity witnesses, the mutual noncontainment, and the corner-character retractions are checked in Lean [13].  $\square$

Noncommutative regional coexistence on this interface therefore passes exactly through character-block structure. The declared overlap algebra is a proper subalgebra of the set intersection of the two block algebras, and genuine coverage semantics and regional-factor receipts for a source-produced net are open.

Dynamics on the private block carries a matching rigidity statement.

**Theorem 2.2** (Finite Stone converse on one full private block). *Let  $\alpha_t$ ,  $t \in \mathbb{R}$ , be star automorphisms of one full finite matrix block  $M_n(\mathbb{C})$  with  $\alpha_0 = \text{id}$ ,  $\alpha_{s+t} = \alpha_s \circ \alpha_t$ , and every orbit map  $t \mapsto \alpha_t(x)$  continuous. Then there is a self-adjoint  $H \in M_n(\mathbb{C})$  with*

$$\alpha_t(x) = e^{-itH} x e^{itH} \quad \text{for all real } t \text{ and all } x \in M_n(\mathbb{C});$$

*the conjugating family  $U_t = e^{-itH}$  is unitary with  $U_0 = 1$  and  $U_{s+t} = U_s U_t$ , and  $H$  is unique up to one additive real scalar multiple of the identity.*

*Proof.* Continuity upgrades to differentiability at parameter zero by interval averaging: the group acts on the matrix-unit basis as a norm-continuous matrix family whose short-interval average is invertible, and the fundamental theorem of calculus recovers the family from its primitive. The derivative at zero is a star-compatible Leibniz endomorphism. Every complex-linear Leibniz endomorphism of the block is an inner derivation, with an explicit matrix-unit witness proved from scratch; star compatibility symmetrizes the witness into one self-adjoint generator; two generators with one commutator action differ by a real scalar multiple of the identity; and the group differential equation identifies the flow with conjugation by the coherent unitary family. Every step is checked in Lean [13].  $\square$

On the finite private block, a pointwise-continuous symmetry flow of the observables therefore has no choice except Hamiltonian form; the additive energy offset is the only freedom. The theorem extends to the general finite private algebra: every unital star subalgebra of a finite complex matrix algebra, containing the ambient identity, is star-isomorphic to a finite direct sum of full matrix blocks, and on that block model a pointwise-continuous star-automorphism group fixes every block projection and is blockwise conjugation by the unitary family of one time-independent self-adjoint Hamiltonian per block. Block fixing is derived rather than assumed; a single automorphism may swap isomorphic blocks, and the swap is realized on the square two-block algebra. The parameter is a real number; no physical clock, source-selected Hamiltonian, or physical time identification is claimed.

## 2.3 Conditional history weights and the variational bridge boundary

Two standard finite-horizon constructions clarify what the observer axioms must supply before an action principle could be claimed.

**Theorem 2.3** (Finite path projection and positive-gap tail). *Let  $\Omega$  be finite, let  $\Gamma = \Omega^{\{0, \dots, N\}}$ , and let  $P, \tau$  be probability laws on  $\Gamma$ , with  $\tau > 0$ . Let  $R > 0$  be a reference weight, let  $F_a : \Gamma \rightarrow \mathbb{R}$  be finitely many declared observables, and suppose*

$$\log \tau(\gamma) = \log R(\gamma) - \sum_a \lambda_a F_a(\gamma) - \log Z$$

*and  $P$  and  $\tau$  have the same  $F_a$ -moments. Then*

$$D(P\|R) = D(P\|\tau) + D(\tau\|R), \quad D(\tau\|R) \leq D(P\|R).$$

*With*

$$A_{\text{eff}}(\gamma) = -\log R(\gamma) + \sum_a \lambda_a F_a(\gamma),$$

*one has  $\tau(\gamma) = \exp[-A_{\text{eff}}(\gamma) - \log Z]$ , so every supplied modal history minimizes  $A_{\text{eff}}$  over  $\Gamma$ .*

Separately, for any nonempty finite set  $\Gamma$ , function  $S : \Gamma \rightarrow \mathbb{R}$ , global minimizer  $\gamma_0$ , and

$$q_\beta(\gamma) = \frac{e^{-\beta S(\gamma)}}{\sum_{\delta} e^{-\beta S(\delta)}},$$

the mass at least  $\Delta > 0$  above that minimum obeys

$$\sum_{S(\gamma) \geq S(\gamma_0) + \Delta} q_\beta(\gamma) \leq |\Gamma| e^{-\beta \Delta}$$

for  $\beta \geq 0$ , and this displayed mass tends to zero as  $\beta \rightarrow \infty$ . Finiteness also gives the stronger concentration statement

$$\lim_{\beta \rightarrow \infty} \sum_{S(\gamma) > S(\gamma_0)} q_\beta(\gamma) = 0,$$

so the total mass outside the full minimizer set vanishes.

*Proof.* Expand  $D(P\|R) - D(P\|\tau)$ . The displayed logarithmic form, normalization, and moment matching make the result independent of  $P$  on the constraint surface; evaluating it at  $P = \tau$  gives the Pythagorean identity. Nonnegativity of  $D(P\|\tau)$  gives minimality. Exponentiating the logarithmic form reverses probability and effective-action order. For the tail bound, the partition sum is at least  $e^{-\beta S(\gamma_0)}$ , while each displayed numerator is at most that quantity times  $e^{-\beta \Delta}$ ; summing and taking the elementary exponential limit proves the fixed-gap statement. Apply that argument to each strictly nonminimal history and sum over the finite set for the final limit. The statements are checked in Lean [13]; they are instances of standard information-projection mathematics [17] and elementary finite exponential estimates.  $\square$

The theorem assumes the exponential form and matching moments. It neither constructs  $R$  from the OPH source nor proves that an Axiom 3 history optimizer exists, has that form, or is unique. Calling  $R$  a probability law additionally requires its normalization.

**Theorem 2.4** (Real finite-horizon variational helpers). *Let  $L : \mathbb{R}^2 \rightarrow \mathbb{R}$  be differentiable and define*

$$\mathcal{S}_L(\gamma) = \sum_{n=0}^{N-1} L(\gamma_n, \gamma_{n+1}) \quad (\gamma : \{0, \dots, N\} \rightarrow \mathbb{R}).$$

*At an interior junction  $k + 1$ , suppose  $\gamma$  minimizes  $\mathcal{S}_L$  under every replacement of  $\gamma_{k+1}$ . Then*

$$\partial_2 L(\gamma_k, \gamma_{k+1}) + \partial_1 L(\gamma_{k+1}, \gamma_{k+2}) = 0.$$

*If a differentiable one-parameter family  $T_s$  satisfies  $T_0 x = x$ , has generator  $\xi(x) = \partial_s T_s x|_{s=0}$ , and obeys  $L(T_s x, T_s y) = L(x, y)$ , then the segment momentum*

$$p_n = \partial_2 L(\gamma_n, \gamma_{n+1}) \xi(\gamma_{n+1})$$

*is equal on the two segments adjacent to that junction. More generally, for a path with  $M + 2$  records, if the same single-site minimization premise holds at all  $M$  interior records, then there is a scalar  $J$  such that  $p_n = J$  on every one of its  $M + 1$  segments.*

*Proof.* Changing one interior record changes exactly the two adjacent summands. Differentiating their sum at its declared minimum gives the first identity. Differentiating the invariance relation at  $s = 0$  and combining it with that identity gives  $p_k = p_{k+1}$ . Applying this identity at every interior junction and inducting along the finite chain gives the scalar  $J$ . These are standard discrete variational and Noether calculations [18]; their precise premises are checked in Lean [13].  $\square$



**Proposition 2.5** (Finite/real variation obstruction). *Fix a real path  $\gamma : \{0, \dots, N\} \rightarrow \mathbb{R}$ , a site  $i$ , and any finite family  $G$  of real paths. There is an  $x \in \mathbb{R}$  for which the path obtained by replacing  $\gamma_i$  with  $x$  does not belong to  $G$ . Hence no finite real-path family contains every single-site real variation of even one supplied path.*

*Proof.* The map from  $x$  to the updated path is injective because evaluation at  $i$  recovers  $x$ . Its image is infinite and therefore cannot be contained in finite  $G$ . This elementary cardinality obstruction is machine checked in Lean [13].  $\square$

The two theorem domains compose only through a certified receipt. The first uses paths in a finite state set; the second uses arbitrary real single-site variations, and Proposition 2.5 rules out closure under all such variations inside any finite real-path family. The composition that survives this obstruction is conditional: an undercut receipt asserts that every real single-site variation of an embedded path is undercut in action by some embedded finite path, and under that receipt finite minimality transfers to real local minimality and to the real Euler–Lagrange packet. Membership receipts are impossible (that is the obstruction restated); undercut receipts are attainable, and the committed quadratic witness carries one, while a one-path control family provably carries none and the transferred conclusion fails there. All three layers, transfer, witness, and control, are checked in Lean [13].

Which action the history law carries is itself a theorem rather than a choice.

**Theorem 2.6** (Derived action of a source chain). *Let  $\Omega$  be a finite nonempty state set,  $P$  a strictly positive row-stochastic kernel, and  $\pi$  a normalized initial law. On paths with  $n$  transitions define the log-transition action  $S_{\log}(\gamma) = -\sum_i \log P(\gamma_i, \gamma_{i+1})$  and the reference  $R(\gamma) = \pi(\gamma_0) |\Omega|^{-n}$ . Then the Markov path law equals the exponential tilt of  $R$  by  $S_{\log}$  at multiplier one, with partition constant  $|\Omega|^{-n}$ ; and a pair  $(S', \lambda')$  reproduces the path law as its tilt of  $R$  if and only if there is a constant  $c$  with  $\lambda' S' = S_{\log} + c$  pointwise.*

*Proof.* Exponentiating the negated action recovers the transition product, and row normalization makes the partition sum telescope to  $|\Omega|^{-n}$ , which cancels against the reference weight. Uniqueness reduces to the statement that two tilts of one strictly positive reference agree exactly when the multiplier-weighted actions differ by an additive constant, proved by taking logarithms at a common path and cancelling the shared normalizer. Both directions of the gauge are realized: per-step additive constants aggregate to a path constant, and the multiplier trades against the action scale, so the normalization convention pins  $\lambda = 1$ . The statements are checked in Lean [13].  $\square$

The committed source chain instantiates the theorem exactly: the tilt of its reference by its log-transition action reproduces its path law with partition constant  $1/4$ , through kernel-decided integer identities, and the committed repair-count action reproduces that law at no multiplier, because it takes equal values on two paths that the reference weighs equally and the chain separates. The action of the realized dynamics is therefore forced up to an additive constant and a multiplier rescaling. This multiplier is an action-normalization gauge for the supplied chain; it is not thereby identified with a gauge-sector coupling coefficient. The separate finite Gibbs-kernel construction for one three-dimensional  $P$ -family factor shows that a common rescaling of its supplied quadratic cost can occupy the same formal multiplier slot, but it neither derives that kernel from source dynamics nor determines relative couplings between simple factors.

The two attained faces of mechanics are joined by a finite Legendre bridge, also checked in Lean [13]. For discrete two-point Lagrangians the Legendre transform, its involutivity on the strictly convex class, the Fenchel–Young inequality with its equality case, and exact degenerate controls are proved; the discrete Euler–Lagrange condition at a junction holds exactly when one step of the

discrete Hamilton flow carries the incoming junction state to the outgoing one, for the quadratic class and for strictly convex Lagrangians with a solver section; single-site minimizers of the log-transition local action coincide with most probable paths of the realized chain, through an exact corner identity; and the constant Noether current of the chain face equals the quadratic Legendre momentum, conserved together with the energy along the free Hamilton orbit that reproduces the committed witness path. One remainder is named: no theorem produces the realized chain from a Hamiltonian flow, and the naive quadratic-Gibbs identification is refuted on the chain literals, so that direction runs through the derived action above or not at all.

The reference measure remains a declared object, and the package supplies no identification of  $A_{\text{eff}}$  with the local action, no identification of the finite scalar chain constant with a vector-valued or physical current, no complex amplitude, interference rule, physical clock,  $\hbar$ , continuum path measure, or laboratory action. It derives the form, the action, and the bridge at the representation level; the physical attachments are open obligations under the attachment doctrine.

## 2.4 Thermodynamics from conditional repair in an observer modular parameter

On one finite regulator stage, the four laws form a conditional theorem package once one repaired distinction and two compatible instantiations of Axiom 3 are supplied. Five named source and physical receipts are unsupplied. Three maps stay distinct. The strict-descent normalizer settles public facts; it is terminating and idempotent and carries no entropy inequality, and the two-point normalizer  $a \mapsto b$ ,  $b \mapsto b$  sends the uniform law to a deterministic law, dropping Shannon entropy from  $\log 2$  to zero. The equilibrium repair operator resamples unresolved degrees inside the fibre of the complete repaired visible datum. The modular flow of a faithful reference supplies the reversible observer ordering and preserves relative entropy. Only the middle map carries the arrow.

Axiom 3 applies separately to states and to transition distributions, and both instantiations are solved exactly. On the state side, with faithful global reference  $\sigma_0$  and observer-visible conserved quantities  $Q_a$ , the information-projection Pythagorean identity  $D(\rho\|\sigma_0) = D(\rho\|\tau_\lambda) + D(\tau_\lambda\|\sigma_0)$  holds on the constrained moment surface, so the optimizer, unique by strict convexity of relative entropy, is the exponential family  $\tau_\lambda = Z^{-1} \exp(\log \sigma_0 - \sum_a \lambda_a Q_a)$ ; on the tracial branch this is the Gibbs family, and equality of intensive multipliers at additive contact is the zeroth law, with transitivity carried by the thermometer theorem: a nondegenerate finite spectrum identifies its inverse temperature, so two Gibbs states that agree as distributions have equal  $\beta$ . On the transition side, the feasible set at  $x$  is the fibre  $F_C(x) = \{y : b_C(y) = b_C(x)\}$  of the complete repaired visible datum  $b_C$ , and the minimizer of  $D(q\|\pi)$  over the fibre, unique by the same strict convexity, is weighted conditional resampling,

$$P_C(x, y) = \mathbf{1}_{\{b_C(y)=b_C(x)\}} \frac{\pi(y)}{\pi(F_C(x))},$$

which is the weighted observation-fiber resampling projector, and any physical identification of a concrete repair law with it passes through the support, equal-fiber-row, and weighted detailed-balance receipt stated above. The kernel is stochastic, idempotent, reversible, and stationary for  $\pi$ , and fixes every fibre-measurable observable, which is dynamical conservation of every protected charge and the content side of the first law; the exact bookkeeping identity  $dU = \delta Q + \delta W$  with its explicit bilinear cross term is the algebraic side. Data processing then gives the transition-side second law  $D(pP_C\|\pi) \leq D(p\|\pi)$ , and the exact split  $D(p\|\tau) = \langle K \rangle_p - S(p)$  with  $K = -\log \tau$  turns it into the modular Clausius inequality  $\Delta S \geq \Delta \langle K \rangle$  for every  $\tau$ -preserving repair channel, which specializes to  $\Delta S \geq \beta Q$  on the identified energy branch. The Landauer bound is a one-line corollary: a repair channel that lowers the record entropy by  $c$  expels at least  $c/\beta$  of energy, so



erasing one bit costs at least  $k_B T \log 2$ . At every finite regulator the excited Gibbs mass obeys  $q_\beta \leq \frac{d-g_0}{g_0} e^{-\beta\Delta}$ , with the quantitative threshold that any target mass  $\varepsilon$  is reached once  $\beta\Delta$  exceeds  $\log \frac{d-g_0}{g_0\varepsilon}$ , and the entropy limit is  $k_B \log g_0$ . The repair kernel preserves full support, since every diagonal entry is positive, while the zero-temperature state is rank deficient for  $g_0 < d$ ; finitely many repair or pinching steps therefore cannot reach it, which is the fixed-regulator third law. Public quantum-record pinching and averaging have explicit normalized Kraus families and preserve trace. A separate module proves these channels completely positive and trace preserving through the Kraus criterion; this result's own packet reports Kraus and trace identities. The record tower  $S(\rho) \leq S(\mathcal{E}_\pi \rho) \leq S(\mathcal{A}_\pi \rho)$  separates conditioned learning from the nonselective commit, so observers convert uncertainty into correlations and durable records without violating the second law.

The second law refines to an exact identity. For one repair step the fluctuating entropy production  $\sigma(x, y) = \log \frac{p(x)}{\pi(x)} - \log \frac{(pP_C)(y)}{\pi(y)}$  obeys three exact relations. The integral identity  $\langle e^{-\sigma} \rangle = 1$  holds for every kernel that preserves the reference; under detailed balance the pointwise relation  $p(x) P_C(x, y) = e^{\sigma(x, y)} (pP_C)(y) P_C(y, x)$  holds at every pair of states; and summing it over a level set gives the forward weight at entropy production  $s$  as  $e^s$  times the reversed weight. The mean of  $\sigma$  is exactly the certified descent  $D(p||\pi) - D(pP_C||\pi)$ , so the second law is the Jensen shadow of an identity rather than a primitive inequality. Detailed balance also gives finite Onsager reciprocity. Because  $e^{-\sigma}$  is a ratio of rationals, the fluctuation certificate closes exactly over the rationals, with no numerical logarithm.

The reversible symmetry and correlation representation below are finite relatives of standard Onsager and Green–Kubo theory [15, 16]. The OPH-specific statement is the resulting model-selection obstruction for the full-fibre publicization projector.

**Theorem 2.7** (Finite transport kernel and repair-projector obstruction). *Let  $K$  be a finite stochastic kernel reversible under a strictly positive weight  $\pi$ , let  $L = I - K$ , and let  $R$  be a linear solver of  $LRj = j$  on the  $\pi$ -centered currents. For any finite current family  $\{j_a\}$ ,*

$$\Gamma_{ab} = \langle j_a, Rj_b \rangle_\pi$$

*is symmetric and positive semidefinite. At every finite cutoff  $N$ ,*

$$\langle j, Rk \rangle_\pi = \sum_{n=0}^N \langle j, K^n k \rangle_\pi + \langle j, K^{N+1} Rk \rangle_\pi.$$

*Thus a Green–Kubo coefficient is a correlation sum plus an exact displayed remainder, rather than a silently assumed infinite series.*

*For the full-fibre conditional-resampling kernel  $P_C$ , idempotence gives  $P_C^n = P_C$  for every  $n \geq 1$ . Positive-lag correlations are therefore constant. A fibre-centered current with  $P_C j = 0$  has no positive-lag memory, while a nonzero one-step correlation makes consecutive partial sums differ by the same nonzero amount. One full-fibre repair projector cannot be a decaying, nonzero-memory transport dynamics.*

*On a finite closed oriented graph, typed positive cell volumes and heat capacities convert concentration to particle amount and temperature to energy. Fick and Fourier fluxes then define canonical one-step updates whose total changes equal the clock increment times the total source and hence vanish when the source sums vanish. Nonnegative conductance makes the flux–gradient pairing nonpositive; explicit negative-conductance controls reverse that sign.*

*Proof.* Detailed balance makes  $L$  self-adjoint, and the exact Dirichlet identity expresses  $\langle f, Lf \rangle_\pi$  as a nonnegative weighted sum of squared differences. Applying this to linear combinations of

the Poisson solutions proves symmetry and the full finite quadratic-form inequality. The cutoff formula telescopes the Poisson equation. Projector idempotence gives the positive-lag dichotomy. Graph summation by parts and cancellation of internal oriented edges give the constitutive and global-balance statements. Every step, including two-state and two-vertex witnesses and negative controls, is checked in Lean [13].  $\square$

The theorem does not select a physical transition generator, distance, clock, boundary condition, conductance, or readout. It proves no stability, hydrodynamic limit, or measured transport coefficient. In particular, the projector obstruction requires a source-derived nonidempotent local or random-scan effective evolution before OPH can make a long-memory transport claim.

The same package carries the entropy-balance shape used by the gravitational branch. For any state and faithful reference the exact identity  $S(p) - S(\tau) = \langle K \rangle_p - \langle K \rangle_\tau - D(p||\tau)$  holds, so the entanglement first law  $\delta S = \delta \langle K \rangle$  is this identity with the quadratic-order deficit dropped; through the central split  $K = 2\pi B + Z$  it reads  $\Delta S = 2\pi \Delta \langle B \rangle + \Delta \langle Z \rangle - D$ , and one repair step with a central charge measurable through the repaired visible datum obeys the cap Clausius inequality  $2\pi \Delta \langle B \rangle \leq \Delta S$  with no further hypothesis, since the kernel conserves every fibre-measurable charge. The finite first-law premise package of the Einstein branch is discharged on this model: on the tangent space of the probability simplex the entropy differential at the reference equals the modular pairing as a theorem, the modular pairing distributes through the central split, and the central pairing is the edge differential, so the branch's finite bulk-edge-central first law applies with no assumed premise beyond the split of the reference. Repair displacements are mass preserving with vanishing central pairing, so the first-order entropy change of one repair step is purely bulk-modular. The physical attachment of the split stays with the energy-clock receipt.

The Lean development carries the exact finite statements: relative entropy nonnegativity, the log-sum inequality, data processing, the conditional-resampling optimizer and its kernel package, the Gibbs Pythagorean identity and minimizer, the first-law split, the low-temperature bound, the fluctuation identities with reciprocity, the finite Green–Kubo matrix and exact cutoff remainder, the repair-projector memory obstruction, the typed Fick and Fourier updates, the exact-remainder cap first law, the discharged Einstein-branch first-law premises, and the Kraus completeness of partition pinching; the modules are `FiniteConditionalRepair`, `FirstLawIdentity`, `FluctuationTheorems`, `GreenKubo`, `GraphDiffusion`, `CapFirstLaw`, `EinsteinPremiseLink`, `StationaryRealization`, and `PartitionPinchingCP` in the Lean thermodynamics and event-algebra libraries, and the exact-rational certificate replays the kernel algebra and fluctuation identities and records the entropy checks [13]. The source-realization interface needs less than reversibility: every stochastic kernel that preserves a faithful reference contracts relative entropy to that reference. Detailed balance is not a premise. The exact lazy directed three-cycle is stationary and violates detailed balance, yet satisfies this second law; the Lean module `StationaryRealization` proves the separation. Detailed fluctuation relations and Onsager reciprocity therefore remain additional receipts rather than consequences of entropy descent.

The pinned source-counted collar table has also been exhausted at its committed resolution. All  $2^4 - 1 = 15$  nonempty field-subset projections of its four committed packet fields were audited on 1028 observers and 31744 counted transitions, without an additional observer-patch run. The repair-load quotient is an eight-state raw chain that is irreducible and aperiodic and has a full-support stationary law. On the committed floating-point table its detailed-balance defect is  $2.009924089 \times 10^{-4}$ ; from the uniform initial law the sampled relative entropy falls from 5.00947 to 1.84009 over sixteen steps. This is a nonreversible finite H-theorem probe, not a microscopic-reversibility receipt. It does not close the realization: the only observed record-family value makes protected-charge preservation vacuous, and the stationary law is not identified with the state optimizer's

source reference. On the fine twenty-state quotient the only closed communicating class is the singleton freezeout state, so the alternative recurrent-class restriction is a trivial equilibrium rather than a nontrivial relaxation model. This exhaustive no-go removes further quotient hunting on the same artifact from the critical path; a nontrivial realization requires changed source dynamics or a bounded run with a nonconstant protected record.

The separately preregistered state table does not repair this failure. Its state-side conditional-resampling action is idempotent, whereas the exact two-state recurrent transition action has the nonconstant eigenpair

$$Pv = \frac{665437}{726948}v, \quad v = (54356, -7155), \quad 0 < \frac{665437}{726948} < 1.$$

If an idempotent action  $H$  and this transition action  $P$  were related by  $HT = TP$ , applying idempotence twice gives  $\lambda(\lambda - 1)Tv = 0$ , hence  $Tv = 0$ . Thus every such intertwiner erases the earned mixing mode. The transition stationary mass 7155/61511 also lies strictly between the adjacent empirical masses 1905/16384 and 1906/16384; no deterministic coarse graining of the 16384-sample state reference produces it. A freely chosen stochastic coupling could evade the denominator test, but would be a new premise rather than source evidence. These exact obstructions close the common-object route for the current artifact, not for every possible source dynamics.

The conditional package retains five typed requirements: the global representation of the weighted local objective for both optimizer instantiations; the common source-derived reference shared by state and transition optimizers, which forbids manufacturing the transition matrix from a desired equilibrium output; the identification of the actual source-derived collar transition matrix with the conditional-resampling kernel through the existing matrix receipt; stochasticity, stationarity, and protected-charge preservation alone certify only the weaker stationary H-theorem branch and do not discharge that full Axiom-3 receipt, while detailed balance is additionally required wherever microscopic reversibility is claimed; the identification of one modular charge with physical energy and a calibrated clock, of which the finite central-interface split  $K_C = 2\pi B_C + Z_C$  is the attained cap-branch part; and refinement-uniform gap control for a continuum third law. The derivation carries a structural point worth stating plainly: every proof in the package is short and elementary once the framework supplies its two inputs, the repair fibre as the transition feasible set and the reference shared with the state optimizer. The four-law identities then require no additional thermodynamic axiom or deep theorem; the work lies in the decomposition, which the certified strict-descent counterexample shows is necessary. On any future source object discharging the named receipts, ordinary finite thermodynamic identities follow from the instantiated observer model rather than being postulated separately.

### 3 Physical Implementation and $A_5/E_8$ Significance

The implementation question is what must exist below the observer-level axioms so they have a physical carrier. OPH's answer is a fixed-cutoff federation of finite patch carriers. Each carrier exposes overlap ports, record registers, repair maps, and checkpoint interfaces. Three geometric objects must stay distinct. A local carrier boundary may have twelve-port icosahedral incidence. The federation screen is the finite federation and its overlap nerve. The support screen is the observer-facing  $S^2$  chart produced from the repaired quotient only on the spherical receipt branch. The implementation is the finite algebraic machinery that makes compare, write, repair, and re-read operations available.

The term “sphere folding” has a restricted technical meaning here. It is the observer-facing spherical presentation of the repaired quotient normal form:

$$\text{Fold}_{S,r}(s) = \chi_{S,r}(n_r(\pi_r(s))) .$$

The quotient map removes hidden carrier presentation, the normal-form map performs accepted repair, and the spherical chart displays caps, collars, cuts, edge sectors, and boundary records. Folding is therefore the geometric face of overlap repair on the screen chart. On the consensus surface this repair is defined first as the quotient operator  $\text{Rep}_\lambda : Q \rightarrow Q$ , with representative-level maps only serving as lifts, so hidden carrier labels do not become additional physical data.

Spherical geometry carries several pieces of the construction at once on the certified branch. An observer-accessible cut has a closed two-dimensional angular chart after the global support incidence, mesh, cross-ratio, and normalization receipts hold. Caps and collars on that chart supply the cut data used by modular flow, entropy variation, and overlap comparison. The orientation-preserving conformal group of the same  $S^2$  chart is the celestial-sphere realization of  $\text{SO}^+(3, 1)$ , so the sphere is the kinematic bridge between finite screen cuts and the emergent  $3 + 1$ -dimensional Lorentz branch once the controlled cap-pair theorem is satisfied. Finite cellulations of the chart provide the regulator surface on which ports, edge sectors, and local comparison data can be made explicit; they are not by themselves a Lorentz invariant continuum.

The echosahedral carrier is a concrete local realization of the observer-patch interface. Its multi-port boundary supplies discrete directions, face and edge incidence, exposed readout channels, record slots, and repair channels. Recurrent toroidal subchannels can expose winding and phase-sensitive observables. A physical phase-locking law would have to produce the accepted overlap repair relation, its confluence receipts, and its noise bounds. No theorem supplies that producer. Phase locking alone leaves BW modular ordering and operational clock calibration unproduced.

A federation of local icosahedral carriers does not automatically produce the global  $S^2$  support screen. That step needs a quotient-visible carrier-to-screen map whose federation nerve has the spherical incidence receipt and whose cap data survive refinement. Local  $A_5$  symmetry can therefore coexist with a nonspherical global nerve. The spacetime and Einstein paper’s topology countermodels mark this boundary.

The  $A_5$  and  $E_8$  labels mark different roles in that machinery.  $A_5$ -icosahedral symmetry is the local finite-screen language used when a patch carrier is made concrete: it organizes ports, faces, and overlap data with enough discrete symmetry to stabilize observer-facing cuts.  $E_8$ -type language is the exceptional closure language used by the high-symmetry branch. It names the root-lattice and affine exceptional structure in which the icosahedral data can sit. The term is representation-closure language for the branches that call for it.

The compact side of that closure language is classified at the bracket level. On the canonical oriented carrier, the compact real locus of the fourteen-parameter  $A_5$ -equivariant Jacobi variety is exactly three families, all inside the derivation-free slice: one closed two-parameter plane, generically  $\mathfrak{so}(3) \oplus \mathfrak{so}(3)$  plus an abelian complement, and two mirror three-parameter cells of  $\mathfrak{su}(3) \oplus \mathfrak{so}(3)$  type, each cut out by one strict sign condition. Every numerical step is exact over  $\mathbb{Q}(\sqrt{5})$ , fifteen certified sample points inhabit the strata, and an independent mutation-tested verifier replays the certificate. The classification is conditional on three named textbook lemmas, the compact-type Killing criterion, the compact semisimple dimension list, and  $A_5$  simplicity; it selects no source, preferred bracket, or holonomy [13]. A compactly closing equivariant bracket therefore lands on the  $\mathfrak{su}(3) \oplus \mathfrak{so}(3)$  shape up to the mirror choice and the closed plane.

This distinction matters for the synthesis paper. The physical claim lives at the observer-visible quotient: finite patches must expose the same records and shared observables after al-

lowed implementation-hiding changes. The microphysics paper gives the concrete reference architecture for that quotient surface, including the federated patch-carrier model, fixed-cutoff edge heat-kernel/Casimir law, central records, the declared algebra-state measurement interface, the declared two-wing Bell/CHSH surface, and the checkpoint/restoration package. See Ref. [5].

The A1–A2 finite theorem and the transportable-sector/Tannaka route are mathematically distinct. The first forces the abstract local Standard Model Lie type. The second is a conditional compact-group classification. Under the declared matrix-current and matter packet, the common  $\mathbb{Z}_6$  kernel and maximal faithful image are exact. The physical global quotient is not source-selected. Physical source binding of the first route and equality as one physical gauge-current object enter as premises. Separate band premises select rank three. Physical family interpretation and extra-sector exclusion require separate identifications. Tensoring the response band with the generation table gives a conditional rank-45 candidate. A distinct local-domain receipt checks a declared tensor-identity operator; it does not source-select the matter action or transport the twelve-port Spin packet. Physical matter-pole, continuum, seam-selection, persistence, and symmetry-descent receipts are required.

## Famous equations and structural outputs recovered

The table below maps headline formulas by logical role. Some rows are core theorem results, some are short corollaries of recovered actions, and some are standard limits once the parent OPH branch is established.

| Recovered result                                                                     | OPH role                        | Famous display                                                                                                                                      | Support boundary                                                                                                                                                                                  |
|--------------------------------------------------------------------------------------|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Three-dimensional observer-frame chart, Lorentz kinematics, and invariant light cone | core theorem                    | $q(\Omega) = (1, \Omega),$<br>$n_C = (\cot \alpha, \csc \alpha \mathbf{c}), n_{gC} = \Lambda_g n_C,$<br>$H^3 \simeq \text{SO}^+(3, 1)/\text{SO}(3)$ | scaling-limit geometric subnet; chart population and neutral bulk are separate premises                                                                                                           |
| Record-conditioned $H^3$ frame estimate                                              | conditional theorem             | $R_i(C, t, O) \rightarrow F(X_i) + e,$<br>$\alpha \bar{d} \leq  F(X) - F(Y) _W,$<br>$S_i(t) = B_H(\hat{X}_i, r_i)$                                  | frame-local calibrated cap responses, compact frame domain, bounded error, quantitative cap frame, and positive gap for unique finite output; no event position is inferred                       |
| Einstein equation                                                                    | conditional composition theorem | $G_{ab} + \Lambda g_{ab} = 8\pi G \langle T_{ab} \rangle$                                                                                           | one source-derived common-domain tower with uniform asymptotics, universal coupling, a vacuum reference, and independent scale readouts; the tower and its certification are consumed as premises |
| Newton-Poisson gravity                                                               | inherited weak-field limit      | $\nabla^2 \Phi = 4\pi G \rho, \ddot{\mathbf{x}} = -\nabla \Phi$                                                                                     | absolute Einstein-branch premises, weak field, and slow motion                                                                                                                                    |

| Recovered result                       | OPH role                                            | Famous display                                                                           | Support boundary                                                                                                                                                                                                                                                                                  |
|----------------------------------------|-----------------------------------------------------|------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Maxwell equations                      | electromagnetic corollary                           | $dF_Q = 0, \quad d*F_Q = g_Q^2 * J_Q$                                                    | ordinary $U(1)_Q$ branch plus explicit Maxwell action/current hypothesis                                                                                                                                                                                                                          |
| Yang–Mills equations                   | compact-gauge corollary                             | $DF = 0, \quad D*F = g^2 * J$                                                            | compact-gauge connection branch plus explicit Yang–Mills action/current hypothesis                                                                                                                                                                                                                |
| Born rule and Lüders update            | fixed-cutoff theorem                                | $\mathbb{P}(E) = \text{Tr}(\rho P_E),$<br>$\rho _E = P_E \rho P_E / \text{Tr}(\rho P_E)$ | declared finite algebra-state embedding of the central record algebra                                                                                                                                                                                                                             |
| Tsirelson bound                        | fixed-cutoff theorem                                | $ S_{\text{CHSH}}  \leq 2\sqrt{2}$                                                       | declared commuting wing algebras, binary settings and readouts, and joint source state; this is an upper bound and supplies no source-derived violation                                                                                                                                           |
| Charge quantization                    | core theorem                                        | $Q = T_3 + Y$ , color singlets have $Q \in \mathbb{Z}$                                   | quarks are confined fractional-charge fields                                                                                                                                                                                                                                                      |
| Classical carrier-mode poles           | conditional action theorem                          | $K_X^{\text{phys}} \propto (\omega^2 - c_*^2  \mathbf{k} ^2) \Pi_X$                      | Maxwell, perturbative pure-Yang–Mills, or pure-Einstein action/phase receipt; the quantum particle claim is a separate premise                                                                                                                                                                    |
| Primitive-port propagation fingerprint | frozen prospective branch prediction                | $\omega^2 = k^2 - (a^2/20)k^4 + (a^4/840)k^6 + (2a^4/7875)k^6 I_6 + O(a^6 k^8)$          | the complete equal-weight twelve-port orbit is the physical directional support; the tested sector realizes the scalar symbol, with equal action on both transverse polarizations for light; intrinsic coefficients must be isolated from source, medium, gravitational, and instrumental effects |
| Source-seam propagation fingerprint    | exact internal action and frozen conditional branch | $\widehat{\Lambda} = q^2 - q^4/20 + (1/840 - I_6/12600)q^6 + O(q^8)$                     | the source fixes the $D_6$ seam carrier and response metric; named naturality and unique-minimizer premises select the homogeneous action; physical position, field, clock, frequency, frame, readout, and nuisance attachments are open                                                          |

| Recovered result                      | OPH role                   | Famous display                                                     | Support boundary                                                                                                                                                                                             |
|---------------------------------------|----------------------------|--------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Coherent-matter scalar susceptibility | conditional scalar theorem | $\chi_\nu^{\text{can}} = 1 - P_\chi/24 = 0.9320429912748350 \dots$ | nondegenerate OPH-coherent material source, Scalar Edge-Center Exhaustion, and protected-reserve collar closure; force additionally requires repair charge, an external gradient, and field-momentum closure |

On the separate global capacity branch, the Bekenstein–Hawking / de Sitter area law keeps the display  $N_{\text{scr}} = A/(4\ell_P^2) = 3\pi/(G\Lambda)$  with screen-capacity closure as its boundary.

## Particle result scope

The particle paper treats the closure matrix explicitly. The non-hadron bundle has the following support levels:

The source chain emits the electroweak chart coordinates (80.330, 91.119) GeV. A separate strict one-loop consumer and interval stack act on an external Standard Model fixture. They exclude scalar zeros on declared principal-sheet boxes and isolate, for each of  $W$  and  $Z$ , one simple scalar zero with derivative and scalar-residue balls in its declared lower-half pole box on a channel-specific algebraic chart. They identify neither chart with the physical resonance sheet and prove no unique continuation, sign bridge, full-matrix Laurent residue, current amplitude, or independent numerical replay. The external fixture is not composed with the OPH chart. A physical  $W/Z$  statement requires an OPH-native source-to-pole packet, a strict source root, an independently physical  $E_\star$ , finite quotient transport, a renormalization-group matching and scheme prescription fixed independently of the comparison, branch rigidity, and propagated uncertainty.

**Exact flavor algebra and its boundary.** The twenty-face orbit has a regular  $C_3$  corner fiber. Every equivariant Hermitian response on that fiber is a circulant  $C = aI + bR + \bar{b}R^2$ . When all three eigenvalues are nonnegative and read as square-root masses, the root-of-unity identities give

$$Q = \frac{1}{3} + \frac{2}{3} \left( \frac{|b|}{a} \right)^2, \quad Q = \frac{2}{3} \iff \frac{|b|}{a} = \frac{1}{\sqrt{2}}.$$

Equal rank-two blocks in the finite tracial-GNS packet give the balanced modulus exactly under the packet premises. The phase, and therefore the two mass ratios it jointly controls, is not fixed. The face fiber also lacks a constructed physical chiral-family attachment. The target-informed response coordinate is a diagnostic.

The five-dimensional traceless-symmetric family space sharpens the dynamical gap. Three-fold and fivefold residual invariance forces a double eigenvalue. The twofold fixed locus has two parameters after overall scaling and admits simple spectrum, leaving exactly enough freedom to carry two mass ratios. Symmetry alone therefore supplies no numerical ratio prediction; a specific screen-derived invariant potential must select the orbit. Ref. [6] gives the proofs and scope.



**Conditional comparison coordinates.** The source-defined zero-selector law outputs the running/chart coordinates (80.330, 91.119) GeV. The separate strict one-loop fixture has principal-sheet scalar zero-exclusion receipts and one simple scalar zero for each of  $W$  and  $Z$  in channel-specific lower-half pole boxes and algebraic charts. No certified continuation identity or composition joins that fixture to the OPH chart, so it supplies no physical  $W/Z$  pole or mass comparison. The Higgs/top pair is read on the double-criticality branch, a zero-continuous-parameter family whose comparison-exposed boundary-scale candidate gives  $(m_H, m_t) = (125.77, 172.63)$  GeV at two loops; its declared calibration surface is a data-anchored fit that validates the formula stack and never predicts. The quark reciprocal-ray candidate fails its common-scale audit with a 21.556% held-out minimax error. The register-Clebsch candidate supplies six assignments and the distinct light-family coefficient-ratio menu  $\{1/9, 1/3, 3, 9\}$  under the declared alphabet and the two invariant channel assignments. The adopted ordering is target-informed and uniquely least discrepant. Both FLAG 2024 rows reject every assignment under the retrospective conservative experimental-only decision rule. The unavailable covariance and absent OPH theory uncertainty preclude a covariance-aware significance, and the rule has no preregistered theory-wide falsification status. This closes only the declared common-transport assignment family. Different coefficient relations, alphabets, charged-family attachments, or generation-dependent threshold transport lie outside this class. The retained results are the conditional channel-pairing theorem, the target-free unordered multiset under the declared rules, and the exact positive-chamber Koide identity. The pairing theorem supplies no physical coefficient equality or source-derived generation order. The corresponding  $\sqrt{m_d/m_s} = 0.2086$  display is an algebraic Gatto–Sartori–Tonin estimate from the same failed ratio, without the up/down matrices and relative eigenbasis needed for a mixing prediction. All of these comparison coordinates, including the selected-carrier, value-law, adapter, two-loop, pole variants, and exact Particle Data Group convention maps, are excluded from the physical particle outputs. Ref. [6] gives the detailed coordinate tables and verification.

## Observable-identification boundary

The arguments use one no-relabeling rule throughout. A finite diagnostic or calibration field does not become a physical observable merely because it is renamed. Capacity bookkeeping needs an independent energy readout before it can count as mass; an archive needs a physical source, propagation, and detector channel before it can count as radiation; a finite repair spectrum needs a continuum operator/readout bridge before it can count as a physical spectrum; and a finite reconstruction threshold needs physical entropy and clock data before it can count as a Page-time claim. The spacetime and Einstein paper records this as the source-separation rule behind the support-level table.

**Sector-dimension shock boundary.** The declared repair generator

$$L^{\text{rep}} = \sum_v (I - P_v)$$

uses single-site heat-bath conditional expectations on  $K_r = L^2(X_r, \pi_r)$  at fixed sector structure. Since the  $P_v$  are orthogonal projections,

$$\langle f, L^{\text{rep}} f \rangle = \sum_v \|(I - P_v)f\|^2,$$

so  $\ker L^{\text{rep}} = \bigcap_v \text{Ran } P_v$ . Identifying this intersection with the constants requires a separate irreducibility result. A horizon shock changes the sector dimensions  $d_\alpha$  in  $L_C = \sum_\alpha (\log d_\alpha) P_\alpha$ , thereby



changing  $X_r$ . It is outside the domain of this fixed-sector repair generator. The repair gap therefore neither constrains nor produces a sector-dimension shock mode.

A separate finite-capacity calculation gives a one-sided sign statement. Uniform transfer of a fraction  $f$  from a fixed horizon–observer budget changes the extremal sector entropy and logarithmic area observable by  $\log(1 - f) < 0$  on every admissible integer depletion, and the positive-real interpolation is strictly decreasing. Its analytic Hessian has negative homogeneous curvature and positive curvature on the fixed-horizon-capacity tangent space, so the relevant maximum is the transfer boundary rather than an interior stationary point. Identifying this capacity transfer with horizon-area transfer gives a conditional mechanism for the de Sitter time-advance sign reported by Chen, Stanford, Tang, and Yang [33, 11]. Identifying the finite capacity-ledger transfer with observer mass is a separate physical dictionary. The normalized icosahedral spectrum additionally requires exact gauge transport of the rotation triplet and identification of the kinetic term with the scaled nearest-neighbour port or edge-sector Laplacian. Its coefficient and physical attachment are separate premises. This mechanism supplies no static-patch trace and leaves the cited trace obstruction in force. That obstruction concerns the positive cyclic trace proposal tested in the cited work. Nontrace horizon-screen descriptions, separately constructed dimension-changing generators, and kinetic operators beyond the scaled nearest-neighbour graph lie outside the tested classes. None of these scope boundaries disproves OPH; the pure-de-Sitter normalization, finite entropy and transfer identities, and regular line-graph theorem survive.

The  $\chi_\nu$  scalar-channel identity is theorem-grade for nondegenerate OPH-coherent material sources once Scalar Edge-Center Exhaustion is imported. Device-scale force, cosmological abundance, and finite covariant dark stress require their own receipt packages.

**Frozen primitive-port propagation prediction.** On the real reciprocal finite-range cosine branch whose complete hop support is the primitive twelve-port orbit, with no independent kinetic term through the displayed order, proper carrier covariance forces equal weights and continuum normalization fixes the displayed propagation symbol. Writing its corrections as  $C_4 k^4 + B_0 k^6 + B_6 k^6 I_6$ , the exact scale-free relations are

$$\frac{B_6}{C_4^2} = \frac{32}{315}, \quad \frac{B_0}{C_4^2} = \frac{10}{21}, \quad \frac{B_6}{B_0} = \frac{16}{75}, \quad C_4 < 0.$$

The intrinsic anisotropic ranks one through five vanish. Rank six has one icosahedral harmonic shape up to an orientation in  $\text{SO}(3)/A_5$ . Here “spin six” names spherical-harmonic rank, unrelated to particle spin. Once a negative  $C_4$  is resolved on this branch, the scale and both sixth-order amplitudes are fixed; the residual freedom is one three-parameter orientation class in  $\text{SO}(3)/A_5$ . For the complete equal-weight cosine kernel and  $0 < |ak| \leq 1$ , an exact three-chart interval cover and local singular-value bounds prove that its stationary set consists of precisely the twelve vertex maxima, twenty face-center minima, and thirty edge-center saddles. This finite-range theorem uses the declared kernel. Source selection and physical readout remain in the branch premises [13].

The certified repair operator acts on thirty internal carrier seams, not on translated field sites. The finite source supplies no map from that operator to a spatial hop stencil or a physical readout. Exact equal-weight stencils on the vertex, face, and edge direction orbits have different rank-six rays. Carrier transitivity fixes equal weights within a selected orbit and does not select the vertex orbit. The displayed prediction therefore remains a named physical-branch statement.

The coefficient manifold and its decision rule are frozen before a qualifying comparison [14]. The dated cosmic-microwave-background template search and every data product examined in it are an excluded exposure class; the linked coefficient manifold has no eligible physical comparison.

If  $C_4$  is negative at five or more standard deviations with enough sensitivity, exclusion of the linked  $B_0/B_6$  terms or the rotated  $I_6$  vector at five or more standard deviations, after the fixed  $\text{SO}(3)/A_5$  profile and with calibrated joint coverage, rejects the primitive-port physical branch. An isolated positive  $C_4$  or nonzero intrinsic anisotropy at ranks one through five also rejects the branch at the same threshold and under the same calibrated coverage. A calibrated joint likelihood that excludes the complete branch manifold at five or more standard deviations also rejects it. Null, underpowered, incomplete-covariance, unresolved-frame, polarization-split, and non-isolated results are inconclusive. Support requires exclusion of the zero-coefficient baseline at five standard deviations or more, agreement with the complete linked branch within two, rejection of the named systematic alternatives, and an independent eligible replication. Minimal locally Lorentz-invariant Standard Model physics with General Relativity gives the zero-coefficient baseline. Nonminimal effective operators can imitate the pattern. A physical test therefore requires the sector bridge, equal action on both transverse polarizations for a photon test, coherent carrier-frame transport, and a nuisance model that isolates the intrinsic coefficient vector. Derivation of these bridges from the repair law is a separate premise. Failure reaches OPH as a whole only if such a derivation proves the branch forced and exclusive.

**Frozen source-seam propagation branch.** The signed source-incidence map sends the complete thirty-seam system onto  $D_6$ , and its pullback response metric completes to the same three-dimensional carrier as the source loads. Under the named naturality, objective, and unique-minimizer premises, the sixty directed seam weights are  $1/60$ . The resulting internal operator is the exact homogeneous Dirichlet generator. With  $q = ak$  and unit direction  $n$ , its normalized spatial character is

$$\hat{\Lambda}(q, n) = \frac{1}{5} \sum_{j=1}^{30} [1 - \cos(q w_j \cdot n)].$$

Its coefficient ray is

$$C_4 = -\frac{a^2}{20}, \quad B_0 = \frac{a^4}{840}, \quad B_6 = -\frac{a^4}{12600},$$

so  $B_0/C_4^2 = 10/21$ ,  $B_6/C_4^2 = -2/63$ , and  $B_6/B_0 = -1/15$ . Intrinsic anisotropies of ranks one through five vanish, while rank six has one icosahedral harmonic up to orientation.

Define

$$P_6(q, n) = q^2 - \frac{q^4}{20} + \left( \frac{1}{840} - \frac{I_6(n)}{12600} \right) q^6.$$

The exact eighth moment and alternating cosine bounds give, uniformly for  $-5/9 \leq I_6(n) \leq 1$  and  $0 \leq q \leq 1$ ,

$$|\hat{\Lambda} - P_6| \leq \frac{7}{388800} q^8, \quad \frac{19}{20} q^2 \leq \hat{\Lambda} \leq q^2.$$

This is a target-free bound for the finite spatial symbol. A separately declared basis-free oscillator completion has a two-dimensional momentum-orthogonal fiber at every nonzero momentum. The scalar symbol acts equally on both directions, and the declared generator gives

$$A_T'' + \Lambda_a A_T = 0, \quad \omega_a^2 = \Lambda_a, \quad \omega_a(0) = 0.$$

The corresponding quadratic energy has zero algebraic first variation on that generator. These are conditional oscillator statements. They do not construct a flow, gauge redundancy, Gauss law, Maxwell dynamics, a physical clock, a photon Hilbert space, or a massless physical photon.

An exact synthetic calibration enumerates all  $2^{12} = 4096$  sign-error vectors in a fixed twelve-row design at noise scale  $1/200$ . It rejects a zero leading coefficient in every replica, and its nominal 95 percent interval covers the injected value in 3904 replicas. The linked higher-order pair is detected in 209 replicas and is unresolved in this stress law. No experimental response or sensitivity follows from this synthetic model.

The nonnegative auxiliary root of the complete positive cosine symbol obeys  $|\Omega(k) - \Omega(p)| \leq |k - p|$  at all momenta in the selected Euclidean carrier chart. The exact theorem uses the full symbol and no Taylor truncation. It supplies no physical position, frequency, clock, field, signal-front, frame, scale, or detector map. If the same seam symbol is identified with physical photon frequency squared, then for  $\mathbf{k} = kn$ ,  $k \geq 0$ , and  $|n| = 1$ , its complete cosine form obeys  $0 \leq \Omega_\gamma(k, n)^2 \leq k^2$  in the Euclidean carrier metric. Ordinary additive energy-momentum conservation and positive-mass electrons and positrons with Lorentz-invariant positive-energy dispersion then exclude photon decay into an electron-positron pair and place the seam-current incoming-energy budget inside the Lorentz-invariant photon budget at fixed incoming momenta. With the soft background photon Lorentz invariant at leading order, use  $E_i^2 = p_i^2 + m_i^2 + \delta_{i,2}E_i^4$ , with  $i \in \{\gamma, +, -\}$ ,  $m_\gamma = 0$ , and  $m_+ = m_- = m_e$ . Independent leading hard-photon, positron, and electron coefficients give a head-on, collinear threshold with fixed positron energy share  $0 < x < 1$  that depends on

$$\delta_{\gamma,2} - x^3\delta_{+,2} - (1-x)^3\delta_{-,2}.$$

At equal sharing this is  $\delta_{\gamma,2} - (\delta_{+,2} + \delta_{-,2})/8$ , a rank-one readout with a two-dimensional coefficient fiber. On the Lorentz-invariant-lepton branch, equal sharing uniquely maximizes the leading head-on, collinear threshold residual. General independent-lepton and full anisotropic optimization are open. These conditional kinematic statements supply no charged-lepton action, pair-production vertex, cross section, opacity, source model, shower response, or detector response.

The source-seam coefficient ray and decision rule are frozen before an eligible physical comparison [13]. Physical position, sector, frequency, clock, frame and boost law, wave-packet propagation, cosmological transport, detector readout, and nuisance isolation are premises. A branch-falsifying null additionally requires a positive lower bound on the physical scale for the same action and preregistered power to exclude the complete admitted manifold. A failed qualifying comparison rejects this physical propagation branch. It reaches OPH as a whole only if a separate theorem makes the branch forced and exclusive.

**Evidence boundary.** The Ward-projected hadronic transport gives  $S_{\text{hadronic}} \in [0.5578, 1.0543]$  for the two electromagnetic endpoint residuals. This interval contains the zero-electroweak diagnostic 0.8954 and spans about  $1.16 \times 10^8$  declared tolerance widths. Its evaluation pixel differs from the comparison pixel, so it is a comparison diagnostic rather than a physical prediction.

The fixed-cutoff direct correctable-public-record map has exact record restrictions, public reachability, publicness, global checkpoint coupling, correctable-code, carrier, extension, refinement, and sewing receipts. A bounded all-rung counterfamily has incompatible zero sets under shared base, positivity, carrier, and executable finite controls. The finite-rung complete-packet lift retains those incompatible zero sets on the six audited rungs. It does not prove that the executable candidates inhabit the complete source class at every rung or bind them to the all-rung Lean completion. A positive direct map requires completion of those bridges or an additional named source law, together with a universe-level carrier attachment. The de Sitter and electroweak interpretations also require horizon-record and common screen/electroweak load identifications. The interval at  $3.5321315434 \times 10^{122}$  belongs to the conditional electroweak bridge. It does not construct the direct map  $F$ , and it differs from the Planck base- $\Lambda$ CDM comparison coordinate by about 6.6

percent. The finite-presence and exponential reserve candidates give  $3.2920978773 \dots \times 10^{122}$  and  $3.3000722254 \dots \times 10^{122}$ . They are separately typed and retrospective. Exact neutral and multiplicative completions of the same local survival datum obey the declared positive composition and regrouping laws but have different global effects. The finite source selects no action and no blocked-event semantics, so the named-law and horizon readings are not evaluable without a stronger source-derived global action.

The fixed 96-entry one-loop electroweak menu excludes its own entries. It says nothing about higher-order, matching, threshold, tadpole, scale, or field-content prescriptions outside that menu. Twelve fixed flavor candidates are likewise excluded without exhausting the broader selector family. These comparisons use known targets and therefore do not count as target-blind predictions.

## Companion Papers

The detailed results used throughout this synthesis are:

1. *Observation-Determined Normal Forms: Stability, Obstructions, and Refinement in Constraint and Rewrite Systems* [1], which gives the presentation-invariant same-source/cross-source quantifier split, endpoint and refinement bounds, collar-repair obstructions, and the finite conditional-resampling receipt used by the repair papers.
2. *Recovering Observer Spacetime and Einstein Dynamics from Overlap Consistency* [2], which gives the exact port-record metric completion together with the support-incidence, modular, event-manifold, stress, generalized-entropy, and Einstein derivation chain. Every finite, continuum, scaling, and physical premise is stated at the arrow where it enters.
3. *Deriving Standard Model Gauge Structure from Observer Overlap Consistency* [3], which gives the categorical and finite-carrier routes. It proves the complete-response Lie-type theorem and gives the conditional matrix-current, matter, and tensor-kernel theorems. Identification of the two routes requires a physical current intertwiner.
4. *Reality as a Consensus Protocol: The Fixed-Point Computation That Implements Physics* [4], which carries the finite patch-net fixed-point, defect, quotient, and operator-record consensus construction. Its semantic-complete transaction theorem derives the local diamond from coherent aggregate gluing and full acceptance dependencies; repair completeness is a separate premise. Quantum error-correcting code/min-cut, spectral, Byzantine fault-tolerance, and hardware-speedup statements require their respective certificates.
5. *Federated Echosphedral Screen Microphysics: Patch Hardware, Records, and Observer Synchronization in OPH* [5], which carries the regulated federated patch-carrier architecture, the fixed-cutoff edge heat-kernel / Casimir theorem, exact finite modular and rate statements, the  $A_5 \times \mathbb{Z}_2$  register classification, and the measurement and observer checkpoint/restoration packages.
6. *Deriving the Particle Zoo from Observer Consistency* [6], which carries the particle derivations, masses, couplings, and sector calculations.
7. *Theoretical Bounds on  $\chi_\nu$  in Observer-Patch Holography and Observer-Patch Holography and Dark Matter* [7, 8], which carry the coherent-matter channel theorem and the conditional repair-charge condensate action with its current, stress, normal-phase, deep-galaxy, and device-force consequences.

8. *Explaining the Yang–Mills Mass Gap with Observer-Patch Repair Dynamics* [9], which isolates the controlled compact-gauge repair mechanism and states the continuum certificate needed before the finite repair gap becomes a Clay-facing four-dimensional Yang–Mills gap.
9. *Observer-Patch Holography as a String-Vacuum Selector* [10], which treats string theory as an effective edge language and applies the OPH acceptance criteria to critical-string candidates; its rank certificate classifies the Bouchard-Donagi row as a structural benchmark rather than a selected witness.

The complete spacetime and Einstein chain is proved in Ref. [2]; the two gauge-reconstruction routes and their identification boundary are proved in Ref. [3]. The synthesis therefore concentrates on their shared observer interpretation, quantitative closures, and cross-branch claim boundary.

## Physical scope of the icosahedral Standard Model and $W/Z$ results

The finite icosahedral package contains an axiom-forced Lie-type theorem and an exact conditional recognition result. Complete reversible response and endogenous overlap transport force the local Standard Model gauge Lie algebra. Incidence determines the antipode  $J$ . Under the explicit contract that an admissible response is a signed central involutive graph automorphism implementing inverse-port readback, the responses are exactly  $\pm J$ ; their common sign is conventional. Conditional also on the matrix current and rank-15 matter contract with its unique charge-conjugate projector pair, the anomaly and tensor-descent certificates fix the  $3 + 2$  block structure, hypercharge lattice, common  $\mathbb{Z}_6$  kernel, and maximal faithful matter image. The kernel is computed on every declared tensor and is insensitive to the projector representative. Source selection of the matrix current, matter action, and physical global quotient is work in progress. This finite calculation supplies no laboratory identification of its current or flux sectors and no physical seam action. Equality with the independently reconstructed Tannaka current, attachment of the band to three physical chiral families, exclusion of extra light sectors, four-dimensional topological attachment, scalar attachment and dynamics, and construction of a chiral quantum field theory remain work in progress.

The conditional field-theory implications are explicit. A finite local action gives an exact finite gauge-invariance and locality theorem, and the familiar electroweak tree kernel is conditional on a separate canonical continuum and action-normalization bridge. An exact finite measure criterion and an exact finite Hamiltonian criterion are two parallel branches over that action. A separate formal perturbative branch carries the strict finite-order  $W/Z$  pole theorem. A nonperturbative continuum completion gives an observable-sector reconstruction implication and a distinct continued-sheet resonance-stability implication. The measure and perturbative branches are parallel descendants of the finite local action. Neither implies the other, and a perturbative pole does not imply the continuum completion.

These quantum field theory (QFT) theorems state what follows from typed action and quantization packets. They do not show that the target-free source emits those packets. The source-selected action and normalization, complete measure, target-clean perturbative matching, physical current amplitudes, source law and covariance, uncertainty enclosure, operational clock, and continuum tower with its continued-sheet packet are work in progress. The declared external Standard Model action and effective-field-theory interval provide bounded validation inputs for the perturbative protocol. Interval receipts exclude scalar zeros in the declared principal-sheet boxes and isolate, for each of  $W$  and  $Z$ , one simple scalar zero with derivative and scalar-residue balls in its declared lower-half pole box on a channel-specific algebraic chart. They identify neither chart with the physical resonance sheet and prove no unique continuation or self-energy sign bridge. Proof-bearing

physical validation also requires two genuinely independent raw loop engines, counterterm and Becchi–Rouet–Stora–Tyutin generation from the complete action, Nielsen identities, an artifact-resolving third verifier, full-matrix contours and Laurent data, dressed-current amplitudes, and complete independent replay. The external packet is a validation input rather than an OPH source producer. No source-native dimensionless or physical-unit  $W/Z$  pole follows from it.

## 4 Consensus, Defects, and Implementation Hiding

The fixed-cutoff consensus package admits an equivalent finite patch-net formulation. Local patch descriptions agree on overlaps, local recovery-derived repair rules remove mismatches, and the physical output is the schedule-independent quotient normal form together with the induced terminal expectation functionals on the declared physical observable algebras. This section records the fixed-point, defect, and gauge-quotient statements in that language.

The consensus branch also carries a simple finite-candidate law-selection model: once one equips candidate repair rules with a fitness functional, replicator dynamics gives a clean toy picture in which more successful reconciliation laws dominate a competing pool. This is a finite-candidate monotonicity result. It supplies no gravity or gauge theorem, universality result, or literal cosmological dynamics.

On its declared fixed-cutoff branch, *Reality as a Consensus Protocol: The Fixed-Point Computation That Implements Physics* [4] supplies the following results:

1. **Constraint-code firewall.** A bare finite overlap net is a finite constraint code: its codewords are exactly the globally consistent states  $C = \Phi^{-1}(0)$ . It is not automatically a QECC/topological code, and its graph min-cut does not determine code distance. Distance/min-cut, Knill–Lafamme resilience, spectral convergence, BFT liveness, and hardware speedup enter only through separate certificates.
2. **Asynchronous confluence.** For the declared accepted repair law, the local-fit contract makes  $\Phi$  a Lyapunov functional, hence gives termination on the finite patch net. The fixed-cutoff union-collar gluing package supplies the local diamond on the physical quotient, and only that confluence condition together with repair completeness yields a unique schedule-independent normal form from a fixed initial quotient state. Same-boundary uniqueness requires the additional unique consistent extension condition in the preserved boundary/sector fiber.
3. **Cycle obstruction and higher-gauge defects.** On the abelian branch, global consistency holds exactly when cycle holonomy vanishes; on the genuinely noncentral branch, the crossed-module gluing orbit

$$q_\Sigma \in \check{H}^2(N_\Sigma, H_\Sigma \rightarrow G_\Sigma)$$

labels the full fixed-cutoff gluing orbit. It can admit multiple strict representatives related by  $H_\Sigma$ -valued edge changes and need not determine one ordinary  $G_\Sigma$ -valued 1-cocycle class. Associator strictifiability is the separate condition that this orbit admit a representative  $(g^{\text{str}}, 1)$ ; strict endpoint-only transport additionally requires at least one allowed strict representative with trivial represented loop holonomy.

4. **Gauge quotient and observable-level confluence.** The repair law descends to the overlap-invariant quotient, and the induced terminal state on the declared physical observable algebras is unique there even when microscopic representatives differ by gauge or sector relabelings inside one quotient-local glued state.



5. **Record algebra and stability.** On the declared fixed-cutoff observer-accessible operator surface, a separately declared algebra-state representation makes the central record projectors carry the Born/Lüders rules, while approximate record projectors inherit explicit  $(\varepsilon, \delta_{\text{rec}})$  stability bounds on that same event surface. This is a theorem about the stated algebraic record interface, not a requirement that OPH first rebuild every mathematical ingredient from operational records alone.

The consensus paper is equally explicit about the imported repair-law data and what sits outside the fixed-cutoff theorems. The declared repair step includes the touched-overlap local-fit contract and the union-collar gluing package on the physical quotient; the branch conditions above that step are repair completeness and, on the Petz branch, the stated support/CPTP clause. On that same fixed-cutoff surface, exact normal-form computation is finite-state and decidable, automatic approximate stability is only collar-local through the splice and record estimates, and long-run noisy approximate consensus is theorem-grade only after a fair-block contraction certificate is supplied for the chosen exported patch-net family. That certificate boundary is not a dependency for the support-visible BW theorem, local Einstein branch, receipt-conditional compact-gauge reconstruction, or realized Standard Model branch.

## 5 Particle-Spectrum Branch

The particle branch follows the same logical order as *Deriving the Particle Zoo from Observer Consistency* [6]. The complete twelve-port response in A1 and endogenous holonomy in A2 first force the abstract local algebra  $\mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(3)$ , without an ambient continuous gauge group. The explicit inverse-port contract separately gives the signed response. Applying the declared matrix-current and rank-15 matter contracts, followed by anomaly balance and tensor descent, gives the exact conditional Standard Model charge lattice and maximal faithful matter image. The one-Higgs clauses leave the window  $3 \leq N_g \leq 5$ . Under the named complete-band and operational-cost premises, an exact screen theorem selects the rank-three response band, and the declared unitary response places its residue at the lowest positive generator frequency. Identifying that finite band with three physical matter families and excluding extra light sectors remain separate physical attachments. The independent transportable-sector and Tannaka reconstruction remains a conditional classification route.

**Structural carrier and physical premises.** Under the declared response and matter contracts, the structural carrier roles and maximal faithful matter image follow from those contracts alone:

$$\frac{\text{SU}(3) \times \text{SU}(2) \times \text{U}(1)}{\mathbb{Z}_6}, \quad N_c = 3, \quad 3 \leq N_g \leq 5.$$

On a declared trace-balanced block carrier  $V = C \oplus W$ , with  $\dim C = 3$ ,  $\dim W = 2$ , and block hypercharges  $(-1/3, 1/2)$ , the selected exterior package

$$\Lambda^2 V \oplus \Lambda^4 V = Q \oplus u^c \oplus e^c \oplus d^c \oplus L$$

is an exact one-generation Standard Model representation witness. It has the three one-Higgs invariant lines, cancels the gauge and mixed anomalies, and contains four weak doublets per generation. Hence every additive isomorphism-invariant load normalized by  $L_P(\mathbb{C}) = P$  gives  $L_P(\text{Hom}_{\text{SU}(2)}(W, M_1)) = 4P$ . A positive unital map between one-dimensional order-unit load lines is unique once the physical order units are identified. Complete reversible response and endogenous

overlap transport force the abstract local Standard Model gauge Lie algebra. Under the conditional matrix-current and rank-15 matter contracts, determinant balance, the non-vacuum exterior package, and the common central kernel are exact finite implications. The scalar scan fixes compatible charges and Yukawa channels, not scalar multiplicity. The finite source does not select the matrix current, matter action, or physical  $\mathbb{Z}_6$  quotient. Laboratory current and flux identification, removal of the omitted  $\Lambda^0 V$  singlet and other light sectors, attachment of the  $A_5$  face module to physical families, scalar attachment and dynamics, four-dimensional instanton normalization, and continuum quantum field theory are work in progress.

**Theorem 5.1** (A1–A2 Lie-type theorem with conditional physical landing). *Assume the declared quotient-visible echosahedral carrier lineage together with its integer atom-counting grammar and normalized central-readback Hilbert–Schmidt cost. Then the counting theorem supplies the twelve unit ports and exact gap, while the oriented-incidence theorem supplies inverse pairing, proper  $A_5$  action, the rank-three frame, and refinement/relabeling naturality. On the named twelve-register realization, atomic signed record events generate integer loads. Conservative whole-unit repairs terminate by strict descent of  $V(N) = \sum_i N_i^2$ , and their minimum move count is natural under rotations and the declared refinements. A half-unit display rescales the same event graph and threshold. The quadratic readback cost, dynamic move cost, load-square Lyapunov function, seam quadratic, and A3 Hessian are distinct objects. Their selection on an arbitrary A1–A3 carrier is not implied by this named realization. The complete-response clause in A1 and endogenous proper-carrier transport in A2 force the abstract local Standard Model gauge Lie algebra  $\mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(3)$ . Assume further the explicit contract that inverse-port readback is a signed central involutive graph automorphism, together with the declared matrix current, rank-15 matter contract, and tensor-descent receipt. Incidence then determines  $J$ , the admissible inverse-port responses are  $\pm J$ , and the conditional finite certificates verify the matrix realization, refinement maps, anomaly-forced determinant balance, Spin lift, rank-15 exterior matter package, and axis/center deck descent. These implications produce the trace-balanced  $3 + 2$  block carrier with hypercharges  $(-1/3, 1/2)$  and fix the common  $\mathbb{Z}_6$  tensor kernel on the declared matter table. The scalar scan identifies compatible charges and Yukawa channels, not  $H = W$  or scalar multiplicity. For a physical realization, assume that a source packet reconstructs the matrix current and matter action; that a complete source character lattice and same-source carrier-loop-to-kernel map select the global quotient; that a source-bound refinement-natural commuting square identifies the  $A_5$  current group and action with the independently reconstructed Tannaka group and its realized matter action; that laboratory current and flux identification passes; that the matter contract is source-bound; that a scalar-multiplicity/one-Higgs receipt lands; that a source-complete no-extra-light-sector receipt lands; and that the  $A_5$  multiplicity is physically attached to three families with the selecting symmetry hidden, broken, or forgotten on the Yukawa surface. On a receipt-certified compact-gauge continuum/QFT landing, the resulting low-energy chiral package has the Standard Model physical global form*

$$\frac{\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)}{\mathbb{Z}_6}$$

*with the exact Standard Model hypercharge lattice, one Higgs doublet, three colors, three attached generations, perturbative anomaly cancellation, even Witten parity, and no simple-GUT  $X/Y$  gauge channel.*

*Proof.* The declared counting and incidence theorems produce the finite port and frame data on the declared carrier lineage. The A1–A2 response clauses and compact classification force the abstract local Standard Model gauge Lie algebra. The declared matrix-current and matter contracts make the exact representation, refinement, rank-15 matter, determinant, Spin, and descent calculations



applicable. Exterior branching gives the five chiral multiplets and the scalar scan gives the compatible cubic invariant lines without a scalar-multiplicity claim. The anomaly and center calculations fix the hypercharge lattice, color triplet, and common  $\mathbb{Z}_6$  tensor kernel from those calculations alone. The source character and loop-to-kernel premises select the quotient. The laboratory flux premise attaches it to the physical spectrum. The source-bound commuting square identifies that group with the group acting through the Tannaka current. The stated completion, sector, and family receipts remove the otherwise available singlet, extra-sector, and family-attachment ambiguities. The continuum quantum-field premise completes the physical implication.  $\square$

This theorem is a logical implication under its premise list. Each named screen-to-current, cross-route identity, carrier, sector, family, and continuum condition is independently removable. Removing one limits the conclusion at the corresponding boundary.

This does not determine quantum-particle masses. The specialist derivation papers separately prove that explicit Maxwell, perturbative pure-Yang–Mills, and pure-Einstein quadratic actions have transverse or TT classical massless modes on their stated backgrounds and phases. Photon, gluon, or graviton particle language additionally requires a positive-energy physical quantization, positive-residue two-point pole, and the appropriate asymptotic/deconfinement receipt. Separate complete-band and cost-order premises select the rank-three response band. Tensoring it with the declared generation table gives a conditional rank-45 candidate; it is not a physical attachment theorem. Chirality and the diagonal  $\mathbb{Z}_6$  action come from that table. A distinct local-domain receipt checks the declared tensor-identity operator and conditional gap inheritance without source-selecting the action or transporting the twelve-port Spin packet. Promotion to three physical families requires matter-pole, continuum Spin/locality, physical seam, persistence, excluded-band, and complement-complete refinement receipts.

**Proof route.** The proof route is the one developed in *Recovering Observer Spacetime and Einstein Dynamics from Overlap Consistency* and *Deriving Standard Model Gauge Structure from Observer Overlap Consistency* [2, 3] and *Deriving the Particle Zoo from Observer Consistency* [6]. Complete reversible response and endogenous overlap transport force the local Standard Model gauge Lie algebra. Incidence determines  $J$ , while the explicit signed-central-involutive inverse-port response contract restricts admissible responses to  $\pm J$ . Together with the conditional matrix current and rank-15 matter contract, anomaly-forced determinant balance and tensor descent give the exact hypercharge lattice, common  $\mathbb{Z}_6$  kernel, maximal faithful matter image, and  $N_c = 3$ . The target-blind producer derives the inverse-port response, without selecting the matrix current. Physical laboratory-current and flux attachment are work in progress. Under the explicit compact-gauge refinement receipt, coherent bosonic pullback ladder, symmetry, and forgetful-fiber conditions of the Standard Model gauge paper, the surviving transportable edge-sector category separately re-constructs a compact internal symmetry group. The one-Higgs clauses give  $3 \leq N_g \leq 5$ . The exact screen-band theorem selects rank three under its complete-band and operational-cost premises, and the declared unitary response places the selected residue at the lowest positive generator frequency. Physical family identification and extra-sector exclusion are open. The particle branch does not bypass the unresolved laboratory current, cross-route identity, or physical family attachment.

| Sector                            | Result class                                                                                                                   | What is fixed here                                                                                                                                                                                                                                                        | Requirements for physical interpretation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Carrier roles and classical modes | conditional group/content theorem plus conditional quadratic-action theorem                                                    | declared quotient, $N_c = 3$ , the conditional window $3 \leq N_g \leq 5$ , and the exact rank-three screen-band selection under its named premises; two Maxwell transverse modes, $2 \dim G$ perturbative Yang–Mills modes, and two Einstein TT modes on stated branches | the rank-three band remains a candidate for three physical families until matter-pole and laboratory attachment; the mode counts are classical kernels and quantum particles require the Hilbert-space, pole-residue, and phase premises                                                                                                                                                                                                                                                                                                                              |
| Electroweak bosons                | exact selected-carrier chart + conditional quotient-transport implication + measured-reference inverse adapter                 | no nonzero source-only physical mass emitted                                                                                                                                                                                                                              | the selected chart, conditional value law, and inverse adapter have different provenance. The finite-carrier certificate is not emitted, the adapter is a separate reference-fitted comparison map, and no physical complex-pole pair is identified                                                                                                                                                                                                                                                                                                                   |
| Fine-structure endpoint           | source witness + mixed diagnostic + empirical closure + measurement                                                            | four distinct coordinates, detailed in the endpoint comparison                                                                                                                                                                                                            | the external-data empirical interval misses the measured endpoint. A physical source-only theorem requires a no-target-leak source-derived hadronic spectral payload, target-independent map selection, and the typed same-quantity bridge                                                                                                                                                                                                                                                                                                                            |
| Higgs/top stage                   | conditional downstream split theorem on the declared running, matching, and threshold surface + compare-only exact calculation | no nonzero source-only physical mass emitted                                                                                                                                                                                                                              | a physical mass requires the source root, physical scale, transport, running, matching, rigidity, provenance, uncertainty, and complex-pole premises. The auxiliary direct-top codomain is comparison-only                                                                                                                                                                                                                                                                                                                                                            |
| Quark family                      | common-scale reciprocal-ray falsification + exact generic interface                                                            | no nonzero source-only physical mass emitted                                                                                                                                                                                                                              | reciprocal-ray closure fails across the tested common scales. Its sub-percent residuals use a target-anchored mixed-convention chart, and a lower-order ablation fits that chart better. The simulator has no Yukawa coupling or Yukawa information. Generation-blind flavor-singlet inputs produce no physical flavor-orbit selector                                                                                                                                                                                                                                 |
| Charged leptons                   | conditional construction + exact witness + face-carrier theorem                                                                | exact $A_5/C_3$ face orbit; conditional contraction theorem; engineered finite model of the declared charged face-quotient schema                                                                                                                                         | no nonzero source-only physical mass is emitted. The bounded model supplies local register state, event readback, and a central accepted/rejected record, closing fixed-cutoff schema existence. Its dimensions, automaton, grading, clock, and response are authored inputs; no frozen receipt excludes target dependence. A separate nature/pole bridge assumes both the physical Yukawa response and singularity readout. Physical source selection, family/Yukawa attachment, interacting kernel, cofinal refinement, and pole-scheme premises remain unsatisfied |
| Neutrinos                         | rejected target-informed weighted-cycle candidate + diagnostic calculations                                                    | no nonzero source-only physical mass emitted; source-level PMNS, physical ordering, Majorana phases, and absolute masses are unformed                                                                                                                                     | the candidate fails the NuFIT 6.1 correlated profile; its family kernel is a template, its charged basis is open and nearly degenerate, and its shared-basis recovery is tautological                                                                                                                                                                                                                                                                                                                                                                                 |

| Sector  | Result class                                     | What is fixed here                                     | Requirements for physical interpretation                                                                                                         |
|---------|--------------------------------------------------|--------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Hadrons | backend-gated<br>nonperturbative<br>continuation | stable-channel and readout<br>architecture are defined | source-only masses require one<br>production backend export bundle, then<br>executed unquenching, runtime receipt,<br>and production systematics |

**Theorem boundary.** Structural carrier roles and the realized Standard Model branch are theorem-bearing outputs; classical propagation is conditional on the displayed action, background, and phase premises, and quantum-particle interpretation requires a physical Hilbert space and pole residue. The electroweak quantitative branch separates an exact selected-carrier chart, a conditional value law, and a reference-fitted inverse adapter. The quotient-transport assumptions imply the value law exactly, but the finite carrier does not emit their certificate. The color-balanced  $(\sqrt{N_c}/2, N_c)$  amplitude/loop construction defines a distinct alternative model rather than the complete transport law. The hierarchy theorem fixes the dimensionless ratio  $v/E_\star$ ; an independently physical  $E_\star$  and a pole receipt govern any mass in GeV. A physical  $W/Z$  mass conclusion also requires a strict source root, a frozen RG/matching/scheme packet, branch rigidity, target-independent precommitment, and uncertainty control. The Higgs/top stage is conditional on the declared downstream surface and inherits the same source-root, transport, scale, scheme, rigidity, provenance, uncertainty, and complex-pole premises. Measured electroweak quantities are comparison data. The executed empirical hadron closure integrates external spectral input and misses the measured Thomson endpoint by a certified same-scheme interval. That payload tests the map; it does not source-close the Ward-projected hadronic spectral transport needed for a source-only theorem.

**The  $\alpha_U(P)$  source proof record.** The electroweak hierarchy used by the clock-scale branch depends first on a source readout of the unified diffusion coupling

$$\alpha_U(P_\star).$$

For a trial pixel  $P$ , define

$$M_U(P) = E_\star e^{-2\pi P^{1/6}}, \quad E_{\text{cell}}(P) = \frac{E_\star}{\sqrt{P}}.$$

On the realized color branch,

$$N_c = 3, \quad \beta_{\text{EW}} := N_c + 1 = 4,$$

and the transmutation scale for a candidate coupling  $a$  is

$$v(P, a) = E_{\text{cell}}(P) \exp\left[-\frac{2\pi}{\beta_{\text{EW}} a}\right].$$

The exterior representation theorem fixes the weak-doublet multiplicity  $3 + 1 = 4$ , which also passes Witten parity on the realized branch. Its use as the electroweak transmutation coefficient  $\beta_{\text{EW}} = 4$  requires the common screen/electroweak load-carrier identification and remains declared. The  $\text{SU}(2)$  one-loop coefficient of the realized one-Higgs branch is  $19/6$  and the declared unification packet's coefficient is 1, and neither equals 4. The integer is exact structural mathematics; the physical load attachment is the open receipt. It is not derived by dividing the 24-slot register by six. The source running family is

$$\alpha_i^{-1}(\mu; P, a) = a^{-1} + \frac{b_i}{2\pi} \log\left(\frac{M_U(P)}{\mu}\right), \quad i = 1, 2, 3.$$

The beta-coefficient packet  $(b_1, b_2, b_3)$ , matching convention, threshold packet, and renormalization convention are part of the source proof record. They are declared before any comparison with measured electroweak or gravitational data.

The hypercharge coupling is

$$\alpha_Y(\mu; P, a) = \frac{3}{5}\alpha_1(\mu; P, a).$$

The source  $Z$ -scale is a fixed point of the source equations, not an inserted measured mass:

$$\mu_Z(P, a) = \frac{v(P, a)}{2} \sqrt{4\pi\alpha_2(\mu_Z; P, a) + 4\pi\alpha_Y(\mu_Z; P, a)}.$$

For  $G = \text{SU}(2), \text{SU}(3)$ , define the finite representation heat-kernel sums

$$Z_G(t) = \sum_R d_R e^{-tC_2(R)}, \quad \bar{\ell}_G(t) = \frac{1}{Z_G(t)} \sum_R d_R e^{-tC_2(R)} \log d_R.$$

The source heat-kernel parameters are

$$t_2(P, a) = 4\pi^2\alpha_2(\mu_Z(P, a); P, a), \quad t_3(P, a) = 4\pi^2\alpha_3(\mu_Z(P, a); P, a).$$

The unified-coupling residual is

$$\Phi_U(P, a) := \bar{\ell}_{\text{SU}(2)}(t_2(P, a)) + \bar{\ell}_{\text{SU}(3)}(t_3(P, a)) - \frac{P}{4}.$$

The source value is the zero

$$\Phi_U(P, \alpha_U(P)) = 0.$$

**Definition 5.2** (No-hidden-free-variable admissibility). *The unified-coupling proof record is admissible only when the representation cutoffs, beta packet, matching prescription, threshold packet, and renormalization convention are frozen before comparison with measured electroweak or gravitational data:*

$N_2, \quad N_3, \quad (b_1, b_2, b_3), \quad \text{matching prescription}, \quad \text{threshold packet}, \quad \text{renormalization convention}.$

*None of these may be selected by minimizing the error against*

$$M_W, \quad M_Z, \quad \alpha_s(M_Z), \quad \sin^2 \theta_W, \quad v_{\text{exp}}, \quad G_{\text{exp}}.$$

*Tuning any of these objects after comparison with observed electroweak or gravitational data makes the hierarchy row a calibration.*

**Theorem 5.3** (No-measured-input unified-coupling proof record). *Let*

$$\mathcal{R}_U = (P_\star, N_2, N_3, I_U, \Phi_U, K_U, a_U^{(0)}, r_U, m_U, \text{DAG}_U)$$

*be an admissible unified-coupling proof record. Here  $N_2, N_3$  are the finite representation cutoffs for  $\text{SU}(2)$  and  $\text{SU}(3)$ ,  $I_U \subset \mathbb{R}_+$  is the declared physical interval for  $\alpha_U$ ,  $\Phi_U$  is the heat-kernel closure residual,  $K_U$  is an interval-Newton or Krawczyk operator,  $a_U^{(0)}$  is the displayed candidate,  $r_U$  is the residual bound,  $m_U$  is a lower derivative bound, and  $\text{DAG}_U$  is the dependency graph.*

*Assume*

$$K_U(I_U) \subset \text{int}(I_U),$$

$$|\Phi_U(P_\star, a_U^{(0)})| \leq r_U,$$

$$|\partial_a \Phi_U(P_\star, a)| \geq m_U > 0 \quad (a \in I_U),$$

and assume that  $\text{DAG}_U$  contains no directed path from measured electroweak masses, measured low-energy gauge couplings, measured  $G$ , Planck area, Planck mass, measured  $\Lambda$ , or any equivalent gravity-calibrated or electroweak-calibrated scale. Then there is a unique source value

$$\alpha_U(P_\star) \in I_U$$

such that

$$\Phi_U(P_\star, \alpha_U(P_\star)) = 0,$$

and

$$|\alpha_U(P_\star) - a_U^{(0)}| \leq \frac{r_U}{m_U}.$$

Consequently, the electroweak hierarchy

$$\frac{v}{E_\star} = P_\star^{-1/2} \exp \left[ -\frac{2\pi}{(N_c + 1)\alpha_U(P_\star)} \right]$$

is no-measured-input on this proof record.

*Proof.* The residual  $\Phi_U(P_\star, a)$  is built from the source-side pixel  $P_\star$ , declared representation cut-offs, source transmutation law, source running law, and source  $Z$ -scale fixed-point equation. The dependency condition excludes measured electroweak data and all gravity-calibrated scales. The interval-Newton/Krawczyk inclusion proves existence and uniqueness of a zero of  $\Phi_U(P_\star, a)$  inside  $I_U$ . The residual estimate and derivative lower bound give

$$|\alpha_U(P_\star) - a_U^{(0)}| \leq \frac{|\Phi_U(P_\star, a_U^{(0)})|}{m_U} \leq \frac{r_U}{m_U}.$$

Substitution into the declared transmutation law gives the displayed hierarchy. The dependency graph contains no measured  $G$ , Planck unit, measured  $M_Z$ , measured  $M_W$ , measured  $\alpha_s(M_Z)$ , or measured low-energy electroweak coupling ancestor, so the emitted hierarchy is source-side on  $\mathcal{R}_U$ .  $\square$

The theorem fixes a dimensionless ratio. It supplies neither an independently physical  $E_\star$  nor a mass in GeV. On the populated branch  $P_\star$  is the unique root of the declared local self-read map, so the downstream source expressions inherit that root. Their physical endpoint interpretation requires target-independent map selection, the typed same-quantity bridge, and the corresponding transport and pole constructions.

**Lemma 5.4** (Exponential sensitivity of the hierarchy row). *On the transmutation branch*

$$\frac{v}{E_\star} = P_\star^{-1/2} \exp \left[ -\frac{2\pi}{4\alpha_U} \right],$$

and on a clock row whose dominant scale dependence enters through  $m_e^2 \propto v^2$  in the leading hyperfine scaling  $\varepsilon_{\text{Cs}} \sim \mathcal{C}_{\text{Cs}} \alpha_\star^4 m_e^2 / (m_p E_\star)$ , the induced gravity readout satisfies approximately

$$\frac{\partial \ln G}{\partial \alpha_U} = \frac{2\pi}{\alpha_U^2}.$$

At  $\alpha_U \simeq 0.0411$ ,

$$\frac{\partial \ln G}{\partial \alpha_U} \simeq 3.7 \times 10^3.$$

Therefore a certified relative gravity precision  $\eta_G$  requires

$$|\delta \alpha_U| \lesssim \frac{\eta_G}{3.7 \times 10^3}.$$

*Proof.* For  $\beta_{\text{EW}} = 4$ ,

$$\ln(v/E_\star) = -\frac{1}{2} \ln P_\star - \frac{2\pi}{4\alpha_U}.$$

Thus

$$\frac{\partial \ln(v/E_\star)}{\partial \alpha_U} = \frac{2\pi}{4\alpha_U^2} = \frac{\pi}{2\alpha_U^2}.$$

The clock gap carries the square of the electroweak scale: in the leading hyperfine scaling  $\varepsilon_{\text{Cs}} \propto m_e^2 \propto v^2$  at fixed Yukawa, fine-structure, and strong-sector ratios, so

$$\frac{\partial \ln \varepsilon_{\text{Cs}}}{\partial \alpha_U} = 2 \cdot \frac{\pi}{2\alpha_U^2} = \frac{\pi}{\alpha_U^2}.$$

Since the gravity row is quadratic in the clock scale,

$$G \propto \varepsilon_{\text{Cs}}^2,$$

one obtains

$$\frac{\partial \ln G}{\partial \alpha_U} = 2 \cdot \frac{\pi}{\alpha_U^2} = \frac{2\pi}{\alpha_U^2}.$$

□

**Remark 5.5** (Digits policy). *The exponential branch allows precision only to the extent that the source proof records carry interval bounds. The number of printed digits in  $G_{\text{SI}}$  may not exceed the precision certified by  $\mathcal{R}_U$ ,  $\mathcal{R}_{\text{QCD}}$ ,  $\mathcal{R}_{\text{flav}}$ , and  $\mathcal{R}_{\text{Cs}}$ . A display such as*

$$\varepsilon_{\text{Cs}} = 2\pi\nu_{\text{Cs}} \sqrt{\hbar G_{\text{ref}}/c^5}$$

*is a calibration checksum unless the full source construction emits the same interval with a no- $G$  dependency graph.*

**No- $G$  clock hierarchy for the gravity scale.** The numerical gravity row uses the dimensionless cesium clock gap

$$\varepsilon_{\text{Cs}} := \frac{\hbar\omega_{\text{Cs}}}{E_\star}, \quad E_\star := \frac{\hbar c}{\ell_\star}, \quad \omega_{\text{Cs}} = 2\pi\nu_{\text{Cs}}.$$

The size of this gap is a hierarchy problem for the OPH scale branch. The gap is represented as a factorized source-side readout. The source-only scale branch is accepted as no- $G$  only when the gap factorizes through source-side dimensionless physics:

$$\mathcal{R}_\gamma = \mathcal{R}_U + \mathcal{R}_\alpha + \mathcal{R}_e^{\text{abs}} + \mathcal{R}_{\text{QCD/nuc}}^{133\text{Cs}} + \mathcal{R}_{\text{atom}}^{133\text{Cs}}.$$

Here  $\mathcal{R}_U$  emits  $\alpha_U(P_\star)$  and the electroweak scale,  $\mathcal{R}_\alpha$  emits the electromagnetic coupling used by the atomic Hamiltonian,  $\mathcal{R}_e^{\text{abs}}$  emits the electron absolute mass ratio,  $\mathcal{R}_{\text{QCD/nuc}}^{133\text{Cs}}$  emits the cesium nuclear source packet, and  $\mathcal{R}_{\text{atom}}^{133\text{Cs}}$  emits the hyperfine spectral gap from those dimensionless data.

At leading OPH source level, the electroweak hierarchy is generated by the declared transmutation law

$$\frac{v}{E_\star} = P_\star^{-1/2} \exp \left[ -\frac{2\pi}{(N_c + 1)\alpha_U(P_\star)} \right], \quad N_c = 3.$$

Thus the weak-to-UV ratio is generated from the dimensionless source data  $P_\star$ ,  $N_c$ , and  $\alpha_U(P_\star)$ . The cesium gap then has the schematic source form

$$\varepsilon_{\text{Cs}} = \mathcal{H}_{\text{Cs}} \left( \alpha_\star, \frac{v}{E_\star}, \frac{\Lambda_{\text{QCD}}}{E_\star}, y_e, y_q, \text{nuclear data, atomic corrections} \right),$$

where all arguments are dimensionless source outputs on the declared branch. In leading hyperfine scaling this contains the familiar structure

$$\varepsilon_{\text{Cs}} \sim \mathcal{C}_{\text{Cs}} \alpha_\star^4 \frac{m_e^2}{m_p E_\star},$$

with relativistic, many-body, recoil, QED, finite-nuclear-size, and nuclear-moment corrections absorbed into  $\mathcal{C}_{\text{Cs}}$ .

**Definition 5.6** (Cesium source packet). *A source-only cesium packet is*

$$\mathcal{D}_{\text{Cs}}^{\text{src}} = \left( \alpha_\star, \frac{m_e}{E_\star}, \frac{\Lambda_{\text{QCD}}}{E_\star}, \{y_q\}, \mathcal{N}_{133}, \mathcal{B}_{\text{atom}}^{\text{Cs}} \right).$$

$\mathcal{N}_{133}$  contains the source-side nuclear data for  $^{133}\text{Cs}$ , including nuclear spin, magnetic moment, magnetization distribution, charge radius, recoil data, and finite-size data.  $\mathcal{B}_{\text{atom}}^{\text{Cs}}$  contains the atomic many-body, relativistic, QED, recoil, and finite-nucleus correction package. Every object in  $\mathcal{D}_{\text{Cs}}^{\text{src}}$  is dimensionless and has no dependency path from measured  $G$ , Planck area, Planck mass, Planck time, measured  $\Lambda$ , or an algebraically equivalent gravity-calibrated scale.

**Definition 5.7** (OPH cesium Hamiltonian). *Given  $\mathcal{D}_{\text{Cs}}^{\text{src}}$ , define the dimensionless cesium Hamiltonian*

$$\hat{H}_{\text{Cs}}^{\text{OPH}} = \hat{H}_{\text{DC}} + \hat{H}_{\text{Breit}} + \hat{H}_{\text{QED}} + \hat{H}_{\text{recoil}} + \hat{H}_{\text{nuc}} + \hat{H}_{\text{manybody}}.$$

The hats indicate that every term is measured in units of  $E_\star$ , not in SI units and not in Planck units inferred from measured  $G$ .

**Theorem 5.8** (No- $G$  OPH clock-hierarchy theorem). *Assume the local pixel proof selects  $P_\star$ . Assume the unified-coupling proof record  $\mathcal{R}_U$  of Theorem 5.3 emits  $\alpha_U(P_\star)$  and*

$$\frac{v}{E_\star} = P_\star^{-1/2} \exp \left[ -\frac{2\pi}{(N_c + 1)\alpha_U(P_\star)} \right].$$

*Assume the electromagnetic endpoint branch  $\mathcal{R}_\alpha$  emits  $\alpha_\star$ , the charged-lepton absolute-scale branch  $\mathcal{R}_e^{\text{abs}}$  emits*

$$\frac{m_e}{E_\star} = \frac{y_e}{\sqrt{2}} \frac{v}{E_\star},$$

*the QCD/nuclear branch  $\mathcal{R}_{\text{QCD}/\text{nuc}}^{133\text{Cs}}$  emits  $\mathcal{N}_{133}$ , and the atomic branch  $\mathcal{R}_{\text{atom}}^{133\text{Cs}}$  supplies isolated spectral enclosures*

$$J_{F=3}, \quad J_{F=4}$$

*for the source-only Hamiltonian*

$$\hat{H}_{\text{Cs}}^{\text{OPH}} = \hat{H}_{\text{DC}} + \hat{H}_{\text{Breit}} + \hat{H}_{\text{QED}} + \hat{H}_{\text{recoil}} + \hat{H}_{\text{nuc}} + \hat{H}_{\text{manybody}}$$

with disjoint intervals and certified hyperfine gap

$$\varepsilon_{\text{Cs}} = \lambda_{F=4}(\hat{H}_{\text{Cs}}^{\text{OPH}}) - \lambda_{F=3}(\hat{H}_{\text{Cs}}^{\text{OPH}}).$$

Assume further that the combined dependency graph

$$\text{DAG}_{\text{Cs}} = \text{DAG}(\mathcal{R}_U, \mathcal{R}_\alpha, \mathcal{R}_e^{\text{abs}}, \mathcal{R}_{\text{QCD/nuc}}^{133\text{Cs}}, \mathcal{R}_{\text{atom}}^{133\text{Cs}})$$

contains no directed path from

$$G_{\text{exp}}, \quad \ell_P^2 = \frac{\hbar G}{c^3}, \quad m_P = \sqrt{\frac{\hbar c}{G}}, \quad t_P = \sqrt{\frac{\hbar G}{c^5}}, \quad \Lambda_{\text{exp}}, \quad \frac{3\pi c^3}{\hbar G},$$

or from any algebraically equivalent gravity-calibrated scale. Then  $\varepsilon_{\text{Cs}}$  is a no- $G$  hierarchy readout. Consequently

$$\gamma_\star = \frac{\ell_\star \nu_{\text{Cs}}}{c} = \frac{\varepsilon_{\text{Cs}}}{2\pi}$$

is non-circular, and

$$G_{\text{SI}} = \frac{c^5}{4\pi^2 \hbar \nu_{\text{Cs}}^2} \varepsilon_{\text{Cs}}^2$$

has no measured- $G$  ancestor.

*Proof.*  $\mathcal{R}_U$  emits the small weak ratio from  $P_\star$ ,  $N_c$ , and  $\alpha_U(P_\star)$ , all of which are dimensionless branch data.  $\mathcal{R}_\alpha$  emits the electromagnetic coupling used by the atomic Hamiltonian.  $\mathcal{R}_e^{\text{abs}}$  emits the electron mass ratio.  $\mathcal{R}_{\text{QCD/nuc}}^{133\text{Cs}}$  emits the cesium nuclear source packet. Therefore every coefficient of  $\hat{H}_{\text{Cs}}^{\text{OPH}}$  has no measured- $G$  ancestor. The atomic branch gives isolated spectral enclosures for the two hyperfine levels, so the gap

$$\varepsilon_{\text{Cs}} = \lambda_{F=4}(\hat{H}_{\text{Cs}}^{\text{OPH}}) - \lambda_{F=3}(\hat{H}_{\text{Cs}}^{\text{OPH}})$$

is well-defined and inherits the no- $G$  dependency condition of the Hamiltonian.

Finally,

$$\varepsilon_{\text{Cs}} = \frac{\hbar(2\pi\nu_{\text{Cs}})}{\hbar c / \ell_\star} = \frac{2\pi\nu_{\text{Cs}}\ell_\star}{c},$$

so

$$\gamma_\star = \frac{\ell_\star \nu_{\text{Cs}}}{c} = \frac{\varepsilon_{\text{Cs}}}{2\pi}.$$

Substitution into  $G_{\text{SI}} = c^3 \ell_\star^2 / \hbar$  gives the displayed expression for  $G_{\text{SI}}$ . The only non-display parent of the expression is the source-side  $\varepsilon_{\text{Cs}}$  readout, so the gravity row is non-circular under the stated hypotheses.  $\square$

**Remark 5.9** (Clock-certificate boundary). *The theorem above is not discharged by printing*

$$\varepsilon_{\text{Cs}} = 3.1139305134 \dots \times 10^{-33}.$$

*That decimal is source-predictive only if it is emitted by*

$$\mathcal{R}_U + \mathcal{R}_\alpha + \mathcal{R}_e^{\text{abs}} + \mathcal{R}_{\text{QCD/nuc}}^{133\text{Cs}} + \mathcal{R}_{\text{atom}}^{133\text{Cs}}.$$

*If instead it is computed as*

$$\varepsilon_{\text{Cs}}^{\text{cal}} = 2\pi\nu_{\text{Cs}} \sqrt{\frac{\hbar G_{\text{exp}}}{c^5}},$$



then it is a gravity-side calibration checksum, not a source-side clock prediction. The paper surface supplies the algebraic certificate shape and the  $\mathcal{R}_U$  numerical witness. The full source-only  $\mathcal{R}_{Cs}$  branch requires the source-only electromagnetic endpoint, charged-lepton absolute-scale premise, source-side cesium nuclear packet, and atomic spectral enclosure.

**Remark 5.10** (Hadronic and nuclear certificate boundary). *The unresolved load-bearing pieces of the no- $G$  SI gravity row are the hadronic/same-scheme endpoint payload in  $\mathcal{R}_\alpha$  and the source-side cesium QCD/nuclear packet in  $\mathcal{R}_{QCD/nuc}^{133Cs}$ . The displayed endpoint and  $\varepsilon_{Cs}$  rows therefore fix those payloads for checksum and comparison purposes. They are not counted as source-only OPH emissions.*

*The source-only completion route is a Ward-projected QCD/nuclear spectral certificate from a dedicated OPH optical-compute backend, or an equivalent nonperturbative backend with manifest provenance and systematics. Ordinary CPU/GPU runs in this paper are not the declared source certificate for that hadronic payload.*

**Direct public-record closure for total capacity.** The global counterpart of the local  $P$ -closure begins with a supplied positive integer capacity-carrier dimension

$$D = \dim \mathcal{H}_{cap,r,D}, \quad N = \log D.$$

The universe-level readback equation is

$$\boxed{N = \log M_0(\mathfrak{U}_N)}.$$

Here the argument fixes the type. In every occurrence,  $M_0$  is a multiplicative code size. Once the complete terminal fiber scalarizes, the universe-level notation means

$$M_0(\mathfrak{U}_N) := \widehat{F}_{r,0}(e^N).$$

Thus  $N$  is the logarithmic capacity read back from the trial universe, while  $M_0(\mathfrak{U}_N)$  is the number of correctable public records. The equation is the logarithmic form of stable direct capacity closure, not a new producer or an appeal to a measured target. The carrier type is frozen in the branch contract. Boundary, total, edge-center, and other Hilbert spaces are not interchangeable. Let  $\tilde{\Omega}_{r,D}$  be the terminal physical quotient fiber reached from the source-derived trial universe  $\mathfrak{U}_{r,D}$ . Its members are repair-normal and pass the declared local, obstruction, and provenance gates. Membership contains no capacity-equals- $D$  predicate.

For  $q \in \tilde{\Omega}_{r,D}$ , observer  $O$  has a finite commutative record algebra  $\mathcal{R}_O(q)$  with atom set  $X_O(q) = \text{At } \mathcal{R}_O(q)$ . Each shared interface  $e$  has an atom set  $X_e(q)$  and a source-derived record-atom readout

$$r_{Oe} : X_O(q) \longrightarrow X_e(q).$$

This atom-level construction is used because the generic visible maps on full observer algebras need not be unital  $*$ -homomorphisms in the direction required by an algebra equalizer. The compatible public global sections are

$$X_{\text{pub}}(q) = \{(x_O)_O : r_{Oe}(x_O) = r_{O'e}(x_{O'}) \text{ on every shared interface}\}, \quad \mathcal{R}_{\text{pub}}(q) = C(X_{\text{pub}}(q)).$$

**Theorem 5.11** (Public global sections). *For every finite observer record diagram,  $X_{\text{pub}}(q)$  is finite and  $C(X_{\text{pub}}(q))$  is a finite commutative unital  $C^*$ -algebra. Every isomorphism of observer record diagrams induces a bijection of public global sections and a  $*$ -isomorphism of their function algebras.*

*Proof.* The public sections form the subset of the finite product  $\prod_O X_O(q)$  satisfying finitely many equalities. A diagram isomorphism transports the atom sets and interface maps, hence compatible tuples bijectively; pullback transports their function algebras.  $\square$

Only endogenously writable records count. Define

$$X_{\text{reach}}(q) = \{x \in X_{\text{pub}}(q) : x \text{ terminates an admissible endogenous semantic history}\}.$$

The history may use declared source data, accepted repair, semantic continuation, and quotient-visible writes. It may not use a target record, supplied-capacity metadata, measured  $\Lambda$ , or the electroweak bridge. The publicness policy is a frozen nonempty family

$$\mathfrak{P}(q) \subseteq 2^{\mathcal{O}(q)} \setminus \{\emptyset\}$$

of authorized observer subsets. Collective, universal-local, and quorum publicness are distinct theorem branches.

For every  $x \in X_{\text{reach}}(q)$ , authorized set  $A \in \mathfrak{P}(q)$ , and allowed same-interface semantic continuation  $\kappa$ , the source construction must emit a joint stochastic kernel

$$K_{A,\kappa}(y \mid x), \quad y \in Y_{A,\kappa},$$

whose observer marginals agree with the local checkpoint packets. The joint kernel is an independent receipt. It cannot be manufactured from local marginals.

**Theorem 5.12** (Local marginals do not determine public capacity). *There are two binary-input joint checkpoint channels with identical observer-by-observer marginals and different public capacities.*

*Proof.* Let  $U$  be uniform. The channel  $(Y_1, Y_2) = (U, U \oplus x)$  has uniform input-independent marginals, while the joint parity recovers  $x$ . Independent uniform outputs have the same marginals and carry no information about  $x$ . Thus the joint capacities are two and one, respectively.  $\square$

For a channel  $K$ , write  $S_K(x) = \{y : K(y \mid x) > 0\}$ . A code  $C \subseteq X_{\text{reach}}(q)$  is zero-error correctable for  $K$  when a decoder  $d_K$  satisfies

$$K(d_K^{-1}(x) \mid x) = 1 \quad (x \in C).$$

**Theorem 5.13** (Support criterion and compound zero-error capacity). *A code is zero-error correctable for  $K$  exactly when the supports  $S_K(x)$ ,  $x \in C$ , are pairwise disjoint. Let  $\mathfrak{K}(q)$  contain every authorized checkpoint channel and every declared finite composition. Define the compound confusability graph  $G_q$  on  $X_{\text{reach}}(q)$  by joining  $x \neq x'$  whenever their output supports overlap for at least one  $K \in \mathfrak{K}(q)$ . Then the exact public-record capacity is*

$$\boxed{M_0(q) = \alpha(G_q)},$$

where  $\alpha$  is the graph independence number.

*Proof.* A successful decoder cannot assign one output to two inputs, which proves the support criterion in one direction. Pairwise disjoint supports admit the decoder that returns the unique supporting input. A code valid for every declared channel is therefore exactly an independent set in the union of the channel confusability graphs.  $\square$

For exact indefinite continuation, replace each kernel by its support relation  $R_K = \{(x, y) : K(y \mid x) > 0\}$ . Boolean relation composition generates at most  $2^{|X_{\text{reach}}|^2}$  relations, so the support semigroup and its compound confusability graph close after finitely many steps.

**Theorem 5.14** (Reversible fast branch). *If every authorized continuation is a known injective deterministic map on  $X_{\text{reach}}(q)$ , then*

$$M_0(q) = |X_{\text{reach}}(q)|.$$

*Proof.* Injectivity keeps every pair of inputs distinguishable for every continuation, so  $G_q$  has no edges.  $\square$

Checkpoint invariance is not the capacity definition. A cyclic permutation of  $m$  labels has a one-dimensional fixed function algebra, while its inverse decodes all  $m$  labels. Thus a fixed-projector count can undercount the correctable capacity arbitrarily.

For a frozen finite channel family or continuation horizon and tolerance  $\varepsilon$ , let  $M_\varepsilon(q)$  be the largest code for which every declared channel has a decoder with worst-input error at most  $\varepsilon$ .

**Theorem 5.15** (Approximate-capacity stability and zero-error discontinuity). *If corresponding channel rows obey*

$$\sup_{K,x} d_{\text{TV}}(K(\cdot \mid x), \widehat{K}(\cdot \mid x)) \leq \delta,$$

*then*

$$M_{\varepsilon+\delta}(\widehat{\mathfrak{K}}) \geq M_\varepsilon(\mathfrak{K}).$$

*Zero-error capacity is discontinuous at every identity channel on more than one symbol: adding arbitrarily small full-support noise collapses  $M_0$  to one.*

*Proof.* Total-variation control changes the probability of each correct-decoding event by at most  $\delta$ . For the second statement,  $(1 - \delta)I + \delta U$ , with  $U$  full support, makes every pair of inputs confusable for every  $\delta > 0$ .  $\square$

The capacity carrier is an exact finite Hilbert space  $\mathcal{H}_{\text{cap},r,D}$  of dimension  $D$ . Each reachable public record class has a nonzero projection  $P_x$  on this carrier, and distinct records have orthogonal projections.

**Theorem 5.16** (Carrier bound and saturation rigidity). *Every correctable public code  $C$  satisfies*

$$|C| \leq D, \quad \log M_\varepsilon(q) \leq \log D = N.$$

*If  $M_0(q) = D$ , the code projections are rank one and sum to the identity.*

*Proof.* The ranks of nonzero orthogonal projections add and their sum has rank at most  $D$ . Equality with  $D$  codewords forces every rank to be one and the sum to have full rank.  $\square$

The primary finite readback remains set-valued:

$$\mathfrak{F}_{r,\varepsilon}(D) = \{M_\varepsilon(q) : q \in \widetilde{\Omega}_{r,D}\}.$$

A scalar  $\widehat{F}_{r,\varepsilon}(D)$  exists exactly when the nonempty terminal fiber has one common defined value. The evaluator must distinguish an empty readback, an ambiguous readback, and a singleton scalar readback. Existential closure means  $D \in \mathfrak{F}_{r,0}(D)$ ; selector-free stable closure means

$$\mathfrak{F}_{r,0}(D) = \{D\}.$$

Equivalently, the unclosed kernel

$$K_r(D, m) = |\{q \in \tilde{\Omega}_{r,D} : M_0(q) = m\}|$$

has support only at  $m = D$ . One saturated branch is insufficient when the terminal fiber is ambiguous.

The full capacity packet consists of observer lineage, record atoms, interface maps, reachable public sections, publicness policy, global kernels, and carrier projections. An isomorphism preserving that packet transports every exact confusability edge, approximate decoder error, and projection rank. Hence  $M_\varepsilon$  is implementation invariant under packet isomorphism.

Once scalarized, the active map  $f(D) = \hat{F}_{r,0}(D)$  is deflationary. It cannot attract a positive fixed point from below. If  $f$  is total on a declared finite admissible chain, monotone, and deflationary, iteration from its top element stabilizes at the greatest fixed point. This is a conditional order theorem, not a uniqueness or cutoff-independence theorem.

**Theorem 5.17** (Bounded capacity-continuation counterfamily). *At rung  $k$ , each of the twenty-four visible base records has  $k$  copies, so  $D_{\text{pub}}(k) = 24k$ . In the spectator control the raw carrier is  $D_{\text{raw}}(k, s) = 24ks$ . A target-clean counterfamily shares base agreement, positivity, the carrier bound, and executable finite incidence, publicness, reversible-action, semantic-projection, composition, extension, and oriented-record fiber-product controls. Its reversible identity completion has*

$$M_{\text{id}}(k) = 24k.$$

*Copy collapse and a two-class completion have*

$$M_{\text{erase}}(k) = 24, \quad M_2(k) = 24 \min(k, 2).$$

*A hidden spectator of multiplicity  $s$  leaves the public capacity at  $24k$ . The corresponding slack-zero sets are every positive  $k$ ,  $\{1\}$ ,  $\{1, 2\}$ , and every positive  $k$  only for  $s = 1$ . The shared bounded structure does not entail a unique zero.*

*Proof.* Each collapse channel yields a disjoint union of complete confusability components. One representative per component is an independent set, and completeness of each component gives the matching upper bound. The collapse maps are idempotent, commute with the forty reversible slot actions, and preserve the old induced graph under rung extension. Lean proves the all-rung arithmetic consequence of typed base agreement, positivity, and the carrier bound, including the disagreement between the displayed zero sets. The executable certificate checks the finite incidence, action, projection, composition, extension, and sewing controls. A separate simulator implementation reconstructs held-out finite sample graphs without importing the producer.  $\square$

This bounded counterfamily is not a complete source-class nonidentifiability theorem. The complete A1–A3 terminal fibers, observer and interface atom maps, joint checkpoint kernels, meaning maps, feasible sets, and regulator controls are absent across capacity. The result does not reject strange-loop self-identification. Equality follows after the simulating and simulated readings are proved to be readings of the same invariant quantity.

Capacity extension and regulator refinement are distinct. For  $D \leq D'$ , an injection of reachable records that reflects confusability,

$$e_{DD'}(x) \sim_{G_{D'}} e_{DD'}(x') \implies x \sim_{G_D} x',$$

maps every coarse independent set to a fine independent set and gives  $M_0(D) \leq M_0(D')$ . At fixed  $D$ , refinement injections with the same property make  $M_{0,r}(D)$  a nondecreasing integer bounded by  $D$ , hence eventually constant along each cofinal sequence.

For regions  $A, B$  sewn along public seam  $S$ , exact public records form the fiber product

$$X_{A \cup B} = X_A \times_{X_S} X_B, \quad |X_{A \cup B}| = \sum_{z \in X_S} |r_A^{-1}(z)| |r_B^{-1}(z)|.$$

This gives an exact CSP, tensor-network, or model-counting route on the reversible branch. A cosmological conclusion additionally requires an exact transfer recurrence, exact fiber-product recursion, or a seam theorem with a certified subleading term.

Define the finite-size slack

$$\boxed{s_r(D) = \log D - \log M_{0,r}(D).}$$

If  $\limsup_{D \rightarrow \infty} \log M_0(D) / \log D < 1$ , closure is excluded at all sufficiently large dimensions. Unit asymptotic density is insufficient:  $M(D) = D$  and  $M(D) = D - 1$  have the same limiting density but opposite fixed-point behavior. Full physical  $N$ -closure requires a source-derived finite-size law proving

$$s(D_\star) = 0, \quad s(D) > 0 \quad (D \neq D_\star)$$

on the declared physical domain, or an equally explicit physical selector. Such a law is a separately named physical premise unless it is derived from a stronger source architecture. The common  $D = 24$  screen packet is not a universe-level selection.

Two independent physical identifications follow only after stable direct closure. The first hypothesis identifies the same capacity carrier with the de Sitter horizon record:

$$N_\star = \log D_\star = \frac{A_{\text{dS}}}{4\ell_\star^2}, \quad A_{\text{dS}} = \frac{12\pi}{\Lambda} \implies \boxed{\Lambda \ell_\star^2 = \frac{3\pi}{N_\star}}.$$

The second hypothesis identifies the screen load with the electroweak load by a positive, unital, refinement-natural map

$$\Xi_r : \mathcal{L}_{\text{screen},r} \longrightarrow \mathcal{L}_{\text{EW},r}.$$

Given the independent screen-sieve and electroweak source laws,

$$\Gamma_{\text{scr}} = \frac{P}{12} \log \frac{N_\star}{\pi}, \quad \log \frac{E_{\text{cell}}}{v} = \frac{\pi}{2\alpha_U(P)}, \quad \Xi_r(\Gamma_{\text{scr}}) = \log \frac{E_{\text{cell}}}{v},$$

it gives the conditional Higgs/electroweak residual and coordinate

$$\boxed{R_{\text{EW}} = \alpha_U(P) \log \frac{N_\star}{\pi} - \frac{6\pi}{P} = 0, \quad N_{\text{bridge}} = \pi \exp \left[ \frac{6\pi}{P\alpha_U(P)} \right].}$$

Neither physical identification constructs the public-record map. The independently frozen operational-resolution residual

$$R_\rho = \log M_0 - \frac{\pi}{\rho_{\text{op}}^2}$$

is likewise a downstream test and never a producer definition.

The finite global-section, correctable-code, compound-capacity, approximate-stability, carrier-bound, scalarization, order, extension, refinement, and sewing implications are theorems. The

fixed- $D = 24$  packet realizes its complete terminal fiber and the associated atom, publicness, joint-kernel, carrier, scalarization, extension, and refinement controls. Physical closure requires a complete capacity-indexed lift of those objects, including the A2 meaning maps, A3 feasible sets, and regulator controls; an exact finite-size slack law with one physical zero; the universe-level carrier attachment; the horizon-record identification; and the common screen/electroweak load-carrier identification. No QCD or hadronic backend is a dependency of this program.

**Outer/inner self-consistency for the pixel ratio.** This synthesis paper is the canonical place where the local pixel ratio  $P$  is defined, so the compact and particle papers can cite one derivation rather than duplicate it. The construction has six steps.

1. **Outer pixel ratio.** The outer-side local screen datum is the dimensionless area ratio

$$P := \frac{a_{\text{cell}}}{\ell_{\star}^2}.$$

Here  $\ell_{\star}^2 = 3\pi/B_{\star}$  is supplied by the selected scale certificate. After that scale is emitted, it is displayed in SI language as the Planck area.

2. **Exact equilibrium benchmark.** The total/bulk/edge hierarchy has one exact self-similar balance point. Writing

$$x(C) := \frac{S_{\text{gen}}(C)}{S_{\text{bulk}}(C)} = 1 + \frac{\langle L_C \rangle}{S_{\text{bulk}}(C)},$$

the self-similar balance condition

$$\frac{S_{\text{gen}}(C)}{S_{\text{bulk}}(C)} = \frac{S_{\text{bulk}}(C)}{\langle L_C \rangle}$$

gives  $x^2 - x - 1 = 0$  and hence  $x = \varphi = (1 + \sqrt{5})/2$ . The realized branch must sit near, but not exactly at, this point because exact equilibrium is too symmetric to support durable records, structure, and dynamics.

3. **Outer detuning variable.** On the declared closure surface we parametrize the outer-side detuning by

$$\alpha_{\text{ext}}(P) := \frac{P - \varphi}{\sqrt{\pi}}, \quad \text{equivalently} \quad P = \varphi + \alpha_{\text{ext}}(P)\sqrt{\pi}.$$

4. **Declared quantitative anchor on the trial domain.** Let  $I \subset \mathbb{R}_+$  be the declared branch interval and let

$$F_{\text{EW}} : I \longrightarrow \mathcal{D}_{\text{EW}}$$

denote the declared electroweak source map. For any trial  $P \in I$ , the forward electroweak branch emits

$$P \longmapsto \alpha_U(P) \longmapsto (t_U(P), t_{\text{tr}}(P)) \longmapsto (t_2(P), t_3(P), v(P)) \longmapsto A_Z(P) = \alpha_{\text{em}}^{-1}(m_Z^2; P).$$

5. **Ward-projected Thomson transport.** Let  $T_Q$  be the required Ward-projected  $U(1)_Q$  transport operator from the  $m_Z$ -anchor to the Thomson limit. The complete source-only operator is not emitted. A comparison branch may insert the measured endpoint  $\alpha^{-1}(0) =$

137.035999177(21), while the separate empirical branch carries the published hadronic compilation and returns 136.3827548175 on [136.3670480603, 136.3984651934]. For either declared branch, define

$$A_T(P) := T_Q(A_Z(P), F_{EW}(P)) = \alpha_{Th}^{-1}(P).$$

The inner coupling used in the outer/inner closure is the coupling, not the inverse coupling:

$$\alpha_{in}(P) := \frac{1}{A_T(P)}.$$

6. **Conditional closure.** If the typed same-quantity bridge, target-independent map-selection premise, and same-scheme endpoint transport hold, the physical pixel ratio is the fixed point for which the outer detuning equals the inner observation coupling:

$$\alpha_{ext}(P) = \alpha_{in}(P).$$

Equivalently,

$$H(P) := P - \varphi - \frac{\sqrt{\pi}}{A_T(P)} = 0, \quad \text{or} \quad P = \varphi + \frac{\sqrt{\pi}}{A_T(P)} = \varphi + \alpha_{in}(P)\sqrt{\pi}.$$

Writing

$$G(P) := \varphi + \frac{\sqrt{\pi}}{A_T(P)},$$

the closure problem is the fixed-point condition  $G(P) = P$ . On any declared branch interval where this induced map is a self-map and a contraction, its mathematical fixed point is locally unique. Equivalently, the solver can use

$$H(P) := P - \varphi - \frac{\sqrt{\pi}}{A_T(P)}$$

and accept the unique zero of  $H$  on the declared branch interval. Observer-facing data may localize that interval. The measured endpoint cannot replace the closure solve.

Conditional on the typed bridge, the closure interpretation assigns one screen cell two coupled roles. On the outer side it is one cell of the screen whose displacement above the exact self-similar equilibrium  $\varphi$  sets the geometric detuning. Conditional on a typed bridge identifying the returned electromagnetic coupling as another reading of that cell, closure requires the outer detuning and inner reading to agree. The pixel ratio  $P$  is the size of that cell in units of the scale-certificate area  $\ell_\star^2$ . The fixed-point equation is the mathematical form of that conditional consistency statement. Neither the exact fixed-point mathematics nor the source certificates construct the bridge or select either certified  $P$  map. On the measured-endpoint comparison branch, the root is

$$P_C \simeq 1.6309682094, \quad \alpha^{-1}(0) = 137.035999177(21).$$

This is a comparison coordinate inferred from the measured endpoint. It does not define the pixel used by the theory. The declared source map has one mathematical fixed point. The same-quantity bridge and target-independent map selection would promote it to a physical closure point. The difference between the source-map root and the comparison coordinate is the loop residual:  $3.0 \times 10^{-4}$  relative on the source chain and  $2.5 \times 10^{-6}$  relative (about  $1.6 \times 10^4$  measurement sigma) for the certified self-consistent gauge-width fixed point  $\alpha^{-1} = 137.035660136946577\dots$ , a

certified fixed point of the declared map with the Ward-projected hadronic transport, map selection, and same-quantity bridge open. Distinct coordinates describe the comparison. The certified source/root witness is 136.994835177413.... The mixed-provenance no-hadron diagnostic is 137.0359595136...; it mixes the inner value at  $P_{\text{fwd}}$  with  $\alpha_U$  at the comparison pixel, is no fixed point, and is excluded from physical output. The external-data empirical closure is 136.3827548175 on [136.3670480603, 136.3984651934], and the measured endpoint is 137.035999177(21). The empirical interval requires a same-scheme correction in [0.6198609041, 0.6505569679] inverse-alpha units, and the standard on-shell reference deficit 0.631 lies inside that certified interval. A zero-momentum laboratory measurement sees the dressed  $U(1)_Q$  current after charged-lepton vacuum polarization, confined-quark/hadron spectral transport, and same-scheme endpoint matching. A physical source-only conclusion requires the missing source-derived hadronic spectral transport and its dependency certificate, target-independent map selection, and the typed same-quantity bridge.

The source-side trunk, mixed diagnostic, empirical endpoint, measurement, and interval have distinct provenance. Downstream quantities inherit the provenance of whichever declared pixel branch they use. A separate hardware note reports an optical-cavity check of the same fixed-point geometry; this is treated as corroborating engineering evidence. The same closure equation also admits an inverse observational use case in which observers infer  $P$  and total screen capacity from inside the world.

One source-only route studies the closure directly at zero momentum. It requires a self-contained hadronic transport law rather than the external empirical compilation used by the comparison.

**Non-hadron results.** The non-hadron results consist of conditional classical carrier modes without quantum mass claims, the exact electroweak selected-carrier chart, the conditional value-law implication, the distinct reference-fitted inverse adapter, the conditional downstream Higgs/top coordinate, a same-family charged witness, and the common-scale rejection of the reciprocal-ray quark texture. The generic quark interface has six scalar coordinates, and no source-derived flavor-orbit selector is emitted. The rejected weighted-cycle neutrino candidate is a comparison-only falsification result. Ref. [6] gives the exact values, premises, diagnostics, and sector boundaries.

**Extended derivation.** The structural-to-family route and the complete per-sector theorem boundaries are developed in *Deriving the Particle Zoo from Observer Consistency* [6].

## 6 Screen Microphysics, Records, and Observer Continuation

The screen-microphysics branch gives a fixed-cutoff engineering realization. It asks whether some microscopic model might exist in principle and gives one explicit regulated carrier architecture: a federation of finite observer patches with echosahedral multi-port interfaces, recurrent toroidal subchannels, exposed overlap data, records, repair instruments, and observer-facing interfaces all made concrete at fixed cutoff. A spherical screen is used there only as an observer-facing regulator chart for support-visible cuts, not as a claim that the universe is a literal spherical quantum computer.

The branch has five theorem-bearing components. First, the reference architecture itself is explicit. Second, the regulated patch-net embedding theorem applies on its fixed-cutoff object/local-interface domain, and the edge-sector thermal/Casimir branch carries an explicit fixed-cutoff law with a separate compact lift. Third, the fixed-cutoff measurement/Born-rule theorem applies on its declared operator and record-event surface. Fourth, the fixed-cutoff Bell/CHSH theorem applies on the declared two-wing surface. Fifth, the checkpoint/restoration observer theorem applies. The



additional premises are packet-level quotient closure where one wants an autonomous exported dynamics, compact-group/Peter–Weyl bookkeeping on the electroweak and compact surfaces, fair-block contraction certificates for long-run noisy approximate consensus on exported packet nets, and model-family universality. The support-visible BW / geometric-modular / local-Einstein closure lives on the companion recovered-core theorems rather than on this fixed-cutoff engineering branch. A reconstruction of Hilbert,  $C^*$ - or von Neumann algebra structure, Born probabilities, trace structure, and entropy from operational records alone is a separate question outside this branch.

**Reference-architecture takeaway.** The synthesis-level point is that the regulated screen-side architecture supplies a concrete habitat in which local observables, overlap observables, record registers, and synchronization maps can all be written without pretending to identify a unique final UV completion. Hardware evidence enters only through a public, hash-stable evidence bundle. The OPH-FPE simulator provides finite-run receipts and replayable experiment bundles [12].

**Imported theorem from Ref. [5] (Fixed-cutoff record algebra and Born–Lüders package).** For each completed compare/write/verify slice of the regulated microphysics, the declared pointer and overlap-sector projectors generate a finite commutative central record algebra

$$\mathcal{Z}_{\text{rec}}.$$

Every observer-accessible event  $E$  in that algebra has probability

$$\mathbb{P}(E) = \text{Tr}(\rho P_E),$$

and conditioning on  $E$  gives the operational post-measurement state

$$\rho|_E = \frac{P_E \rho P_E}{\text{Tr}(\rho P_E)}.$$

If a practical readout instead uses projectors  $Q_a$  with commuting central reference projectors  $\hat{Q}_a$  on the same declared slots and

$$\delta_{\text{rec}} = \max_a \|Q_a - \hat{Q}_a\|,$$

then

$$\|[Q_a, Q_b]\| \leq 4 \delta_{\text{rec}},$$

and any accessible-state perturbation  $\|\tilde{\rho} - \rho\|_1 \leq \varepsilon$  changes each declared elementary record-event probability by at most  $\varepsilon + \delta_{\text{rec}}$ .

The charged-family construction contains one engineered witness for this distinction. Inside a bounded finite patch, eight local register graphs and eight declared paths contain 6,467 matrix units. A noncentral event projection is read into a central two-valued accepted/rejected record, and sixty  $A_5$  charts verify the declared equivariance. The evidence bundle proves that the stipulated fixed-cutoff schema is nonempty. Its register sizes, path automaton, signs, clock, and response are authored, while inert ancillary stabilization does not supply a cofinal physical refinement. The witness therefore connects event readback to a public record at fixed cutoff; it does not select the charged source, recovery law, family attachment, or physical pole.

**Imported theorem from Ref. [5] (Fixed-cutoff Bell / CHSH package).** On the declared fixed-cutoff physical observable algebra, fix commuting left/right wing subalgebras, central binary setting registers, binary projective readouts on each wing, and a physical source state on the joint wing algebra. Then the compare slice carries the joint law

$$p(a, b \mid x, y) = \text{Tr}(\rho_{LR} P_{a|x}^{(L)} P_{b|y}^{(R)}),$$

the local marginals are independent of the remote setting, the Bell correlators satisfy

$$E(x, y) = \text{Tr}(\rho_{LR} A_x B_y),$$

and the CHSH combination obeys

$$|E(0, 0) + E(0, 1) + E(1, 0) - E(1, 1)| \leq 2\sqrt{2}.$$

If the source family contains an explicit two-qubit branch with the stated Pauli readouts, the same fixed-cutoff surface saturates  $2\sqrt{2}$  exactly on that branch. The source-state input is explicit: the theorem derives the Bell law and the Tsirelson bound from the stated fixed-cutoff operator surface, while Bell-pair preparation and two-qubit factor structure are stated branch conditions.

**Imported theorem from Ref. [5] (Checkpoint/restoration and observer backup).** For an observer patch  $P_O$ , let the checkpoint data consist of the observer-facing record algebra, the accessible local state, the future update schedule, and the externally visible overlap interface data. Exact restoration reproduces the full future law of observer-accessible events, while an  $\varepsilon$ -accurate restoration changes that future law by at most  $\varepsilon$  in total variation.

**Observer-facing conclusion.** The combined theorems supply a fixed-cutoff measurement interface, a fixed-cutoff checkpoint/restoration package, and an operational observer-identity criterion on observer-accessible event algebras. Stronger substrate-selection or strange-loop closure claims are outside the recovered theorem package.

**Extended derivation.** The fixed-cutoff edge-law, measurement, Born-rule, Bell/CHSH, and checkpoint/restoration packages are developed in *Federated Echosahedral Screen Microphysics: Patch Hardware, Records, and Observer Synchronization in OPH* [5].

## 7 Worksheet/String Branch

The string/worldsheet result is a theorem-level bridge from the fixed-cutoff edge-sector law to the two-dimensional Yang–Mills partition function, followed by a controlled large- $N_{\text{edge}}$  effective-description theorem on the stated branch. The conditional support-visible four-dimensional Yang–Mills form and repair-gap theorems are separate results whose continuum identification and gap transport require their stated receipts.

**Imported theorem (Heat-kernel edge-sector bridge).** Import the fixed-cutoff same-overlap edge-law package from *Federated Echosahedral Screen Microphysics* [5]. Under the same overlap-gauge realization and local-Gibbs/MaxEnt edge branch, together with the edge-Hamiltonian and one-sided-algebra hypotheses (EH-1)/(EH-2) of Theorem 6.20 (the sector probabilities are the one-sided  $d_R$  convention; the closed partition below is the two-sided  $d_R^2$  convention), the edge-sector weights satisfy

$$p_R(t) \propto d_R e^{-tC_2(R)}.$$

Consequently the edge partition function matches the standard two-dimensional Yang-Mills heat-kernel form. Osterwalder–Schrader and Wightman-strength continuum reconstruction for the four-dimensional Euclidean Yang–Mills form and the mass-gap statement require the separate support-visible compact-gauge premises.

**Imported proof route.** The microphysics source surface carries the fixed-cutoff thermal/Casimir law on the declared overlap interface. Peter–Weyl decomposition [56] then identifies the resulting sum with the standard heat-kernel expression for two-dimensional Yang-Mills on the compact-group refinement lift.

**Compact-carrier theorem.** Writing the closed edge partition function as

$$Z_{\text{edge}}(t) = \sum_R d_R^2 e^{-tC_2(R)},$$

the compact carrier states the exact bridge itself as a named theorem: Peter–Weyl identifies it with the compact-group heat kernel at the identity,

$$Z_{\text{edge}}(t) = K_t(1).$$

The Chapman–Kolmogorov gluing law for  $K_t$  is the precise sense in which collar sewing matches the two-dimensional Yang–Mills heat-kernel surface before any large- $N_{\text{edge}}$  continuation is invoked. The compact carrier is a partition-identity theorem on its stated branch. The four-dimensional OPH claim identifies the compact-gauge branch with the Euclidean Yang–Mills form and applies the repair-dynamics theorem on that same support-visible branch.

**Controlled large- $N_{\text{edge}}$  branch.** Fix a distinct large- $N_{\text{edge}}$  realization, with  $N_{\text{edge}} \neq N_c = 3$ , and the fixed- $\tau$  variable

$$\tau = tN_{\text{edge}}$$

on a compact window  $I$ . If the resulting edge free energy satisfies the Standard Model gauge paper’s large- $N_{\text{edge}}$  criterion, with remainder control on  $I$ , then the Gross-Taylor rewriting [55] is a controlled theorem-level worldsheet effective description of the edge dynamics on that branch. Critical-superstring claims, worldsheet CFT closure, anomaly cancellation, and full massless-spectrum matching are outside the recovered core theorem package.

**Higher-gauge continuation slot.** The genuinely noncentral crossed-module gluing data on cuts provide the natural continuation-level slot for  $B$ -field / gerbe data in any string-style reorganization of the OPH edge sector. This is a structural analogy only; it is not an open/closed-string boundary formalism or a critical-string theorem.

## 8 Theorem Boundaries and Conditional Continuations

The synthesis has the following theorem boundary:

1. The fixed-cutoff collar, higher-gauge, consensus, record, Bell/CHSH, checkpoint, and restoration packages are theorem-bearing on their stated finite-regulator surfaces.
2. Physical UV uniqueness is quotient-level and stable under inert ancillas. OPH fixes terminal values on declared physical observables, not a unique microscopic representative.

3. The Lorentz and null-modular chain is closed on the support-visible refinement/scaling branch by the finite cap-normal BW scaling theorem on the geometric subnet. Its two inputs are the finite cap-normal support/flow certificate and an independently complete multiresolution algebra-state comparison package on the same tower. Fixed-collar Markov replacement with carried  $r_{\text{FR}}$ ,  $\delta^{\text{M}}$ , and regularized modular remainders, support-readable modular covariance, support-order faithfulness, BW framing, held-out oriented cross-ratio rigidity, and independently normalized geometric  $2\pi$ -KMS convergence make the cap modular automorphism geometric. The compact cap-normal theorem then identifies  $S^2$  with projective future null rays  $q(\Omega) = (1, \Omega)$ , represents oriented round caps by  $n_C = (\cot \alpha, \csc \alpha \mathbf{c})$ , proves the signed incidence formula, and obtains  $n_{gC} = \Lambda_g n_C$  and  $H^3 \simeq \text{SO}^+(3, 1)/\text{SO}(3)$ . This is not a finite-cell Lorentz-invariance claim, a one-shot upgrade from small CMI to exact Markov geometry, a proof that bare finite consensus produces the cap-normal certificate, or a populated/neutral-bulk claim. The absolute Einstein implication additionally requires one source-derived common-domain tower carrying the event, stress, entropy, uniform-asymptotic, universal-coupling, vacuum-reference, and independent-scale premises. Construction and certification of that tower are work in progress.
4. Record-conditioned observer-frame estimation in  $H^3$  is a separate conditional theorem: calibrated modular cap responses select a frame value only when they factor through frame-local  $H^3$  data on a compact domain and the finite cap frame has  $\alpha > 0$ . Exact data identify one frame value; finite noisy data produce a frame ball, and a unique finite value requires a positive residual gap  $\Delta_{\text{loc}}$ . This does not locate an event, populate the event base, derive particle species or stress-energy, or construct chart-blind neutral bulk.
5. The cosmological constant branch has a closed conditional implication layer and a bounded capacity counterfamily. The descended semantic checkpoint packet defines exact public capacity as the independence number of the compound confusability graph at finite integer dimension. Under a faithful carrier representation and whole-fiber scalarization it is deflationary; under a confusability-reflecting capacity embedding, top-down iteration selects its greatest fixed point, while fixed-capacity refinement injections imply eventual exact stabilization. A source-derived packet realizes the finite receipts at  $D = 24$  inside its declared finite source category. Reversible, copy-collapse, two-class, and hidden-spectator continuations share base agreement, positivity, the carrier bound, and executable finite controls while giving different exact fixed sets. The exact verdict applies to this bounded completion class. Universal all-rung membership in the complete A1–A3 capacity-source contract and the executable-to-Lean bridge are open. Direct  $N$  is not evaluable on the incomplete source antecedent. A positive cosmic value requires a complete source antecedent, one physical zero, and a universe-level carrier attachment. The horizon–record identification then gives  $\Lambda_{\text{CRC}} \ell_{\star}^2 = 3\pi/N_{\text{CRC}}$ , and the common screen/electroweak load-carrier map separately gives the electroweak bridge. A diagonal normal-form count constructs neither the off-diagonal map nor deterministic closure. The operational resolution experiment is an independent test rather than the producer.
6. Complete reversible response and endogenous overlap transport force the abstract local Standard Model gauge Lie algebra. Under the explicit inverse-port contract, incidence and target-blind port readback derive the signed response. Under the conditional matrix current and matter representation, anomaly-forced determinant balance and tensor descent give exact hypercharge,  $N_c = 3$ , the common  $\mathbb{Z}_6$  kernel, and the maximal faithful matter image. Source selection of the matrix current, matter action, and physical  $\mathbb{Z}_6$  quotient is work in progress. Laboratory identification of the current and flux sectors is work in progress. Compact gauge

reconstruction uses the theorem-produced combined transportability criterion and fixed-cutoff tensor-generated bosonic sector category. On a cofinal tail carrying the explicit compact-gauge refinement receipt, the bosonic refinement ladder and compatible forgetful fibers reconstruct a compact group from sectors with strictifiable central or higher-associator defects and at least one allowed trivial-holonomy strict representative. That is a distinct conditional classification route. Physical family identification and extra-sector exclusion require separate identifications. Laboratory current identification, equality with the Tannaka current, and the physical matter-pole and continuum interpretation of the conditional rank-45 family candidate are open.

7. The particle branch is sector-split. Electroweak  $W/Z$  has an exact selected-carrier chart, a conditional value law whose finite-carrier certificate is not emitted, and a separate reference-fitted inverse adapter. An external strict one-loop fixture has principal-sheet scalar zero-exclusion receipts and one simple scalar zero for each of  $W$  and  $Z$  in channel-specific lower-half pole boxes and algebraic charts. No certified continuation identity or composition joins it to the OPH chart, so it supplies no physical pole or mass comparison. Higgs/top is conditional on its declared downstream surface. The reciprocal-ray quark candidate fails on dimensionless common-scale Yukawas, while the exact generic interface has six scalar coordinates and no source-derived flavor-orbit selector. Its sub-percent residuals use a target-anchored mixed-convention chart. The particle simulator contains no Yukawa coupling and returns a null calibration result. The weighted-cycle neutrino candidate is rejected, charged-lepton source landing is open despite the engineered fixed-cutoff record witness, and source-only hadron masses require a production OPH backend. The empirical hadron endpoint is an external-data validation surface rather than that backend.
8. The string/worldsheet branch is a controlled heat-kernel effective-description branch. A critical-string conclusion requires a separate selector premise.

**Local unification boundary.** One local unification identity is explicit. There is no common theorem for  $c$ ,  $G$ ,  $W$ ,  $Z$ , and  $H$ . The declared local input  $P$  governs the electroweak/Higgs trunk and fixes the matching between cell area and edge entropy. The gravity normalization is emitted by the selected scale certificate, while the invariant causal speed  $c$  is structural from the Lorentz branch and gains its SI label only on the local readout package.

| Quantity | Value                                                                                                                | Derivation chain                                                                                                                                              | Caveat                                                                                                                                                              |
|----------|----------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $c_*$    | 299792458 m/s by SI definition                                                                                       | structural Lorentz/BW branch $\rightarrow$ common invariant null cone/speed $\rightarrow$ SI unit convention                                                  | the common cone and dimensionless speed equality are structural; the decimal magnitude is not predicted                                                             |
| $G$      | $6.674299995910528 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$ on the stated local extension surface | $\gamma_* = \ell_* \nu_{Cs}/c$ , $B_* = 3\pi/\ell_*^2$ ,<br>$a_{\text{cell}} = P\ell_*^2$ , $\ell_{\text{shared}} = P/4$ ,<br>$G_{\text{SI}} = c^3\ell_*^2/h$ | the pixel fixes the shared cell/edge identity and cancels in the Newton area law; the strict classical-regime clause is explicit                                    |
| $M_W$    | no nonzero source-only physical mass emitted                                                                         | selected carrier $\rightarrow$ exact chart; quotient-transport assumptions $\Rightarrow$ conditional value law; inverse target adapter separate               | the finite-carrier certificate is not emitted, and no pole mass is identified                                                                                       |
| $M_Z$    | no nonzero source-only physical mass emitted                                                                         | same chart, conditional law, and separate adapter                                                                                                             | same source-certificate and pole-mass boundary                                                                                                                      |
| $M_H$    | no nonzero source-only physical mass emitted                                                                         | double-criticality family from the gauge sector $\rightarrow$ frozen boundary-scale candidate<br>$(m_H, m_t) = (125.77, 172.63) \text{ GeV}$                  | inherits the boundary-scale selection theorem plus the source-root, physical-scale, running, matching, rigidity, provenance, uncertainty, and complex-pole premises |

**Conditional electroweak comparison coordinates.** The strictest target-free  $W/Z$  branch is the source-side zero-selector law, (80.330, 91.119) GeV, with no target input and a single discrete two-law choice. This is a running/chart coordinate, not a physical pole-mass prediction. The source does not close the renormalized-vev, tadpole, field-content, threshold, matching, running, complex-pole, or theory-covariance map needed to compare it with measured resonance parameters, so no physical comparison is defined. For the PDG-2026 reference pair, the exact convention map gives energy-pole masses  $(M_W, M_Z) = (80.3411410, 91.1623040)$  GeV, while the separate legacy coordinates  $\sqrt{\text{Re } s}$  are (80.3340218, 91.1537725) GeV. The running/chart coordinates have no OPH theory covariance, and the physical readout contract is open. A separate strict one-loop fixture has principal-sheet scalar zero-exclusion receipts and one simple scalar zero for each of  $W$  and  $Z$  in channel-specific lower-half pole boxes and algebraic charts. No certified continuation identity or composition joins that fixture to the OPH chart. The selected carrier, conditional value law, and reference-fitted inverse adapter give the downstream comparison coordinates

|                                  |                                     |
|----------------------------------|-------------------------------------|
| selected carrier                 | (80.3862916924, 91.1829044467) GeV, |
| conditional value law            | (80.3770000154, 91.1879780779) GeV, |
| reference-fitted inverse adapter | (80.3625, 91.1879) GeV.             |

The Higgs/top pair is read on the double-criticality branch, whose frozen boundary-scale candidate  $E_* e^{-\pi} P^{-1/6}$  gives  $(m_H, m_t) = (125.77, 172.63)$  GeV at two loops, with  $m_H = 125.72$  GeV on the fit-free curve at the measured top; the declared calibration surface (125.1995304097, 172.3523553288) GeV is a target-anchored fit. These coordinates are comparison-only and are not physical particle outputs.

The local finite package supplies the strict classical-regime clause and the exact selected-carrier chart. The electroweak value law is conditional on the finite quotient-transport certificate. The familiar-unit display split on that same surface is

$$L_{\text{loc}} = \sqrt{a_{\text{cell}}} \hat{L}(P), \quad t_{\text{loc}} = \frac{\sqrt{a_{\text{cell}}}}{c} \hat{T}(P),$$

$$E_{\text{loc}} = \frac{\hbar c}{\sqrt{a_{\text{cell}}}} \hat{E}(P), \quad \Theta_{\text{loc}} = \frac{\hbar c}{k_B \sqrt{a_{\text{cell}}}} \hat{\Theta}(P),$$

with dimensionless  $\hat{L}, \hat{T}, \hat{E}, \hat{\Theta}$ . The gravity-side shared-edge identity fixes the cancellation of  $P$ , while the local SI readout comes from  $\ell_*^2$ . Meters, seconds, GeV, and Kelvin are downstream display conventions built from the structural  $c$  row and the familiar constants  $\hbar, k_B$ .

**Companion papers.** Detailed derivations appear in Refs. [2, 3, 4, 5, 6, 9, 10].

## 9 Conclusion

The three-axiom basis, gravity and gauge branches, consensus formulation, particle-spectrum calculation, regulated screen microphysics, fixed-cutoff measurement and observer results, and controlled large- $N_{\text{edge}}$  worldsheet effective description fit into one typed construction.

The finite analytic core establishes schedule-independent consensus on the declared quotient, the conditional relativity and Einstein branches, and the local Standard Model gauge Lie algebra. The finite matrix current, matter image, and common  $\mathbb{Z}_6$  kernel are exact conditional realizations. The fixed-cutoff microphysics gives explicit record, measurement, Bell, checkpoint, and restoration

interfaces. The worldsheet result is a controlled heat-kernel effective description under its stated large- $N_{\text{edge}}$  premises.

The numerical maps have no continuously tunable fit dials. Interval arithmetic proves that each declared fixed-point map has exactly one fixed point on its stated domain. A source-only audit covers the declared seventeen-row menu for the executed capacity families, the two declared  $P$  maps, one direct fixed-cutoff control, the common-load baseline, reserve maps, RC-LOAD, the archived construction aggregate, and five hierarchy packets. Fifteen of its thirty entries carry exact fixed-point progress, while none closes target-independent selection, same-quantity construction, and the complete source return map together. Four of five preflight-completion gates are undischarged. This finite inventory does not exhaust other source laws. Physical conclusions require target-independent map selection, the relevant same-quantity bridges, the Ward-projected hadronic transport, the capacity readback map  $F$ , the forward electroweak source chain, a source-derived flavor-orbit selector, and the corresponding pole, scale, family, and nonperturbative attachments. Strong CP, full flavor and CKM closure, charged-lepton source landing, and source-only hadron masses are work in progress. Refs. [2, 3, 4, 5, 6] give the sector-by-sector proofs and quantitative details. The shared numerical surface is concise: the fixed point  $P_\star$  organizes the bosonic mass trunk and the gravity-side shared edge-entropy identity beneath the emitted local  $G$  row, while the same familiar-unit package reads meters and seconds from the single ruler  $\sqrt{a_{\text{cell}}}$  together with the structural invariant speed  $c$ , and reads GeV and Kelvin from the inverse local ruler through  $\hbar$  and  $k_B$ .

## A Candidate Microphysics Finite Reference Checks

The synthesis paper records the finite echosahedral reference-model checks in an appendix. The reference object is a typed carrier/federation data structure, not a physical federation source and not a realization of observer patches, overlap observables, or a support  $S^2$ . Its verified content is finite and structural: one shared twelve-port 12/30/20 carrier template with antipode and  $A_5$  action, typed collar bijections, orientation reversal, canonical finite matrix-algebra maps on individual seams, external-boundary coverage, presentation relabeling invariance, and a finite reference-tower commutation check.

**Proposition (Finite carrier and seam conformance).** For the declared finite reference objects, the checker recomputes local carrier conformance, seam bijectivity and orientation, endpoint algebra-schema agreement, the permutation-induced matrix-algebra isomorphism on each seam, boundary coverage, and the presentation firewall. The two-carrier reference fixture has no composable seam triangle. Its higher-overlap Čech condition is therefore true only vacuously, and its nonvacuous higher-overlap witness is false. A separate coherent-triangle control exercises the cocycle checker, but deliberately fails the positive federation-sewing conditions and cannot be combined with the reference fixture as a physical witness.

Consequently, the physical echosahedral-federation realization and source-instrument premises are false. The level-zero through level-two naturality calculation is a finite diagram for the shared regular-icosahedron template; it selects no physical support chart and supplies no carrier-to-support  $S^2$ ,  $H^3$ , event, BW, or KMS receipt.

**Proof sketch.** Each positive finite statement follows by exact recomputation on the declared template, seam maps, canonical matrix schemas, boundary declarations, and reference refinement maps. The absent composable seam triangle prevents a nonvacuous higher-overlap witness. No measured

physical carrier-to-patch identification or quotient-visible spherical-support map is present. The physical federation, source, and support-screen premises therefore remain unsatisfied.

**Proposition (Fixed-cutoff edge heat-kernel calculation).** For a separately declared finite sector kernel satisfying weighted proposal symmetry and the Casimir acceptance rule, the stationary sector law is

$$\pi_\beta(\alpha) \propto d_\alpha e^{-\beta C_2(\alpha)},$$

and along a separately assumed refinement ladder converging to a compact-group Peter–Weyl decomposition [56] the same weights converge cylinderwise to the heat-kernel/Casimir law used in the OPH edge-sector branch.

**Proof sketch.** The finite proposal kernel is chosen so that the weighted proposal symmetry  $d_\alpha q_{\alpha\beta} = d_\beta q_{\beta\alpha}$  holds. Detailed balance with the Casimir acceptance rule then fixes the stationary distribution in finite dimension. The compact-group heat-kernel law is the conditional refinement lift. Neither calculation supplies the missing physical federation, source-instrument, or support- $S^2$  receipt.

**Extended derivation.** The finite candidate model, validation package, and conditional theorems are developed in *Federated Echosahedral Screen Microphysics* [5].

## B Interpretive Epilogue: State-and-Law Habitat

This appendix records the OPH state-and-law habitat available inside the framework. A strange-loop closure theorem is outside this appendix.

The OPH inputs used here are:

1. the patch net  $P \mapsto A(P)$  of von Neumann algebras together with isotony and overlap restriction maps;
2. a compatible local state family, plus the separately declared global-extension interface used by this appendix to represent it on the inductive-limit algebra;
3. the finite Axiom-3 information-projection branch and the separate optimizer-pushforward interface, which together supply a common finite family of gauge-invariant local constraint observables across cutoffs;
4. on the fixed-cutoff branch, the derived finite type-I presentations and compact boundary gluing groups used elsewhere in the OPH package.

The law slot is encoded by the finite expectation-value coordinates of the retained A1-generated constraint family used by Axiom 3. That choice furnishes a compact convex state-and-law habitat for the framework.

**Theorem (Internal state-and-law habitat theorem).** Fix finitely many screen patches  $P_1, \dots, P_N$  in the standard OPH setup. For each  $i$ , let

$$M_i := A(P_i), \quad M_{ij} := A(P_i \cap P_j),$$



where  $M_i$  is the patch von Neumann algebra and  $M_{ij} \subset M_i, M_j$  is the overlap algebra. Let

$$S_i := S(M_i) \subset M_i^*$$

be the full state space of  $M_i$ , endowed with the weak\* topology  $\sigma(M_i^*, M_i)$ .

Choose a finite family of self-adjoint gauge-invariant local constraint observables

$$\mathcal{O} = \{O_1, \dots, O_m\}$$

from the A1-generated Axiom-3 constraint grammar, with each  $O_a$  supported in one of the chosen patches; write  $i(a)$  for an index such that  $O_a \in M_{i(a)}$ .

Define the overlap-consistent state sector

$$X_{\text{ov}} := \left\{ (\omega_1, \dots, \omega_N) \in \prod_{i=1}^N S_i : \omega_i|_{M_{ij}} = \omega_j|_{M_{ij}} \text{ for all } i, j \right\}.$$

Define the OPH law-coordinate map

$$c : X_{\text{ov}} \rightarrow \mathbb{R}^m, \quad c(\omega_1, \dots, \omega_N) := (\omega_{i(1)}(O_1), \dots, \omega_{i(m)}(O_m)).$$

Let

$$X_{\mathcal{O}} := \left\{ (\omega_1, \dots, \omega_N, \ell) \in \left( \prod_{i=1}^N S_i \right) \times \mathbb{R}^m : (\omega_1, \dots, \omega_N) \in X_{\text{ov}}, \ell = c(\omega_1, \dots, \omega_N) \right\}.$$

Then:

1. the ambient product

$$E := \left( \prod_{i=1}^N M_i^* \right) \times \mathbb{R}^m$$

is a Banach space for any product norm;

2.  $X_{\mathcal{O}}$  is nonempty and convex;
3.  $X_{\mathcal{O}}$  is norm-closed in  $E$ ;
4.  $X_{\mathcal{O}}$  is compact in the product topology

$$\tau := \left( \prod_{i=1}^N \sigma(M_i^*, M_i) \right) \times \text{Euclidean topology on } \mathbb{R}^m;$$

*Proof.* Each  $M_i$  is a von Neumann algebra in the OPH patch net, hence a Banach space, so each dual  $M_i^*$  is Banach. A finite product of Banach spaces is Banach, hence so is  $E$ .

For each  $i$ , the state space

$$S_i = \{\omega \in M_i^* : \omega \geq 0, \omega(1) = 1\}$$

is convex and weak\* compact by Banach–Alaoglu, because it is a weak\* closed subset of the dual unit ball. For every pair  $(i, j)$ , restriction along  $M_{ij} \subset M_i$  and  $M_{ij} \subset M_j$  defines affine weak\* continuous maps

$$r_{ij} : S_i \rightarrow S(M_{ij}), \quad r_{ji} : S_j \rightarrow S(M_{ij}).$$

Hence  $X_{\text{ov}}$  is an intersection of affine equalizer sets, so it is convex and  $\tau$ -closed inside  $\prod_i S_i$ . Since  $\prod_i S_i$  is compact and convex in the product weak\* topology,  $X_{\text{ov}}$  is compact and convex as well.

Nonemptiness comes from the OPH global state on the inductive-limit algebra: its restrictions

$$(\omega|_{M_1}, \dots, \omega|_{M_N})$$

belong to  $X_{\text{ov}}$  because the restrictions agree on every overlap by construction.

Each coordinate  $\omega_{i(a)} \mapsto \omega_{i(a)}(O_a)$  is affine and weak\* continuous, since evaluation at a fixed algebra element is weak\* continuous. Thus  $c$  is affine and  $\tau$ -continuous. The graph map

$$G : X_{\text{ov}} \rightarrow E, \quad G(x) := (x, c(x)),$$

is affine and continuous, so its image  $X_{\mathcal{O}} = G(X_{\text{ov}})$  is convex and  $\tau$ -compact.

To see norm-closedness in  $E$ , let  $(x_n, c(x_n)) \rightarrow (x, \ell)$  in norm. Norm convergence implies weak\* convergence on each coordinate, so  $x \in X_{\text{ov}}$  because  $X_{\text{ov}}$  is weak\* closed. Since  $c$  is weak\* continuous,  $c(x_n) \rightarrow c(x)$ , while norm convergence in  $\mathbb{R}^m$  also gives  $c(x_n) \rightarrow \ell$ . Hence  $\ell = c(x)$ , so  $(x, \ell) \in X_{\mathcal{O}}$ .

Finally,  $X_{\mathcal{O}}$  is a compact convex subset of the locally convex topological vector space  $E$  equipped with  $\tau$ .  $\square$

The additional inputs needed for a strange-loop closure theorem are:

1. a closure map  $T$  built from internal operations such as overlap repair, Axiom-3 information reprojection, an explicit collar-recovery interface, and record-sector coarse-graining;
2. a proof that the stronger “observer-supporting” or “selected-world” criteria carve out a nonempty  $T$ -invariant observer-supporting subset of  $X_{\mathcal{O}}$ ;
3. any uniqueness, contraction, or Lyapunov-type stability estimate for that closure map.

The habitat theorem supplies only the ambient Banach/compact-convex setting. It does not define a canonical internal self-map, it does not identify any nonempty invariant observer-supporting subset, and it does not prove uniqueness or stability. In particular, nonemptiness and compact-convexity of the ambient habitat do not by themselves imply the existence of a nonempty invariant observer-supporting sector. This does not erase the exact finite packet branch proved on the consensus surface: there the quotient normal-form map pushes forward to a continuous affine idempotent self-map of the finite packet simplex, with fixed points exactly the packets supported on quotient normal forms. Any further strange-loop reading of the ambient habitat as a closure story for existence is interpretive only and is not part of the recovered theorem package.

## B.1 Additional Conditional Results

Using the theorem classification of Section 3, the broader claims read as follows. “Declared-branch corollary” means proved under the named branch conditions and cited companion results.

| Claim                       | Classification                              | Boundary                                                                                                                                                                                                                                                   |
|-----------------------------|---------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Quantum gravity consistency | Conditional theorem under explicit receipts | The Einstein relation follows from one source-derived common-domain tower carrying the geometric modular, null, event, entropy, certified-tail, coupling, vacuum, and scale premises. Construction and certification of such a tower are work in progress. |

| Claim                                                   | Classification            | Boundary                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|---------------------------------------------------------|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| UV completion / microscopic uniqueness                  | Separate scope boundary   | Fixed-cutoff theorem packages exist, but a unique microscopic representative is not identified. The support-visible BW scaling theorem and the receipt-certified realized combined-zero-obstruction bosonic compact-gauge branch (central or higher-associator strictification plus an allowed trivial-holonomy strict representative) hold under their declared premises; the separate boundary is microscopic representative uniqueness rather than a missing recovered-core lift.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Measurement problem                                     | Separate scope boundary   | Ref. [5] closes the fixed-cutoff central-record / Born-Lüders interface, but a full philosophical or continuum-level measurement closure is not derived here.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Cosmological principle and horizon homogeneity          | Separate scope boundary   | MaxEnt symmetry inheritance is suggestive only; an independent derivation of cosmological isotropy and homogeneity sits outside the recovered-core theorems.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Observer-relative modular ordering                      | Declared-branch corollary | On the $BW_{S_2}$ geometric branch, the cap modular automorphism parameter supplies a dimensionless ordering for the observer's accessible algebra/state pair. Physical time additionally requires an observer-readable transition, event correspondence, and calibrated clock instrument.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Global / operational problem of time                    | Separate scope boundary   | OPH does not prove a separate operational-clock theorem or a global solution of the problem of time.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Black-hole information paradox                          | Separate scope boundary   | Edge-center sectorization and recoverability motivate an internal resolution template. A physical black-hole-information theorem requires the radiation algebra, exterior time calibration, evaporation dynamics, and entropy readout.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Cosmological-capacity closure                           | Conditional corollary     | Conditional on a complete capacity-source antecedent selecting one physical zero of the finite-size slack, a universe-level carrier attachment, stable whole-fiber closure $\mathfrak{F}_{r,0}(D_\star) = \{D_\star\}$ , and horizon-record identification $N_{\text{CRC}} = S_{\text{ds}}$ , one gets $\Lambda_{\text{CRC}} \ell_\star^2 = 3\pi/N_{\text{CRC}}$ . The fixed- $D = 24$ packet is exact inside its declared finite source category. The bounded completion class has incompatible fixed sets. Universal all-rung membership in a complete A1–A3 capacity-source contract and an executable-to-Lean bridge are absent, so direct $N$ is not evaluable on the incomplete source antecedent and the stronger source-class verdict is open. Exact positive compositional completions separately show that the declared finite reserve datum selects no global action or blocked-event semantics. The static-patch SI display additionally uses the selected OPH scale certificate. The uncorrected electroweak/Higgs bridge differs from the weighted Planck base- $\Lambda$ CDM coordinate by 6.6 percent. Separately premised finite-presence and Poisson formulas give retrospective residuals of $-0.63$ and $-0.39$ percent. The finite source selects neither formula, and neither comparison establishes the common-carrier or horizon receipt. |
| Bare cosmological-constant problem from local null data | Separate scope boundary   | The local null-data route determines the Einstein branch only modulo $\Lambda g_{ab}$ and does not derive the screen-capacity identification from bare OPH axioms.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

| Claim                                                                 | Classification                                       | Boundary                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-----------------------------------------------------------------------|------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Magnetic cocharacter and simple-GUT monopole boundary                 | Conditional matter-image and global-form arithmetic  | Complete reversible response and endogenous overlap transport force the local Standard Model gauge Lie algebra. Incidence and target-blind port readback derive the signed response. Under the conditional matrix current and rank-15 matter representation, the common central kernel is $\mathbb{Z}_6$ and the quotient by it is the maximal faithful image. The cover and its $\mathbb{Z}_2$ , $\mathbb{Z}_3$ , and $\mathbb{Z}_6$ quotients carry the same local tensors. The six-axis menu matches the $\mathbb{Z}_6$ quotient only after its coefficient relations are declared, so it does not select the physical global form. The cocharacter $(1, 1, 1/6)$ and its electromagnetic multiple apply to the declared image. The Dirac-pairing commutant theorem selects the electric polarization uniquely inside the declared line lattice. Theta periodicity requires four-dimensional instanton-sector data. The product adjoint removes the simple-GUT X/Y production route, not all possible 't Hooft lines. |
| Gauge-mediated proton-decay boundary                                  | Product-adjoint corollary                            | The connected product adjoint has no X/Y generator, so the standard simple-GUT channel is absent; general proton stability is not claimed.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Proton spin fraction                                                  | Separate scope boundary                              | The proton-spin claim requires nonperturbative quantum chromodynamics.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Dark matter phenomenology                                             | Separate scope boundary                              | Modular-anomaly response laws are conjectural beyond the recovered gravity chain.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Baryon asymmetry scale                                                | Separate scope boundary                              | Suppression-counting estimates are not a substitute for a derived out-of-equilibrium mechanism.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Three generations                                                     | Conditional window and rank-three response candidate | CKM capability and the weak-sector clause give $3 \leq N_g \leq 5$ . Under the named band premises, rank three is selected. Tensoring it with the declared generation table gives a conditional rank-45 candidate whose chirality and diagonal $\mathbb{Z}_6$ action come from the table. A separate local-domain receipt checks a declared tensor-identity operator and conditional gap inheritance without selecting the matter action or transporting the Spin packet. Physical matter-pole and continuum identification, extra-sector completeness, and the charged-lepton and electroweak quantitative branches carry their own declared premises.                                                                                                                                                                                                                                                                                                                                                                  |
| Downstream flavor structure, Koide, and charged-lepton fits           | Separate scope boundary                              | Companion flavor constructions require extra ansätze and sit outside the recovered core.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Edge-to-2D-YM / controlled large- $N_{\text{edge}}$ worldsheet branch | Declared-branch corollary                            | On the stated overlap-gauge, local-Gibbs/MaxEnt, compact-group, and large- $N_{\text{edge}}$ conditions, the edge-sector heat-kernel bridge and controlled Gross–Taylor worldsheet effective description are theorem-level. This is a two-dimensional partition and worldsheet-effective branch, separate from the four-dimensional compact-gauge theorem.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Support-visible compact-gauge Yang–Mills form and mass gap            | Conditional theorem under explicit receipts          | The finite compact-gauge cylinder system has a proof-bearing projective weak-* / GNS extraction. Identification with the four-dimensional Euclidean Yang–Mills state and the equality $\Delta_{\text{YM}} = \Delta_{\text{rep}}$ additionally require the renormalized Yang–Mills, finite ground-state-transform/cross-fiber, transfer/vacuum, OS-regularity/noncollapse, and uniform-gap receipts; extraction alone does not transport the gap.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Critical superstring / worldsheet CFT lift                            | Separate scope boundary                              | Critical-superstring claims, worldsheet CFT closure, anomaly cancellation, and full massless-spectrum matching require extra ingredients outside the recovered core.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Why anything exists / strange-loop closure                            | Separate scope boundary                              | Appendix B provides only the habitat theorem and fixed-point setting. The OPH closure map, invariant observer-supporting sector, and uniqueness/stability proofs are not part of this theorem package.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Fixed-cutoff check-point/restoration/backup                           | Fixed-cutoff theorem                                 | Ref. [5] proves same-interface checkpoint/restoration and backup for observer-accessible event algebras with explicit future-law error control.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Substrate-transfer / stronger observer continuation                   | Separate scope boundary                              | Stronger substrate-selection, redesigned-environment continuation, strange-loop closure, uniqueness, and stability claims are outside this theorem package.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

## C Observer Continuation and Backup

This appendix uses the fixed-cutoff checkpoint/restoration/error/identity theorem package proved in Ref. [5] and summarizes its algebraic interface in observer language. It does not define a strange-loop closure map on an invariant observer-supporting sector or establish uniqueness and stability for such a map. Nor does it extend the fixed-cutoff theorems to continuation across redesigned environments: the proved package is same-interface restoration on observer-accessible event algebras with explicit error control.

### C.1 Observer as Algebraic Pattern

For this appendix, use the support-local algebra-state-record reduct

$$O_{\text{red}} = (P, \mathcal{A}(P), \rho, R),$$

where  $P$  is a support-screen patch,  $\mathcal{A}(P)$  is its local algebra,  $\rho$  is the local state, and  $R$  is the record algebra. The full operational observer also requires overlap interface algebras and restriction maps, allowed update or repair instruments, durable checkpointed records, and feedback/readback that can affect subsequent admissible behavior. Records are carried exactly by central record projectors and, on practical readout surfaces, by approximately commuting projectors in overlap centers, so they are shareable without violating no-cloning constraints for generic quantum states.

Ref. [5] proves a fixed-cutoff observer-facing measurement interface: a finite central record algebra, Born probabilities for its event projectors, and Lüders conditioning on that same commuting record algebra. This appendix uses that fixed-cutoff measurement package in observer language. If a practical readout uses projectors that are  $\delta_{\text{rec}}$ -close in operator norm to that central reference algebra and the accessible state is perturbed by at most  $\varepsilon$  in trace norm, each declared elementary record-event probability shifts by at most  $\varepsilon + \delta_{\text{rec}}$ .

### C.2 Markov Collar Factorization

For a collar tripartition  $A$ - $B$ - $D$ , the exact Markov normal form is used only when  $I(A : D|B) = 0$  holds literally, or along a controlled fixed-collar family for which the exact-Markov replacement modulus  $\delta_{A:B:D}^{\text{M}}(\varepsilon) \rightarrow 0$ . In that exact or controlled limit one obtains

$$\rho_{ABD} = \bigoplus_{\alpha} p_{\alpha} \rho_{Ab_L}^{(\alpha)} \otimes \rho_{b_R D}^{(\alpha)}.$$

The sector label  $\alpha$  is classical center data. This decomposition is the mathematical basis for extracting an interior observer state with a controlled boundary interface. Small CMI by itself supplies a recovered comparison state, while an exact Markov state requires the exact or controlled limiting premise stated above.

### C.3 Checkpoint and Restoration Map

A checkpoint is the tuple

$$\mathcal{C} = (R, \alpha, \rho_{\text{int}}^{(\alpha)}),$$

with  $\rho_{\text{int}}^{(\alpha)}$  the interior state after fixing  $\alpha$ . Given a compatible target environment state  $\sigma_{\text{env}}^{(\alpha)}$ , a restored state is

$$\rho_{\text{new}}^{(\alpha)} = \rho_{\text{int}}^{(\alpha)} \otimes \sigma_{\text{env}}^{(\alpha)}.$$

For approximate Markov collars, recovery is controlled by the standard bound

$$\|\rho_{ABD} - (\text{id}_A \otimes \mathcal{R}_{B \rightarrow BD})(\rho_{AB})\|_1 \leq 2\sqrt{1 - e^{-\varepsilon}} \leq 2\sqrt{\varepsilon}.$$

Here  $\varepsilon$  is measured in nats. This gives quantitative trace-distance control on the recovered comparison state under finite CMI (finite collar error). The exact HJPW splice form instead requires literal exact Markovity or the separate fixed-collar replacement modulus tending to zero, in each case together with the Markov-split alignment hypothesis of Section 2.3 identifying the HJPW factors with the preselected edge factors. Interpreting either control as observer continuation requires additional modeling assumptions beyond the bound itself.

The fixed-cutoff microphysics claim is stronger than the bare recoverability estimate written above. In Ref. [5], exact checkpoint restoration preserves the full future law of observer-accessible events, while an  $\varepsilon$ -accurate restoration changes that future law by at most  $\varepsilon$  in total variation. The stronger strange-loop / substrate-transfer closure story is outside this theorem.

## C.4 Physical Meaning

At fixed cutoff, Ref. [5] proves a checkpoint/restoration theorem and backup corollary for observer-accessible event algebras. The proved package stops at same-interface backup/restoration with explicit future-law error control; stronger continuation and substrate-selection claims are outside it. This appendix states the observer-facing meaning of that theorem and its algebraic prerequisites. The stronger substrate-selection, strange-loop closure, uniqueness, and stability package is separate from the fixed-cutoff theorem.

# D Cosmology, Horizons, and Modular-Anomaly Continuations

This appendix carries the cosmology- and continuation-facing technical statements for the synthesis paper. They do not enlarge the recovered core; they make the structural continuation boundary explicit on the synthesis surface.

## D.1 Observer Registry and Clock-Naturality Certificate Contract

For finite exported observer systems, a theorem-faithful evidence bundle separates executor meta-data from observer semantics. It contains:

- (i) a semantic history DAG or labeled poset  $H_O = (E_O, \preceq_O, \ell_O)$ , with event keys computed from semantic payload, observer token, visible footprint, and semantic parents;
- (ii) a provenance envelope containing worker ID, repair iteration, retry count, queue position, timestamp, and packet latency, none of which may enter the semantic event key unless declared as physical input;
- (iii) local observer registry groupoids with disjoint patch-observer, cap-observer, and future-observer namespaces, explicit birth-event classes, continuation arrows, split/merge lineage arrows, and overlap functors preserving those data;
- (iv) a global registry descent certificate: cocycle agreement, trivial monodromy, no duplicate global observer identifiers, no mixed patch/cap/future namespace collisions, and no reuse of local anchor-patch identifiers as local observer indices;

- (v) observer-algebra extraction data

$$\text{ObsAlg}(\hat{q}_{\text{nf}}) = (\mathcal{A}_O, \omega_O, \mathcal{R}_O, \dots)$$

whose outputs are state-preserving isomorphic for history-equivalent or implementation-equivalent terminal augmented normal forms;

- (vi) a support-cap chart map  $J_O$  natural with those observer-algebra isomorphisms; and
- (vii) a clock instrument  $\Theta_O$  with affine reparameterization and residual bound for state, calibration, and record-readout errors.

**Theorem D.1** (Finite observer-clock certificate soundness). *On a finite exported model, suppose the augmented repair relation on  $\hat{Q} = \{(q, [H])\}$  terminates, terminal states are exactly the consistent physical states with completed semantic records, every augmented critical pair has a history-coherent join, implementation steps are linearizable to semantic commits or stutters, registry descent passes the groupoid cocycle and monodromy checks, and the observer-algebra/chart/clock maps satisfy the state-preserving and affine-residual certificates above. Then worker count, queue order, retry delivery, scheduler counters, and packet latency that is not declared physical input do not change the observer-readable history class, registry class, or clock law beyond the declared affine clock residual.*

*Proof.* Termination plus the augmented local diamond gives confluence by Newman’s lemma for terminal history classes rather than only physical quotient states. Linearizability says every implementation execution projects to the same semantic relation up to stutters, so executor metadata cannot create new semantic events. Registry descent identifies the same global observer groupoid across local presentations. State-preserving observer-algebra extraction and natural support-cap charts transport the modular flow along the same observer identity, while the clock instrument certificate bounds the operational readout after the declared affine reparameterization.  $\square$

**Remark D.2** (Falsifiers). *The certificate fails if semantic event identifiers depend on worker IDs, timestamps, queue positions, repair iteration counters, or retries; if duplicate retry delivery creates two visible record commits; if two namespaces collide in the global registry; if registry cocycles or lineage arrows fail on overlaps; if histories from two executors are not labeled-poset isomorphic; or if clock residuals exceed the declared bound. These are public evidence-bundle failures that exceed field-name mismatches.*

## D.2 Cosmological Principle

**Lemma 4.1 (MaxEnt symmetry inheritance).** Let  $G$  act on the screen by automorphisms  $\alpha_g$ . If the constraint set is  $G$ -invariant and von Neumann entropy is  $G$ -invariant, then the MaxEnt optimizer  $\omega$  can be taken  $G$ -invariant. If the optimizer is unique, it is automatically  $G$ -invariant.

**Theorem 4.2 (Isotropy from  $\text{SO}(3)$ -invariant constraints).** Under  $\text{SO}(3)$ -invariant constraints and MaxEnt uniqueness, the reference state satisfies

$$\omega \circ \alpha_g = \omega \quad \forall g \in \text{SO}(3).$$

The corresponding stress tensor has perfect-fluid form:

$$\langle T_{ab} \rangle = \rho u_a u_b + p(g_{ab} + u_a u_b).$$

**Theorem 4.3 (Schur-type homogeneity).** Let  $(\Sigma, h_{ij})$  be a three-dimensional Riemannian manifold. If at every point  $p \in \Sigma$ , the curvature tensor is  $\text{SO}(3)$ -invariant, then

$$R_{ijkl}(p) = K(p)(h_{ik}h_{jl} - h_{il}h_{jk}),$$

and the second Bianchi identity forces  $\nabla_m K = 0$ , so  $K$  is constant.

**Corollary 4.4.** If OPH supplies isotropy for all observers, spatial geometry is a constant-curvature space form:  $S^3$ ,  $\mathbb{R}^3$ , or  $H^3$ .

**Theorem 4.5 (FLRW emergence).** With the semiclassical Einstein equation from the declared finite generalized-entropy, stationarity, stress, and continuum interfaces, a perfect-fluid stress tensor from the rotationally invariant information-projection branch, and positive  $\Lambda$  from finite screen capacity, the emergent geometry is FLRW:

$$ds^2 = -dt^2 + a(t)^2 \left( \frac{dr^2}{1 - kr^2} + r^2 d\Omega^2 \right),$$

with Friedmann equations

$$H^2 + \frac{k}{a^2} = \frac{8\pi G}{3}\rho + \frac{\Lambda}{3}, \quad \dot{H} - \frac{k}{a^2} = -4\pi G(\rho + p).$$

### D.3 Horizon-Problem Control

**Theorem 6.1 (Conditional collar-CMI bound).** For a finite-range Gibbs tripartition  $A$ - $B$ - $D$  with collar width  $\delta$ , assume the uniform strong conditional matrix-mixing premise of Theorem 2.5. Then

$$I(A : D \mid B) \leq c \cdot |\partial C|_{\text{UV}} \cdot e^{-\delta/\xi},$$

with the explicit constants stated there. Ordinary two-point clustering does not supply the premise. On the exact central-interface branch,  $I(A : D \mid B) = 0$ .

**Corollary 6.2.** For bounded observable  $O$  with  $\|O\| \sim 1$ ,

$$|\Delta\langle O \rangle| \leq 2\sqrt{1 - e^{-\varepsilon}} \leq 2\sqrt{\varepsilon}$$

when  $\varepsilon$  is CMI in nats. On the CMB anisotropy target  $\delta T/T \lesssim 10^{-5}$ , this gives the familiar  $\varepsilon$ -scale bound used in the horizon-problem bookkeeping:

$$\varepsilon \lesssim 2.5 \times 10^{-11} \text{ nats}, \quad \frac{\delta}{\xi} \gtrsim \ln \left( \frac{c \cdot |\partial C|_{\text{UV}}}{\varepsilon_{\text{max}}} \right) \approx 24.4 + \ln(c|\partial C|_{\text{UV}}).$$

The number 24.4 is only the unit-prefactor value. The boundary count cannot be dropped. With  $\xi \approx \sqrt{P} \ell_\star \approx 1.28 \ell_\star$ , where  $\ell_\star$  is displayed as the usual Planck length after the selected scale certificate, the corresponding collar condition is

$$\delta_{\text{CMB}} \gtrsim 1.28 \ell_\star [24.4 + \ln(c|\partial C|_{\text{UV}})].$$

This is a state-recovery benchmark for a bounded observable, not a derivation of CMB anisotropy. On a continuum family the separate rate condition  $\delta/\xi - \log |\partial C|_{\text{UV}} \rightarrow +\infty$  is required.



**Theorem 6.3 (Homogeneity as the MaxEnt default).** If MaxEnt constraints are uniform across UV cells with no marked points, the MaxEnt state is invariant under triangulation automorphisms. In the continuum limit this gives  $\text{SO}(3)$  invariance.

**Reading rule.** The cosmological-principle and horizon-homogeneity package is a branch-local theorem package on the realized symmetric MaxEnt branch. The exact theorem-level content is:

1.  $\text{SO}(3)$ -invariant constraint data plus MaxEnt uniqueness force an isotropic reference state and a perfect-fluid stress tensor.
2. If the same isotropy condition holds for all observers, the Schur/Bianchi argument forces constant-curvature spatial slices.
3. Combined with the semiclassical Einstein branch and positive  $\Lambda$  from finite screen capacity, the metric takes FLRW form.
4. Markov control supplies an explicit patch-overlap anisotropy bound and the collar benchmark used in the CMB homogeneity bookkeeping.

The package does not prove from the bare OPH axioms alone that the realized cosmological branch must satisfy  $\text{SO}(3)$ -invariant constraint data, that MaxEnt uniqueness holds in every cosmological sector, or that the observed universe saturates the displayed CMB benchmark.

## D.4 Conditional Screen-Spectrum Continuation

**Theorem 6.4 (Conditional OPH screen spectrum).** Let  $(\mathcal{S}_r)$  be a cofinal finite spherical OPH screen system with a schedule-independent quotient-normal-form scalar  $q_r$ , removal of the background and dipole sector, and a target-free positive quadratic repair operator  $K_r$ . If  $K_r \rightarrow K$ , the source-selected finite scalar release energy satisfies  $2E_{q,r}^{\text{src}}/d_r \rightarrow A_q$ , and local MaxEnt is imposed at fixed expected quadratic release energy, then  $q_r$  converges in finite harmonic distributions to the centered Gaussian screen field with covariance  $A_q K^{-1}$ . Exact per-sample release energy would instead give a microcanonical ellipsoid. If the certified repair-scale measure is  $t^{\theta/2} dt/\Gamma(1 + \theta/2)$ , then

$$K_{\text{asy}} = (-\Delta_{S^2})^{1+\theta/2}, \quad C_{\ell,\text{asy}}^q = A_q[\ell(\ell+1)]^{-1-\theta/2},$$

as the large- $\ell$  asymptotic model. The theorem-grade finite- $\ell$  conformal-shell family uses the normalized gamma-ratio precision

$$\kappa_\ell(\theta) = \frac{\Gamma(\ell+2+\theta/2)}{\Gamma(\ell-\theta/2)},$$

with

$$C_\ell^q = A_q \frac{\Gamma(\ell-\theta/2)}{\Gamma(\ell+2+\theta/2)}.$$

If  $q$  is additionally certified as the thin-shell pullback of a homogeneous curvature field with  $\Delta_\zeta^2(k) = A_\zeta(k/k_\star)^{-\theta}$ , this same operator supplies the exact finite- $\ell$  lift; the pure fractional Laplacian is only the asymptotic scaffold. Suppose the same source construction emits a strongly continuous full-collar survival cocycle with generator density  $P_\star/24$ , together with the orientation-reversal half-collar identity. Then

$$\theta = \frac{P_\star}{48}, \quad n_s = 1 - \frac{P_\star}{48}, \quad \kappa_{\text{rep}}^{\text{edge}} = \frac{P_\star}{48(P_\star - \varphi)}.$$

A finite one-step survival value determines  $\theta$  through  $-\log u_q(\log b)/\log b$ ; it does not supply the infinitesimal generator receipt. If a scale-natural source embedding transports this cocycle to the physical covariance through  $D_s^{-1}C_\zeta D_s = e^{-\theta s}C_\zeta$ , the source family is  $\Delta_\zeta^2(k) = A_\zeta(k/k_\star)^{-\theta}$ . A single shell has an infinite-dimensional radial kernel, including positive ambiguities. Complete radial cross-covariances provide the independent tomography route.

**Receipt boundary.** The conditional theorem fixes the exact angular family. Its radial theorem gives physical source-dilation and cross-covariance tomography as separate uniqueness routes. A finite OPH source construction that emits the primitive collar ensemble, full-collar generator density, half-collar identity, conformal precision, and physical dilation-intertwiner or tomography receipt is work in progress. Fawzi–Renner/Markov-collar observable control supplies upper bounds on release observables; it is not an amplitude equality. Finite-width windows require an exact Bessel-kernel error certificate. Physical temperature and polarization transfer and a frozen likelihood contract are work in progress.

## D.5 Cosmological-Constant / Screen-Capacity Closure

**Theorem 6.5 (Local/global cosmological-capacity closure).** On the gravity branch, assume the local Einstein equation has been recovered only modulo  $\Lambda g_{ab}$ , stable direct public-record closure at a carrier dimension

$$\mathfrak{F}_{r,0}(D_\star) = \{D_\star\}, \quad N_{\text{CRC}} = \log D_\star,$$

and the independent horizon–record identification readout

$$N_{\text{CRC}} = S_{\text{dS}},$$

the standard de Sitter entropy relation

$$S_{\text{dS}} = \frac{A_{\text{dS}}}{4G} = \frac{3\pi}{G\Lambda},$$

and the standard de Sitter static-patch formulas

$$r_{\text{dS}} = \sqrt{\frac{3}{\Lambda}}, \quad t_\Lambda = \frac{r_{\text{dS}}}{c}.$$

Then:

1. local null data determine the Einstein branch only modulo  $\Lambda g_{ab}$ ;
2. with the selected scale certificate, the same branch has the global display

$$G_{ab} + \frac{3\pi}{GN_{\text{CRC}}} g_{ab} = 8\pi G \langle T_{ab} \rangle;$$

3. the same capacity closure fixes the entropy and dimensionless capacity relations

$$S_{\text{dS}} = N_{\text{CRC}}, \quad A_{\text{dS}} = 4GN_{\text{CRC}}, \quad r_{\text{dS}} = \sqrt{\frac{3}{\Lambda}}, \quad t_\Lambda = \frac{r_{\text{dS}}}{c};$$

4. the observed cosmic age is a downstream FLRW benchmark rather than an additional theorem output.

So the cosmological-constant package is one local/global theorem chain rather than a split local argument plus an unrelated global readout.

**Scope boundary.** The hypotheses are the local Einstein branch, stable whole-fiber public-record closure, horizon-record identification, the de Sitter entropy relation, and the standard static-patch formulas. The local null-data route does not by itself determine the global capacity; that value is fixed only on a stable direct-capacity branch with a physical finite-size selector and horizon-record identification. The conditional capacity closure fixes  $\Lambda_{\text{CRC}}\ell_\star^2 = 3\pi/N_{\text{CRC}}$ ; the SI static-patch scale additionally requires the selected scale certificate.

**Capacity self-closure target.** At finite regulator use the frozen capacity-carrier dimension  $D$ , with  $N = \log D$ . For each terminal state in the unclosed fiber, form compatible global sections of the local record-atom diagram, retain the endogenously reachable sections, freeze the authorized publicness policy, and use the source-derived joint checkpoint kernels. Their compound confusability graph  $G_q$  gives

$$M_0(q) = \alpha(G_q), \quad \mathfrak{F}_{r,\varepsilon}(D) = \{M_\varepsilon(q) : q \in \tilde{\Omega}_{r,D}\}.$$

The scalar active map exists only when the whole nonempty terminal fiber has one common defined value. Fixed checkpoint projectors are not the capacity: a cyclic permutation fixes only constant functions while preserving every record label. Local checkpoint marginals are also insufficient to determine the required joint channel.

A faithful capacity-carrier representation gives  $M_\varepsilon(q) \leq D$ , and stable closure is

$$\mathfrak{F}_{r,0}(D_\star) = \{D_\star\}.$$

Equality at zero error forces a rank-one complete public record basis. With a confusability-reflecting embedding, capacity extension is monotone, so a total scalar deflationary map on a declared finite chain reaches its greatest fixed point from the top. Fixed- $D$  refinement then stabilizes exactly. None of these implications supplies cutoff independence or a unique physical zero of

$$s(D) = \log D - \log M_0(D).$$

The fixed- $D = 24$  packet realizes the complete terminal fiber inside its declared finite source category. A bounded all-rung counterfamily shares base agreement, positivity, the carrier bound, and executable finite incidence, action, projection, composition, extension, and sewing controls while giving different exact zero sets. Lean proves the all-rung arithmetic consequence of the base, positivity, and carrier-bound structure. The executable certificate checks the remaining finite controls, and an independent simulator reconstructs held-out finite rungs. The complete capacity-indexed A1–A3 packet lift is absent, so this result does not settle the full source class. A positive direct result also requires a source law selecting the continuation when necessary and a universe-level carrier attachment.

The finite kernel

$$K_r(D, m) = |\{q \in \tilde{\Omega}_{r,D} : M_0(q) = m\}|$$

distinguishes one closed state from deterministic whole-fiber closure. A diagonal argmax is an extra selector. After direct active closure, the horizon-record identification yields the area-law display, while the common screen/electroweak load-carrier identification yields the electroweak bridge. The operational scale  $\rho_{\text{op}}$  is an independent estimator tested through  $\log M_0 - \pi/\rho_{\text{op}}^2$ ; it does not define the map. The  $\Lambda$ -located value remains a measured-side comparison coordinate.

**Capacity normalization.** The branch uses  $N_{\text{scr}}$  as the de Sitter entropy capacity. The bare horizon ratio is

$$N_{\text{patch}} = \left( \frac{r_{\text{dS}}}{\ell_P} \right)^2,$$

so

$$N_{\text{scr}} = \pi N_{\text{patch}} = \frac{3\pi}{\Lambda \ell_P^2}.$$

For the observed late-time scale,  $N_{\text{patch}} \simeq 1.05 \times 10^{122}$  and  $N_{\text{scr}} \simeq 3.31 \times 10^{122}$ . These are benchmark displays, not high-precision inputs for the Newton row.

## D.6 Black-Hole Structural Package and Continuation Boundary

The AMPS-style information trilemma assumes:

1. an outgoing mode  $B$  entangled with an interior partner  $A$  (smooth horizon),
2. late radiation  $B$  entangled with early radiation  $R$  (unitarity), and
3. monogamy of entanglement.

The false assumption in this setting is the naive tensor factorization

$$\mathcal{H} \stackrel{?}{=} \mathcal{H}_{\text{inside}} \otimes \mathcal{H}_{\text{outside}} \otimes \mathcal{H}_{\text{radiation}}.$$

**Theorem 7.1 (Edge-center black-hole decomposition).** On the fixed-cutoff edge-center collar branch, assume the black-hole collar state lies in the exact Markov class or in the idealized exact-recoverability limit used elsewhere in the paper. Then the collar decomposition gives

$$\rho_{A_\delta B_\delta D_\delta} = \bigoplus_{\alpha} p_{\alpha} \left( \rho_{A_\delta b_L^\alpha} \otimes \rho_{b_R^\alpha D_\delta} \right).$$

The glue between inside and outside is the edge-sector label  $\alpha$  in the center. Given  $\alpha$ , inside and outside factorize.

**Theorem 7.2 (Hawking/KMS normalization).** On the geometric modular branch where the horizon generator carries the standard  $2\pi$  KMS normalization, the outside algebra is in a KMS state at inverse temperature

$$\beta = \frac{2\pi}{\kappa}$$

with respect to the horizon Killing generator.

**Corollary 7.3 (Recoverability-style interior encoding).** On the same fixed-cutoff collar carrier, when  $I(A_\delta : D_\delta \mid B_\delta) \leq \varepsilon$ , there exists a recovery channel  $\mathcal{R}_{B_\delta \rightarrow A_\delta B_\delta}$  such that

$$\|\rho_{A_\delta B_\delta D_\delta} - (\mathcal{R}_{B_\delta \rightarrow A_\delta B_\delta} \otimes \text{id}_{D_\delta})(\rho_{B_\delta D_\delta})\|_1 \leq 2\sqrt{1 - e^{-\varepsilon}} \leq 2\sqrt{\varepsilon}.$$

Here  $\varepsilon$  is in nats. So the interior collar data are encoded in the exterior collar together with the shared cut data, rather than supplied by an independent tensor factor.

**Corollary 7.4 (Discrete area spectrum and Schwarzschild transition identity).** The same edge-center package carries a discrete area operator

$$L_C = \sum_{\alpha} (\log d_{\alpha}) P_{\alpha}, \quad A_{\alpha} = 4G_{\text{geom}} \log d_{\alpha} = 4\ell_{\star}^2 \ln d_{\alpha},$$

and for a Schwarzschild sector transition  $d \rightarrow d'$  one has

$$\Delta(Mc^2) = k_B T_H \ln(d'/d).$$

On the emission branch  $d' = d/k$  with  $k > 1$ , the emitted line energy magnitude is

$$E_{\text{emit}} = k_B T_H \ln k.$$

**Scope boundary.** The retained structural theorem package is exactly Theorem 7.1, Theorem 7.2, Corollary 7.3, and Corollary 7.4. Integer-transition combs, QNM selectors, Page-type linewidth estimates, PBH burst templates, Kerr/LIGO horizon spectroscopy templates, and any Page-curve or island closure are outside that structural core. Those items require continuation inputs such as an integer-transition selection rule, a semiclassical evaporation-power model, a QNM/transition identification, greybody matching, or an explicit island prescription.

The same interpretation rule applies here. A finite capacity register is not an exterior mass, a stored record is not Hawking radiation, a finite repair spectrum is not a GR quasinormal-mode spectrum, and an exact finite recovery threshold is not a Page time unless the corresponding independent physical bridge supplies source-separated readout, calibration, residuals, controls, and fixed validation references.

## D.7 Modular-Anomaly Continuation

**Theorem 9.1 (Modular additivity defect).** Define the collar modular additivity defect by

$$\Delta K_{\delta} := K_{ABD} - K_{AB} - K_{BD} + K_B.$$

Then

$$\langle \Delta K_{\delta} \rangle_{\omega} = -I(A : D \mid B)_{\omega}.$$

This is a scalar recovery identity. It can enter the modular-anomaly dark-sector continuation only after a separate matching theorem or hypothesis identifies it with an anomalous local modular-energy source. By itself it constructs neither a rank-two tensor nor a dark-matter theorem.

The continuation therefore reserves an anomalous tensor slot only for a finite covariant source packet that supplies source localization, CMI-to-modular-source matching when that route is used, rank-two reconstruction, conservation, normalization, and universal coupling. Once those receipts pass, the modified Einstein equation is

$$G_{ab} + \Lambda g_{ab} = 8\pi G (\langle T_{ab} \rangle + \langle T_{ab}^{\text{anom}} \rangle),$$

with rest-frame normalization

$$\langle T_{00}^{\text{anom}} \rangle := \frac{15}{8\pi^2} \frac{\delta \langle K_C^{\text{anom}} \rangle}{\ell^4}.$$

On that receipt-certified continuation surface the tensor gravitates and is covariantly conserved. Calling it dark additionally requires the no-electromagnetic-coupling receipt. Central or recoverability data alone do not establish any of these properties.

The dark-sector continuation maps scalar repair occupation to a canonical pair  $(n, \theta)$ . Conditional on the proposed action, its dilute homogeneous phase scales as pressureless matter and its cubic condensed phase gives  $a_R = \sqrt{a_b a_0}$  in the spherical deep regime. The action supplies a repair current and metric stress rather than a virtual source.

**Reading rule.** The scalar modular-additivity identity is a recovery theorem. The canonical repair pair, baryonic source map, dimensional couplings, complete constitutive law, relativistic refinement limit, abundance, and physical likelihoods are work in progress. Every cosmological quantity on this surface is diagnostic.

## References

- [1] B. Müller, D. Matscheko, and J. Hill, *Observation-Determined Normal Forms: Stability, Obstructions, and Refinement in Constraint and Rewrite Systems*, 2026. Available at [https://github.com/FloatingPragma/observer-patch-holography/blob/main/extra/observable\\_normal\\_forms.pdf](https://github.com/FloatingPragma/observer-patch-holography/blob/main/extra/observable_normal_forms.pdf).
- [2] B. Müller, A. Osika, M. Ponder, K. Xue, P. Nguyen, M. A. Visser, and D. Matscheko, *Recovering Observer Spacetime and Einstein Dynamics from Overlap Consistency*, 2026. Available at [GitHub PDF](#).
- [3] B. Müller, A. Osika, M. Ponder, K. Xue, P. Nguyen, M. A. Visser, and D. Matscheko, *Deriving Standard Model Gauge Structure from Observer Overlap Consistency*, 2026. Available at [GitHub PDF](#).
- [4] B. Müller, K. Xue, K. A. Anirudha, D. Matscheko, and J. Hill, *Reality as a Consensus Protocol: The Fixed-Point Computation That Implements Physics*, 2026. Available at [https://github.com/FloatingPragma/observer-patch-holography/blob/main/paper/reality\\_as\\_consensus\\_protocol.pdf](https://github.com/FloatingPragma/observer-patch-holography/blob/main/paper/reality_as_consensus_protocol.pdf).
- [5] B. Müller, A. Osika, K. Xue, B. Cassie, M. A. Visser, and D. Matscheko, *Federated Echoshedral Screen Microphysics: Patch Hardware, Records, and Observer Synchronization in OPH*, 2026. Available at [https://github.com/FloatingPragma/observer-patch-holography/blob/main/paper/screen\\_microphysics\\_and\\_observer\\_synchronization.pdf](https://github.com/FloatingPragma/observer-patch-holography/blob/main/paper/screen_microphysics_and_observer_synchronization.pdf).
- [6] B. Müller, A. Osika, M. Ponder, K. Xue, M. A. Visser, and D. Matscheko, *Deriving the Particle Zoo from Observer Consistency*, 2026. Available at [https://github.com/FloatingPragma/observer-patch-holography/blob/main/paper/deriving\\_the\\_particle\\_zoo\\_from\\_observer\\_consistency.pdf](https://github.com/FloatingPragma/observer-patch-holography/blob/main/paper/deriving_the_particle_zoo_from_observer_consistency.pdf).
- [7] B. Müller, A. Osika, and D. Matscheko, *Theoretical Bounds on  $\chi_\nu$  in Observer-Patch Holography*, 2026. Available at [https://github.com/FloatingPragma/observer-patch-holography/blob/main/extra/chi\\_nu\\_susceptibility\\_bounds.pdf](https://github.com/FloatingPragma/observer-patch-holography/blob/main/extra/chi_nu_susceptibility_bounds.pdf).
- [8] B. Müller and D. Matscheko, *Observer-Patch Holography and the Dark Matter Phenomenon*, 2026. Available at [https://github.com/FloatingPragma/observer-patch-holography/blob/main/cosmology/oph\\_dark\\_matter\\_paper.pdf](https://github.com/FloatingPragma/observer-patch-holography/blob/main/cosmology/oph_dark_matter_paper.pdf).
- [9] B. Müller and J. Hill, *Explaining the Yang–Mills Mass Gap with Observer-Patch Repair Dynamics*, 2026. Available at [https://github.com/FloatingPragma/observer-patch-holography/blob/main/extra/yang\\_mills\\_gap\\_clay\\_problem.pdf](https://github.com/FloatingPragma/observer-patch-holography/blob/main/extra/yang_mills_gap_clay_problem.pdf).
- [10] B. Müller, *Observer-Patch Holography as a String-Vacuum Selector: Observers, Clocks, Edge Strings, and the Bouchard-Donagi Witness*, 2026. Available at [https://github.com/FloatingPragma/observer-patch-holography/blob/main/extra/observer\\_patch\\_holography\\_as\\_string\\_vacuum\\_selector.pdf](https://github.com/FloatingPragma/observer-patch-holography/blob/main/extra/observer_patch_holography_as_string_vacuum_selector.pdf).

- [11] B. Müller, *The de Sitter Time-Advance Sign from a Finite Screen with Fixed Capacity*, 2026. Available at [https://github.com/FloatingPragma/observer-patch-holography/blob/main/extra/de\\_sitter\\_time\\_advance\\_sign\\_from\\_fixed\\_screen\\_capacity.pdf](https://github.com/FloatingPragma/observer-patch-holography/blob/main/extra/de_sitter_time_advance_sign_from_fixed_screen_capacity.pdf).
- [12] B. Müller, *OPH-FPE: finite simulator and receipt engine for Observer-Patch Holography physics experiments*, 2026. Available at <https://github.com/muellerberndt/oph-physics-sim>.
- [13] Pragma Research, *Observer-Patch Holography: machine-checked proofs and executable certificates*, 2026. Available at <https://github.com/FloatingPragma/observer-patch-holography>.
- [14] Pragma Research, *Frozen Prediction Register and Primitive Twelve-Port Prediction Receipt*, 2026. Immutable source receipt: [https://github.com/FloatingPragma/observer-patch-holography/blob/66176656dc1143f9ec50ba1a6e409c403545857f/code/a5\\_fingerprint/runtime/spin\\_six\\_primitive\\_port\\_prediction\\_receipt.json](https://github.com/FloatingPragma/observer-patch-holography/blob/66176656dc1143f9ec50ba1a6e409c403545857f/code/a5_fingerprint/runtime/spin_six_primitive_port_prediction_receipt.json). Append-only custody packet: [https://github.com/FloatingPragma/oph-meta/tree/8cc5261653e37cbca0e6017fcc95a9fe7f649963/falsification/frozen\\_targets/fz11\\_2026-07-31](https://github.com/FloatingPragma/oph-meta/tree/8cc5261653e37cbca0e6017fcc95a9fe7f649963/falsification/frozen_targets/fz11_2026-07-31).
- [15] L. Onsager, “Reciprocal Relations in Irreversible Processes. I,” *Physical Review* **37**, 405–426 (1931), <https://doi.org/10.1103/PhysRev.37.405>.
- [16] M. S. Green, “Markoff Random Processes and the Statistical Mechanics of Time-Dependent Phenomena. II,” *Journal of Chemical Physics* **22**, 398–413 (1954), <https://doi.org/10.1063/1.1740082>; R. Kubo, “Statistical-Mechanical Theory of Irreversible Processes. I,” *Journal of the Physical Society of Japan* **12**, 570–586 (1957), <https://doi.org/10.1143/JPSJ.12.570>.
- [17] I. Csiszár, “I-Divergence Geometry of Probability Distributions and Minimization Problems,” *The Annals of Probability* **3**, 146–158 (1975), <https://doi.org/10.1214/aop/1176996454>.
- [18] J. E. Marsden and M. West, “Discrete Mechanics and Variational Integrators,” *Acta Numerica* **10**, 357–514 (2001), <https://doi.org/10.1017/S096249290100006X>.
- [19] I. Esteban, M. C. Gonzalez-Garcia, M. Maltoni, I. Martinez-Soler, J. P. Pinheiro, and T. Schwetz, *NuFit-6.0: Updated global analysis of three-flavor neutrino oscillations*, JHEP **12** (2024) 216, arXiv:2410.05380, with the NuFIT 6.1 (2025) profile-table release at <https://www.nu-fit.org/?q=node/309>.
- [20] M. H. A. Newman, “On theories with a combinatorial definition of ‘equivalence’,” *Ann. of Math. (2)* **43** (1942), 223–243.
- [21] E. H. Lieb and D. W. Robinson, “The finite group velocity of quantum spin systems,” *Commun. Math. Phys.* **28** (1972), 251–257.
- [22] J. J. Bisognano and E. H. Wichmann, “On the duality condition for a Hermitian scalar field,” *J. Math. Phys.* **16** (1975), 985–1007.
- [23] J. J. Bisognano and E. H. Wichmann, “On the duality condition for quantum fields,” *J. Math. Phys.* **17** (1976), 303–321.
- [24] R. Brunetti, D. Guido, and R. Longo, “Modular Structure and Duality in Conformal Quantum Field Theory,” *Commun. Math. Phys.* **156** (1993), 201–219, arXiv:funct-an/9302008.

- [25] H.-W. Wiesbrock, “Half-Sided modular inclusions of von-Neumann-Algebras,” *Commun. Math. Phys.* **157** (1993), 83–92.
- [26] H. Araki and L. Zsidó, “Extension of the structure theorem of Borchers and its application to half-sided modular inclusions,” *Rev. Math. Phys.* **17** (2005), 491–543, arXiv:math/0412061.
- [27] D. Guido, R. Longo, and H.-W. Wiesbrock, “Extensions of Conformal Nets and Superselection Structures,” *Commun. Math. Phys.* **192** (1998), 217–244, arXiv:hep-th/9703129.
- [28] H.-W. Wiesbrock, “Modular Intersections of von-Neumann-Algebras in Quantum Field Theory,” *Commun. Math. Phys.* **193** (1998), 269–285.
- [29] W. G. Unruh, “Notes on black-hole evaporation,” *Phys. Rev. D* **14** (1976), 870–892.
- [30] T. Jacobson, “Thermodynamics of spacetime: The Einstein equation of state,” *Phys. Rev. Lett.* **75** (1995), 1260–1263, doi:10.1103/PhysRevLett.75.1260, arXiv:gr-qc/9504004.
- [31] T. Jacobson, “Entanglement equilibrium and the Einstein equation,” *Phys. Rev. Lett.* **116** (2016), 201101, doi:10.1103/PhysRevLett.116.201101, arXiv:1505.04753.
- [32] G. W. Gibbons and S. W. Hawking, “Cosmological event horizons, thermodynamics, and particle creation,” *Phys. Rev. D* **15** (1977), 2738–2751, doi:10.1103/PhysRevD.15.2738.
- [33] Y. Chen, D. Stanford, H. Tang, and Z. Yang, “Negative shocks versus static patch holography,” arXiv:2607.14042 [hep-th] (2026). <https://arxiv.org/abs/2607.14042>
- [34] Planck Collaboration, “Planck 2018 results. VI. Cosmological parameters,” *Astron. Astrophys.* **641** (2020), A6, doi:10.1051/0004-6361/201833910, arXiv:1807.06209.
- [35] R. Bousso, Z. Fisher, J. Koeller, S. Leichenauer, and A. C. Wall, “Proof of the quantum null energy condition,” *Phys. Rev. D* **93** (2016), 024017, arXiv:1509.02542.
- [36] R. Bousso, Z. Fisher, S. Leichenauer, and A. C. Wall, “Quantum focusing conjecture,” *Phys. Rev. D* **93** (2016), 064044, arXiv:1506.02669.
- [37] S. Balakrishnan, T. Faulkner, Z. U. Khandker, and H. Wang, “A general proof of the quantum null energy condition,” *J. High Energy Phys.* **2019** (2019), 20, arXiv:1706.09432.
- [38] E. H. Lieb and M. B. Ruskai, “Proof of the strong subadditivity of quantum-mechanical entropy,” *J. Math. Phys.* **14** (1973), 1938–1941.
- [39] E. H. Wichmann, “Density matrices arising from incomplete measurements,” *J. Math. Phys.* **4** (1963), 884–896.
- [40] F. Hiai, M. Ohya, and M. Tsukada, “Sufficiency, KMS condition and relative entropy in von Neumann algebras,” *Pacific J. Math.* **96** (1981), 99–109.
- [41] M. Takesaki, “Conditional expectations in von Neumann algebras,” *J. Funct. Anal.* **9** (1972), 306–321.
- [42] D. Petz, “Sufficient subalgebras and the relative entropy of states of a von Neumann algebra,” *Commun. Math. Phys.* **105** (1986), 123–131.
- [43] D. Petz, “Sufficiency of channels over von Neumann algebras,” *Quart. J. Math.* **39** (1988), 97–108.



- [44] O. Fawzi and R. Renner, “Quantum conditional mutual information and approximate Markov chains,” *Commun. Math. Phys.* **340** (2015), 575–611, arXiv:1410.0664.
- [45] P. Hayden, R. Jozsa, D. Petz, and A. Winter, “Structure of states which satisfy strong subadditivity of quantum entropy with equality,” *Commun. Math. Phys.* **246** (2004), 359–374.
- [46] S. Doplicher and J. E. Roberts, “A new duality theory for compact groups,” *Invent. Math.* **98** (1989), 157–218.
- [47] S. Doplicher and J. E. Roberts, “Why there is a field algebra with a compact gauge group describing the superselection structure in particle physics,” *Commun. Math. Phys.* **131** (1990), 51–107.
- [48] T. Tannaka, “Über den Dualitätssatz der nichtkommutativen topologischen Gruppen,” *Tohoku Math. J.* **45** (1938), 1–12.
- [49] M. G. Krein, “A principle of duality for a bicomact group and a square block algebra,” *Dokl. Akad. Nauk SSSR* **69** (1949), 725–728.
- [50] H. Georgi and S. L. Glashow, “Unity of all elementary-particle forces,” *Phys. Rev. Lett.* **32** (1974), 438–441.
- [51] S. L. Glashow, J. Iliopoulos, and L. Maiani, “Weak interactions with lepton-hadron symmetry,” *Phys. Rev. D* **2** (1970), 1285–1292.
- [52] E. Witten, “An  $SU(2)$  anomaly,” *Phys. Lett. B* **117** (1982), 324–328.
- [53] S. Dimopoulos, S. Raby, and F. Wilczek, “Supersymmetry and the scale of unification,” *Phys. Rev. D* **24** (1981), 1681–1683.
- [54] U. Amaldi, W. de Boer, and H. Fürstenau, “Comparison of grand unified theories with electroweak and strong coupling constants measured at LEP,” *Phys. Lett. B* **260** (1991), 447–455.
- [55] D. J. Gross and W. Taylor, “Two-dimensional QCD is a string theory,” *Nucl. Phys. B* **400** (1993), 181–208, arXiv:hep-th/9301068.
- [56] F. Peter and H. Weyl, “Die Vollständigkeit der primitiven Darstellungen einer geschlossenen kontinuierlichen Gruppe,” *Math. Ann.* **97** (1927), 737–755.
- [57] A. W. Knap, *Lie Groups Beyond an Introduction*, second edition, Progress in Mathematics **140**, Birkhäuser, 2002.
- [58] A. Bullivant, M. Calçada, Z. Kádár, P. Martin, and J. Faria Martins, “Topological phases from higher gauge symmetry in 3+1D,” *Phys. Rev. B* **95** (2017), 155118, arXiv:1606.06639.
- [59] S. Chandrasekharan and U.-J. Wiese, “Quantum link models: A discrete approach to gauge theories,” *Nucl. Phys. B* **492** (1997), 455–471, arXiv:hep-lat/9609042.
- [60] W. Donnelly and A. C. Wall, “Entanglement entropy of electromagnetic edge modes,” *Phys. Rev. Lett.* **114** (2015), 111603, arXiv:1412.1895.
- [61] F. Pastawski, B. Yoshida, D. Harlow, and J. Preskill, “Holographic quantum error-correcting codes: Toy models for the bulk/boundary correspondence,” *JHEP* **06** (2015), 149, arXiv:1503.06237.

- [62] M. A. Levin and X.-G. Wen, “String-net condensation: A physical mechanism for topological phases,” *Phys. Rev. B* **71** (2005), 045110, arXiv:cond-mat/0404617.
- [63] D. Laghi, G. Carullo, J. Veitch, and W. Del Pozzo, “Quantum black hole spectroscopy: probing the quantum nature of the black hole area using LIGO-Virgo ringdown detections,” *Class. Quantum Grav.* **38** (2021), 095005, arXiv:2011.03816.
- [64] Clay Mathematics Institute, “Yang–Mills & the Mass Gap.” Available at <https://www.claymath.org/millennium/yang-mills-the-maths-gap/>.
- [65] A. Jaffe and E. Witten, “Quantum Yang–Mills Theory,” official Clay Mathematics Institute problem description. Available at <https://www.claymath.org/wp-content/uploads/2022/06/yangmills.pdf>.
- [66] L. Wu, “Poincaré and transportation inequalities for Gibbs measures under the Dobrushin uniqueness condition,” *Annals of Probability* **34** (2006), 1960–1989, arXiv:math/0611635.
- [67] K. Osterwalder and R. Schrader, “Axioms for Euclidean Green’s functions,” *Communications in Mathematical Physics* **31** (1973), 83–112.
- [68] K. Osterwalder and R. Schrader, “Axioms for Euclidean Green’s functions II,” *Communications in Mathematical Physics* **42** (1975), 281–305.
- [69] Particle Data Group, “Review of Particle Physics,” *Phys. Rev. D* **110** (2024), 030001. Available at <https://pdg.lbl.gov/2024/download/db2024.pdf>.
- [70] Particle Data Group, “Electroweak Model and Constraints on New Physics,” Available at <https://pdg.lbl.gov/2024/reviews/rpp2024-rev-standard-model.pdf>.

## Statements and Declarations

**Funding declaration:** The authors declare that no external grants or dedicated third-party research funding were received for the preparation of this manuscript.

**Author affiliations declaration:** Bernhard Mueller is affiliated with Pragma Inc., Washington, USA. Alexander Osika is affiliated with SNRGY Inc., Gothenburg, Sweden. No institutional affiliations were declared for Mario Ponder, Kai Xue, Ben Cassie, Peter Nguyen, Maarten Antoine Visser, Kale Arnav Anirudha, David Matscheko, Jonathan Hill, or William T. Glynn in this submission. Correspondence email addresses are provided in the correspondence declaration.

**Consent to Participate declaration:** not applicable.

**Consent to Publish declaration:** not applicable.

**Author Contribution declaration:** Bernhard Mueller developed the OPH concept, mathematical formulation, synthesis, main manuscript text, and submission package. Alexander Osika contributed to the OPH/Karma and OMEGA hardware path, experimental prototype framing, hardware evidence boundary, and submission review. Mario Ponder contributed the exact conditional port-current construction, verifier, proof packet, receipt, and tests, as well as proof and validation work on the hierarchy certificates, the edge-entropy area-law route to the Einstein/Newton coupling, coupling normalization, particle and neutrino certificates, and manuscript/book consistency. Kai Xue contributed consistency checks, simulator review, visualization review, and prototype engineering review. Ben Cassie contributed hardware-evidence review, implementation-claim review, experimental-prototype framing, and submission-package review. Peter Nguyen contributed

proof review, branch-boundary analysis, Yang–Mills and compact-gauge analysis, finite packet-quotient closure, fine-structure certification, and manuscript consistency. Maarten Antoine Visser contributed to the  $P/N=24$  repair-resonance material, consensus-protocol consistency review, and physics feedback on the foundations and emergent-spacetime framing. Kale Arnav Anirudha contributed consensus-paper material, formal review of the consensus-protocol surface, QBFT assumptions, FLP citation handling, repair-map definitions, QECC notation, asynchronous bound wording, and the BFT/QECC appendix material. David Matscheko provided the finite proof review for consensus repair, Rule-90 boundary reconstruction, Route-A assembly, finite modular-flow, Einstein-algebra, hypercharge/ $Z_6$ , scalar-channel/collar/channel-bridge, dark-sector, energy-cage, and run-matrix theorems; he also reviewed the dark-sector and coherent-matter susceptibility arguments and contributed to the anti-gravity experiment path. William T. Glynn contributed simulator development, computational implementation, reproducible-build infrastructure, and release-validation work in the main OPH research repository. All authors reviewed and approved the manuscript.

**Correspondence declaration:** Correspondence should be addressed to Bernhard Mueller, [bernhard@floatingpragma.ai](mailto:bernhard@floatingpragma.ai), and Alexander Osika, [alex@snrgystudios.com](mailto:alex@snrgystudios.com).

**Data Availability declaration:** Source manuscripts, public PDFs, simulation code, and implementation notes are available through the OPH repository and its linked public artifacts. Some hardware and prototype evidence requires a sanitized ancillary audit bundle before public disclosure because raw logs may include device configuration, operational metadata, or private infrastructure details. The intended audit bundle contains source hashes, simulator configs, finite-receipt JSON, verifier code, hardware acceptance records, calibration logs, and selected run traces sufficient to replay the stated implementation claims.

**Code Availability declaration:** The OPH simulator, finite-consensus harnesses, visualization payload schemas, and public paper build sources are maintained in the OPH project repositories. Submission artifacts include a standalone source file and a source bundle so that the manuscript can be rebuilt without relying on private local paths.

**Ethics declaration:** not applicable.

**Competing Interests declaration:** The authors participate in OPH-related research and implementation programs, including prototype hardware and simulator development. The authors declare no external competing financial interests for this manuscript.

**Use of AI-assisted tools declaration:** AI-assisted tools were used for language editing, drafting assistance, formatting support, and submission-package preparation. The authors reviewed and are responsible for all scientific content, claims, references, and declarations.

| Domain          | Output                                                                                       | Mathematical role                                                                                                                                                                                                                     | Physical boundary                                                                                                                                                                                                                                                                                         |
|-----------------|----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Carrier modes   | two transverse Maxwell modes; 2 dim $G$ perturbative Yang–Mills modes; two Einstein TT modes | symmetry-protected massless classical kernels on the stated action branches                                                                                                                                                           | no quantum zero-mass particle row without the physical Hilbert-space, pole-residue, and phase receipts                                                                                                                                                                                                    |
| $W/Z$           | source-only chart coordinates (80.330, 91.119) GeV; no source-only physical mass emitted     | strict source-defined zero-selector branch; exact selected-carrier chart; separate strict scalar consumer with principal-sheet zero exclusion and one simple channel-specific lower-half algebraic-chart zero for each of $W$ and $Z$ | the external fixture is not composed with the OPH chart. Neither declared chart is identified with the physical resonance sheet; unique continuation, sign bridge, full-matrix Laurent residue, current amplitude, independent replay, and $W/Z$ mass comparison remain absent                            |
| Fine structure  | four distinct coordinates                                                                    | source witness, mixed diagnostic, empirical closure, and measurement                                                                                                                                                                  | the empirical interval misses the measured endpoint. A physical source-only statement requires the source-derived hadronic spectral backend, target-independent map selection, and the typed same-quantity bridge                                                                                         |
| Higgs/top       | no nonzero source-only physical mass emitted                                                 | double-criticality family from the gauge sector; boundary-scale candidate fixed before comparison; declared-surface calibration fit kept separate                                                                                     | a physical statement requires the boundary-scale selection theorem together with the source root, physical scale, running, matching, rigidity, provenance, uncertainty, and complex-pole construction                                                                                                     |
| Charged leptons | no nonzero source-only physical mass emitted                                                 | exact positive-chamber $C_3$ circulant identity; conditional finite tracial-GNS balance; target-informed response diagnostic; conditional nature/pole transport                                                                       | the exact identity gives $Q = 1/3 + (2/3)( b/a )^2$ , and the finite packet gives $ b/a  = 1/\sqrt{2}$ under its declared event-block hypotheses. Physical family attachment, phase, mass ratios, source selection, interacting kernel, infrared completion, and cofinal refinement are separate premises |
| Quarks          | no nonzero source-only physical mass emitted                                                 | common-scale reciprocal-ray falsification; rejected register-Clebsch candidate; exact six-scalar interface; restricted source-spread lower bound; scoped real-axis no-go                                                              | the reciprocal-ray candidate fails across the tested common scales. The register-Clebsch pairing result permits separate invariant channels, and its restricted unordered multiset is exact and runtime-target-free conditional on the declared alphabet. Those facts neither                             |